

=====

PRONOTES - COMPLETE FULL CODEBASE EXPORT (ALL CODE + TESTS)
Repository: C:\Users\idree\Documents\pronotesdevelop
Commit: 6a369da (Shape recognition: turn hand-drawn shapes into exact ones)
Total Files: 194
Total Lines of Code: 75751

=====

TABLE OF CONTENTS:

1. android/app/src/main/java/com/idrees/pronotes/GallerySaverPlugin.java	(148 lines,	6416 bytes)
2. android/app/src/main/java/com/idrees/pronotes/MainActivity.java	(484 lines,	22208 bytes)
3. capacitor.config.ts	(23 lines,	684 bytes)
4. components.json	(21 lines,	430 bytes)
5. electron/main.js	(151 lines,	5275 bytes)
6. eslint.config.mjs	(55 lines,	1944 bytes)
7. next.config.ts	(83 lines,	3874 bytes)
8. package.json	(120 lines,	3415 bytes)
9. postcss.config.mjs	(5 lines,	81 bytes)
10. public/manifest.json	(75 lines,	1680 bytes)
11. public/sw.js	(131 lines,	3947 bytes)
12. scripts/audit.mjs	(83 lines,	3102 bytes)
13. scripts/build-apk.mjs	(61 lines,	2891 bytes)
14. scripts/build-desktop.mjs	(31 lines,	1008 bytes)
15. scripts/build-static.mjs	(171 lines,	7455 bytes)
16. scripts/build.mjs	(33 lines,	1162 bytes)
17. scripts/deploy-android.mjs	(119 lines,	4162 bytes)
18. scripts/generate-icons.js	(156 lines,	5334 bytes)
19. scripts/make-android-icons.mjs	(51 lines,	1779 bytes)
20. scripts/make-ico.mjs	(48 lines,	1724 bytes)
21. scripts/sync-pdf-worker.mjs	(32 lines,	1298 bytes)
22. scripts/test-runner.mjs	(222 lines,	6673 bytes)
23. src/app/globals.css	(195 lines,	7525 bytes)
24. src/app/layout.tsx	(102 lines,	2576 bytes)
25. src/app/page.tsx	(23 lines,	560 bytes)
26. src/components/pwa/PWARegister.tsx	(134 lines,	5068 bytes)
27. src/components/theme-provider.tsx	(11 lines,	299 bytes)
28. src/components/ui/button.tsx	(59 lines,	2123 bytes)
29. src/components/ui/context-menu.tsx	(252 lines,	8222 bytes)
30. src/components/ui/dialog.tsx	(143 lines,	3982 bytes)
31. src/components/ui/dropdown-menu.tsx	(257 lines,	8284 bytes)
32. src/components/ui/input.tsx	(21 lines,	967 bytes)
33. src/components/ui/popover.tsx	(48 lines,	1635 bytes)
34. src/components/ui/scroll-area.tsx	(58 lines,	1645 bytes)
35. src/components/ui/separator.tsx	(28 lines,	699 bytes)
36. src/components/ui/slider.tsx	(63 lines,	2001 bytes)
37. src/components/ui/toast.tsx	(129 lines,	4829 bytes)
38. src/components/ui/toaster.tsx	(35 lines,	785 bytes)
39. src/components/ui/tooltip.tsx	(61 lines,	1891 bytes)
40. src/components/whiteboard/AccountSection.tsx	(243 lines,	8097 bytes)
41. src/components/whiteboard/BacklinksPanel.tsx	(207 lines,	8497 bytes)
42. src/components/whiteboard/CanvasContextMenu.tsx	(102 lines,	3377 bytes)
43. src/components/whiteboard/CaptureHud.tsx	(112 lines,	5066 bytes)
44. src/components/whiteboard/CaptureMarquee.tsx	(122 lines,	4244 bytes)
45. src/components/whiteboard/CapturePanel.tsx	(834 lines,	33802 bytes)
46. src/components/whiteboard/ColorSwatches.tsx	(147 lines,	4142 bytes)
47. src/components/whiteboard/DockControls.tsx	(122 lines,	4535 bytes)
48. src/components/whiteboard/DuplicateNotebookModal.tsx	(532 lines,	19640 bytes)
49. src/components/whiteboard/ElementRenderer.tsx	(1133 lines,	35757 bytes)
50. src/components/whiteboard/ExportModal.tsx	(630 lines,	24413 bytes)
51. src/components/whiteboard/FavoritePensBar.tsx	(772 lines,	30539 bytes)
52. src/components/whiteboard/FileDropZone.tsx	(305 lines,	12358 bytes)
53. src/components/whiteboard/GraphView.tsx	(1519 lines,	60179 bytes)
54. src/components/whiteboard/ImageActions.tsx	(146 lines,	5960 bytes)
55. src/components/whiteboard/ImageInsertSheet.tsx	(472 lines,	19106 bytes)
56. src/components/whiteboard/ImageStudio.tsx	(477 lines,	18628 bytes)
57. src/components/whiteboard/InlineTextEditor.tsx	(1577 lines,	61903 bytes)
58. src/components/whiteboard/InsertPagesModal.tsx	(438 lines,	17647 bytes)
59. src/components/whiteboard/KeyboardShortcutsModal.tsx	(258 lines,	12290 bytes)
60. src/components/whiteboard/LaserPointer.tsx	(332 lines,	10664 bytes)
61. src/components/whiteboard/LinkLegend.tsx	(73 lines,	2795 bytes)
62. src/components/whiteboard/LinkManager.tsx	(447 lines,	18380 bytes)
63. src/components/whiteboard/LockedModeBar.tsx	(134 lines,	5696 bytes)
64. src/components/whiteboard/Minimap.tsx	(223 lines,	8008 bytes)
65. src/components/whiteboard/MoveToDialog.tsx	(549 lines,	19666 bytes)
66. src/components/whiteboard/NotebookPages.tsx	(910 lines,	35396 bytes)
67. src/components/whiteboard/PaperPattern.tsx	(57 lines,	1816 bytes)
68. src/components/whiteboard/PdfPageLayer.tsx	(118 lines,	3643 bytes)

69. src/components/whiteboard/PdfTextSelect.tsx	(772 lines,	29912 bytes)
70. src/components/whiteboard/PenDiagnostics.tsx	(283 lines,	10530 bytes)
71. src/components/whiteboard/PropertiesPanel.tsx	(635 lines,	27793 bytes)
72. src/components/whiteboard/SearchModal.tsx	(965 lines,	38449 bytes)
73. src/components/whiteboard/SelectToolButton.tsx	(401 lines,	16789 bytes)
74. src/components/whiteboard/SelectionActions.tsx	(217 lines,	9270 bytes)
75. src/components/whiteboard/SelectionOverlay.tsx	(189 lines,	6007 bytes)
76. src/components/whiteboard/SettingsPanel.tsx	(797 lines,	36425 bytes)
77. src/components/whiteboard/ShapeAssist.tsx	(90 lines,	2787 bytes)
78. src/components/whiteboard/Sidebar.tsx	(2049 lines,	81914 bytes)
79. src/components/whiteboard/SplitPane.tsx	(626 lines,	24977 bytes)
80. src/components/whiteboard/SplitPicker.tsx	(242 lines,	9322 bytes)
81. src/components/whiteboard/StepperControl.tsx	(151 lines,	4910 bytes)
82. src/components/whiteboard/SyncPanel.tsx	(327 lines,	12565 bytes)
83. src/components/whiteboard/ThemeToggle.tsx	(99 lines,	3362 bytes)
84. src/components/whiteboard/ToolProperties.tsx	(940 lines,	36962 bytes)
85. src/components/whiteboard/Toolbar.tsx	(995 lines,	39301 bytes)
86. src/components/whiteboard/ToolbarSettings.tsx	(176 lines,	6249 bytes)
87. src/components/whiteboard/TutorialModal.tsx	(348 lines,	12659 bytes)
88. src/components/whiteboard/WhiteboardApp.tsx	(724 lines,	31767 bytes)
89. src/components/whiteboard/WhiteboardCanvas.tsx	(4614 lines,	188043 bytes)
90. src/components/whiteboard/trashToast.tsx	(73 lines,	2329 bytes)
91. src/hooks/use-long-press.ts	(51 lines,	1809 bytes)
92. src/hooks/use-toast.ts	(194 lines,	3917 bytes)
93. src/lib/utils.ts	(6 lines,	166 bytes)
94. src/lib/whiteboard/account.ts	(183 lines,	6439 bytes)
95. src/lib/whiteboard/appClipboard.ts	(88 lines,	3047 bytes)
96. src/lib/whiteboard/capture.ts	(626 lines,	25165 bytes)
97. src/lib/whiteboard/export.ts	(1229 lines,	52326 bytes)
98. src/lib/whiteboard/fflate.d.ts	(10 lines,	439 bytes)
99. src/lib/whiteboard/gallery.ts	(53 lines,	2227 bytes)
100. src/lib/whiteboard/geometry.ts	(1026 lines,	33044 bytes)
101. src/lib/whiteboard/grid.ts	(31 lines,	843 bytes)
102. src/lib/whiteboard/imageEdit.ts	(253 lines,	9737 bytes)
103. src/lib/whiteboard/imageImport.ts	(59 lines,	2346 bytes)
104. src/lib/whiteboard/imagestore.ts	(147 lines,	5782 bytes)
105. src/lib/whiteboard/input.ts	(241 lines,	9286 bytes)
106. src/lib/whiteboard/links.ts	(725 lines,	22114 bytes)
107. src/lib/whiteboard/notebook.ts	(1034 lines,	37369 bytes)
108. src/lib/whiteboard/officeImport.ts	(738 lines,	30204 bytes)
109. src/lib/whiteboard/onedrive.ts	(353 lines,	11737 bytes)
110. src/lib/whiteboard/palm.ts	(158 lines,	6510 bytes)
111. src/lib/whiteboard/pdf.ts	(161 lines,	6166 bytes)
112. src/lib/whiteboard/pdfText.ts	(801 lines,	29209 bytes)
113. src/lib/whiteboard/pdfTextIndex.ts	(246 lines,	9847 bytes)
114. src/lib/whiteboard/pdfTextStore.ts	(135 lines,	5039 bytes)
115. src/lib/whiteboard/pdfstore.ts	(115 lines,	4481 bytes)
116. src/lib/whiteboard/persistence.ts	(916 lines,	34852 bytes)
117. src/lib/whiteboard/placement.ts	(279 lines,	10183 bytes)
118. src/lib/whiteboard/pointerHandoff.ts	(90 lines,	3723 bytes)
119. src/lib/whiteboard/search.ts	(678 lines,	24167 bytes)
120. src/lib/whiteboard/shapeRecognition.ts	(894 lines,	34400 bytes)
121. src/lib/whiteboard/share.ts	(193 lines,	7449 bytes)
122. src/lib/whiteboard/shortcuts.ts	(32 lines,	1059 bytes)
123. src/lib/whiteboard/sort.ts	(25 lines,	895 bytes)
124. src/lib/whiteboard/speechRecognition.ts	(866 lines,	32496 bytes)
125. src/lib/whiteboard/split.ts	(16 lines,	587 bytes)
126. src/lib/whiteboard/stroke.ts	(760 lines,	26481 bytes)
127. src/lib/whiteboard/textParser.ts	(550 lines,	17455 bytes)
128. src/lib/whiteboard/textbox.ts	(25 lines,	1377 bytes)
129. src/lib/whiteboard/types.ts	(1407 lines,	45512 bytes)
130. src/lib/whiteboard/workspace.ts	(237 lines,	8023 bytes)
131. src/lib/whiteboard/workspaceDb.ts	(233 lines,	9269 bytes)
132. src/store/capture.ts	(55 lines,	1958 bytes)
133. src/store/whiteboard.ts	(3538 lines,	126991 bytes)
134. tailwind.config.ts	(64 lines,	1696 bytes)
135. tests/adversarial_challenger1.test.ts	(835 lines,	27885 bytes)
136. tests/adversarial_zoom_gestures.test.ts	(493 lines,	16375 bytes)
137. tests/capture.test.ts	(361 lines,	15960 bytes)
138. tests/challenger_2_empirical.test.ts	(461 lines,	23372 bytes)
139. tests/challenger_geometry_stress.test.ts	(450 lines,	18190 bytes)
140. tests/challenger_stroke_stress.test.ts	(337 lines,	16834 bytes)
141. tests/dictation_continuation.test.ts	(295 lines,	11340 bytes)
142. tests/dictation_robustness.test.ts	(304 lines,	9958 bytes)
143. tests/duplicate_notebook.test.ts	(496 lines,	17813 bytes)
144. tests/e2e/m3-pwa-branding.test.js	(1080 lines,	37790 bytes)
145. tests/e2e/mutation-challenge.js	(183 lines,	6994 bytes)
146. tests/e2e/runner.js	(183 lines,	6969 bytes)

147. tests/e2e/test-helpers.js	(459 lines,	13379 bytes)
148. tests/e2e/tier1-features.test.js	(1580 lines,	75932 bytes)
149. tests/e2e/tier2-boundaries.test.js	(1158 lines,	57539 bytes)
150. tests/e2e/tier3-pairwise.test.js	(459 lines,	27112 bytes)
151. tests/e2e/tier4-scenarios.test.js	(603 lines,	23107 bytes)
152. tests/eraser_settle.test.ts	(104 lines,	3596 bytes)
153. tests/export_geometry.test.ts	(199 lines,	7814 bytes)
154. tests/fixtures/shape_user_strokes.json	(1 lines,	48172 bytes)
155. tests/history.test.ts	(140 lines,	4503 bytes)
156. tests/image_placement.test.ts	(146 lines,	4919 bytes)
157. tests/image_studio.test.ts	(160 lines,	6186 bytes)
158. tests/interactive_knowledge_graph.test.ts	(233 lines,	8698 bytes)
159. tests/m1_adversarial_spen_stress.test.ts	(667 lines,	27897 bytes)
160. tests/m1_challenger_2_adversarial.test.ts	(739 lines,	27474 bytes)
161. tests/milestone1.test.ts	(407 lines,	16015 bytes)
162. tests/milestone2_tools_qa.test.ts	(429 lines,	13772 bytes)
163. tests/milestone3_features_r3.test.ts	(306 lines,	9830 bytes)
164. tests/notebook.test.ts	(802 lines,	31288 bytes)
165. tests/notebook_camera.test.ts	(348 lines,	15778 bytes)
166. tests/notebook_store.test.ts	(1055 lines,	37538 bytes)
167. tests/office_import.test.ts	(150 lines,	7318 bytes)
168. tests/onedrive.test.ts	(214 lines,	6655 bytes)
169. tests/pdf_text.test.ts	(289 lines,	12793 bytes)
170. tests/pen_dynamics_and_stabilization.test.ts	(321 lines,	11702 bytes)
171. tests/pens_bar_placement.test.ts	(92 lines,	3318 bytes)
172. tests/pointer_handoff.test.ts	(313 lines,	10925 bytes)
173. tests/pro_text_tool.test.ts	(319 lines,	11825 bytes)
174. tests/regression_fix_pass.test.ts	(1447 lines,	66989 bytes)
175. tests/review_fixes.test.ts	(179 lines,	7037 bytes)
176. tests/search_scopes.test.ts	(260 lines,	11771 bytes)
177. tests/select_modes.test.ts	(199 lines,	5801 bytes)
178. tests/shape_recognition.test.ts	(408 lines,	16276 bytes)
179. tests/shape_tools.test.ts	(148 lines,	6024 bytes)
180. tests/share_insert.test.ts	(497 lines,	18482 bytes)
181. tests/sidebar_batch.test.ts	(130 lines,	3975 bytes)
182. tests/sidebar_trash.test.ts	(164 lines,	5723 bytes)
183. tests/spen_deep_audit.test.ts	(514 lines,	19930 bytes)
184. tests/spen_review.test.ts	(114 lines,	4512 bytes)
185. tests/split_view.test.ts	(435 lines,	16237 bytes)
186. tests/stylus_button.test.ts	(116 lines,	4788 bytes)
187. tests/text_editing.test.ts	(323 lines,	11868 bytes)
188. tests/toolbar_fixes.test.ts	(152 lines,	5867 bytes)
189. tests/toolbar_laser.test.ts	(90 lines,	3637 bytes)
190. tests/top_tools_and_sidebar.test.ts	(550 lines,	21796 bytes)
191. tests/wheel_device.test.ts	(210 lines,	8213 bytes)
192. tests/workspace_store_v2.test.ts	(434 lines,	16691 bytes)
193. tsconfig.json	(42 lines,	732 bytes)
194. tsconfig.test.json	(21 lines,	402 bytes)

```
=====
// === START FILE: android/app/src/main/java/com/idrees/pronotes/GallerySaverPlugin.java ===
// File 1 of 194 | Lines: 148 | Size: 6416 bytes
=====
```

```
package com.idrees.pronotes;

import android.Manifest;
import android.content.ContentResolver;
import android.content.ContentValues;
import android.media.MediaScannerConnection;
import android.net.Uri;
import android.os.Build;
import android.os.Environment;
import android.provider.MediaStore;
import android.util.Base64;

import com.getcapacitor.JSObject;
import com.getcapacitor.PermissionState;
import com.getcapacitor.Plugin;
import com.getcapacitor.PluginCall;
import com.getcapacitor.PluginMethod;
import com.getcapacitor.annotation.CapacitorPlugin;
import com.getcapacitor.annotation.Permission;
import com.getcapacitor.annotation.PermissionCallback;

import java.io.File;
```

```

import java.io.FileOutputStream;
import java.io.OutputStream;

/**
 * Saves a picture where the Gallery / Photos app will show it.
 *
 * Why this is native
 * -----
 * The export path (share.ts) writes to the app's CACHE and opens the share
 * sheet. That is right for sending a file, but nothing written there - or
 * anywhere else @capacitor/filesystem can reach on Android 10+ - is indexed
 * by the media store, so the picture never appears in Gallery. Scoped storage
 * leaves exactly one sanctioned way in: insert the image into MediaStore.
 *
 * Android 10+ (API 29+) MediaStore insert into Pictures/<album>, marked
 * IS_PENDING while the bytes are written so Gallery
 * never shows a half-written file. Needs NO
 * permission: an app may always add its own media.
 * Android 5-9 The public Pictures/<album> folder plus a media
 * scan, which needs WRITE_EXTERNAL_STORAGE (declared
 * with maxSdkVersion 28, so newer devices never see
 * the prompt).
 *
 * JS: registerPlugin("GallerySaver").saveImage({ data, filename, album })
 * where `data` is base64 PNG/JPEG bytes (no data: prefix). Resolves with
 * { uri } of the saved picture.
 */
@CapacitorPlugin(
  name = "GallerySaver",
  permissions = {
    @Permission(strings = { Manifest.permission.WRITE_EXTERNAL_STORAGE }, alias = "storage")
  }
)
public class GallerySaverPlugin extends Plugin {

  @PluginMethod
  public void saveImage(PluginCall call) {
    if (Build.VERSION.SDK_INT < Build.VERSION_CODES.Q && getPermissionState("storage") != PermissionState.
      GRANTED) {
      requestPermissionForAlias("storage", call, "storagePermissionCallback");
      return;
    }
    save(call);
  }

  @PermissionCallback
  private void storagePermissionCallback(PluginCall call) {
    if (getPermissionState("storage") == PermissionState.GRANTED) {
      save(call);
    } else {
      call.reject("Permission to save pictures was not given.", "PERMISSION_DENIED");
    }
  }

  private void save(PluginCall call) {
    String data = call.getString("data");
    String filename = call.getString("filename", "ProNotes.png");
    String album = call.getString("album", "ProNotes");
    if (data == null || data.isEmpty()) {
      call.reject("No picture was given.");
      return;
    }
    int comma = data.indexOf(',');
    if (data.startsWith("data:") && comma >= 0) data = data.substring(comma + 1);

    final byte[] bytes;
    try {
      bytes = Base64.decode(data, Base64.DEFAULT);
    } catch (IllegalArgumentException e) {
      call.reject("The picture data could not be read.", e);
      return;
    }
    String mime = filename.toLowerCase().endsWith(".jpg") || filename.toLowerCase().endsWith(".jpeg")
      ? "image/jpeg"
      : "image/png";

    try {
      Uri uri = Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q

```

```

        ? saveViaMediaStore(bytes, filename, album, mime)
        : saveLegacy(bytes, filename, album, mime);
    JSONObject ret = new JSONObject();
    ret.put("uri", uri != null ? uri.toString() : "");
    call.resolve(ret);
} catch (Exception e) {
    call.reject("The picture could not be saved to the gallery: " + e.getMessage(), e);
}
}

private Uri saveViaMediaStore(byte[] bytes, String filename, String album, String mime) throws Exception {
    ContentResolver resolver = getContext().getContentResolver();
    ContentValues values = new ContentValues();
    values.put(MediaStore.Images.Media.DISPLAY_NAME, filename);
    values.put(MediaStore.Images.Media.MIME_TYPE, mime);
    values.put(MediaStore.Images.Media.RELATIVE_PATH, Environment.DIRECTORY_PICTURES + File.separator + al
        bum);
    values.put(MediaStore.Images.Media.DATE_TAKEN, System.currentTimeMillis());
    values.put(MediaStore.Images.Media.IS_PENDING, 1);

    Uri collection = MediaStore.Images.Media.getContentUri(MediaStore.VOLUME_EXTERNAL_PRIMARY);
    Uri uri = resolver.insert(collection, values);
    if (uri == null) throw new IllegalStateException("MediaStore refused the new picture");
    try (OutputStream out = resolver.openOutputStream(uri)) {
        if (out == null) throw new IllegalStateException("could not open the new picture for writing");
        out.write(bytes);
    } catch (Exception e) {
        // Leave no empty, pending entry behind in the gallery.
        resolver.delete(uri, null, null);
        throw e;
    }
    ContentValues done = new ContentValues();
    done.put(MediaStore.Images.Media.IS_PENDING, 0);
    resolver.update(uri, done, null, null);
    return uri;
}

@SuppressWarnings("deprecation")
private Uri saveLegacy(byte[] bytes, String filename, String album, String mime) throws Exception {
    File dir = new File(Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_PICTURES), alb
        um);
    if (!dir.exists() && !dir.mkdirs()) throw new IllegalStateException("could not create " + dir);
    File file = new File(dir, filename);
    try (FileOutputStream out = new FileOutputStream(file)) {
        out.write(bytes);
    }
    // Without a scan the file exists but the gallery does not know it.
    MediaScannerConnection.scanFile(getContext(), new String[] { file.getAbsolutePath() }, new String[] {
        mime }, null);
    return Uri.fromFile(file);
}
}

```

```

=====
// === END FILE: android/app/src/main/java/com/idrees/pronotes/GallerySaverPlugin.java ===
=====

```

```

=====
// === START FILE: android/app/src/main/java/com/idrees/pronotes/MainActivity.java ===
// File 2 of 194 | Lines: 484 | Size: 22208 bytes
=====

```

```

package com.idrees.pronotes;

import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.database.Cursor;
import android.net.Uri;
import android.provider.OpenableColumns;
import android.os.Build;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.MotionEvent;
import android.webkit.PermissionRequest;
import android.webkit.WebView;

```

```

import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

import java.io.File;
import java.io.FileOutputStream;
import java.io.InputStream;
import java.io.OutputStream;
import java.util.ArrayList;
import java.util.List;
import org.json.JSONArray;
import org.json.JSONObject;

import com.getcapacitor.BridgeActivity;
import com.getcapacitor.BridgeWebChromeClient;

/**
 * Bridges the stylus side button and eraser tip into the web layer.
 *
 * Why this is native
 * -----
 * Chromium's WebView does not surface the S Pen side button to web content at
 * all – the in-app diagnostic recorded nothing while it was held, so there is
 * no JavaScript fix available. Android itself does report it, on every
 * MotionEvent, and because this is a Capacitor app the Activity sees those
 * events before the WebView consumes them.
 *
 * Reading the button correctly, though, is not just "check getButtonState()".
 * The first version did exactly that on every event and was visibly buggy:
 *
 * 1. Android CLEARS the button state on ACTION_UP. Lifting the nib while
 *    still holding the button therefore read as a release – the tool flipped
 *    back to the pen, and the next hover flipped it to the eraser again. That
 *    is the flicker.
 *
 * 2. ACTION_HOVER_EXIT does NOT only mean "the pen left". It also fires
 *    immediately before every ACTION_DOWN, as the pen crosses from hovering
 *    into contact. Treating it as "pen gone" cancels the gesture mid-stroke.
 *
 * 3. The eraser TIP is a tool type, not a button (Android's stylus guide is
 *    explicit: `TOOL_TYPE_ERASER == event.getToolType(i)`). Folding it into
 *    the button test made one latch answer two different questions.
 *
 * So the rules below are deliberate:
 *
 * ACTION_BUTTON_PRESS / ACTION_BUTTON_RELEASE authoritative – these exist
 *                                     for precisely this purpose
 * DOWN / MOVE / HOVER_ENTER / HOVER_MOVE sync from getButtonState()
 * UP / CANCEL IGNORE the button state; it
 *                                     has been cleared and does not
 *                                     mean the button came up
 * HOVER_EXIT maybe the pen left, maybe the
 *                                     nib is landing – decided by a
 *                                     short timer any further
 *                                     stylus event cancels
 *
 * Not covered: the S Pen's Bluetooth remote button (Air Actions). That is
 * delivered over BLE to Samsung's S Pen Remote SDK and never reaches an
 * ordinary Activity.
 */
public class MainActivity extends BridgeActivity {

    private static final int REQUEST_RECORD_AUDIO_PERMISSION = 2001;
    private PermissionRequest pendingAudioPermissionRequest = null;

    @Override
    public void onCreate(Bundle savedInstanceState) {
        // App-local plugins must be registered BEFORE the bridge starts.
        // GallerySaver puts screenshots where the Gallery app shows them.
        registerPlugin(GallerySaverPlugin.class);
        super.onCreate(savedInstanceState);

        // Configure WebView to grant audio capture permissions for Web Speech / getUserMedia
        if (getBridge() != null && getBridge().getWebView() != null) {
            getBridge().getWebView().setWebChromeClient(new BridgeWebChromeClient(getBridge())) {
                @Override
                public void onPermissionRequest(final PermissionRequest request) {
                    boolean audioRequested = false;

```

```

        for (String r : request.getResources()) {
            if (PermissionRequest.RESOURCE_AUDIO_CAPTURE.equals(r)) {
                audioRequested = true;
                break;
            }
        }
        if (!audioRequested) {
            // Not ours to decide: Capacitor's own handling asks
            // Android for whatever the page wants (camera...).
            // Granting blindly handed out access nobody approved.
            super.onPermissionRequest(request);
            return;
        }
        runOnUiThread(() -> {
            // Only the microphone is granted here, even when the
            // page asked for more in the same request.
            final String[] audioOnly = new String[]{PermissionRequest.RESOURCE_AUDIO_CAPTURE};
            if (ContextCompat.checkSelfPermission(MainActivity.this, Manifest.permission.RECORD_AUDIO)
DIO)
                == PackageManager.PERMISSION_GRANTED) {
                request.grant(audioOnly);
            } else {
                // A request still waiting is answered, never dropped:
                // an unanswered one leaves the page waiting forever.
                if (pendingAudioPermissionRequest != null) {
                    pendingAudioPermissionRequest.deny();
                }
                pendingAudioPermissionRequest = request;
                ActivityCompat.requestPermissions(
                    MainActivity.this,
                    new String[]{Manifest.permission.RECORD_AUDIO},
                    REQUEST_RECORD_AUDIO_PERMISSION
                );
            }
        });
    }
});
}

handleIncoming(getIntent());
}

// -----
// Files opened or shared from other apps ("Open with" / "Share")
// -----
// Android hands us a content:// URI that only this Activity may read, and
// only for a while. So the bytes are copied into the app's cache first,
// then the page is told the file's path; it fetches it through Capacitor's
// local server (/capacitor_file_/...) and imports it like a dropped file.
// The page may still be loading (a cold start from WhatsApp), so delivery
// is retried until the page says it took the files.

private static final long DELIVER_RETRY_MS = 400;
private static final int DELIVER_MAX_TRIES = 150; // ~60 s

@Override
protected void onNewIntent(Intent intent) {
    super.onNewIntent(intent);
    setIntent(intent);
    handleIncoming(intent);
}

private void handleIncoming(Intent intent) {
    if (intent == null) return;
    final String action = intent.getAction();
    final List<Uri> uris = new ArrayList<>();
    if (Intent.ACTION_VIEW.equals(action) && intent.getData() != null) {
        uris.add(intent.getData());
    } else if (Intent.ACTION_SEND.equals(action)) {
        Uri u = intent.getParcelableExtra(Intent.EXTRA_STREAM);
        if (u != null) uris.add(u);
    } else if (Intent.ACTION_SEND_MULTIPLE.equals(action)) {
        ArrayList<Uri> list = intent.getParcelableArrayListExtra(Intent.EXTRA_STREAM);
        if (list != null) uris.addAll(list);
    }
    if (uris.isEmpty()) return;
    final String fallbackType = intent.getType();
    // Handled once: rotating the screen must not import the file again.

```

```

intent.setAction(Intent.ACTION_MAIN);
intent.setData(null);

new Thread() -> {
    JSONArray files = new JSONArray();
    File dir = new File(getCacheDir(), "incoming");
    //noinspection ResultOfMethodCallIgnored
    dir.mkdirs();
    // Copies are only needed until the page has imported them (it
    // keeps its own copy in IndexedDB), so a day-old one is garbage.
    File[] old = dir.listFiles();
    if (old != null) {
        long cutoff = System.currentTimeMillis() - 24L * 60 * 60 * 1000;
        for (File f : old) {
            //noinspection ResultOfMethodCallIgnored
            if (f.lastModified() < cutoff) f.delete();
        }
    }
    for (Uri uri : uris) {
        try {
            String name = displayName(uri);
            String type = getContentResolver().getType(uri);
            if (type == null) type = fallbackType;
            if (type == null && name.toLowerCase().endsWith(".pdf")) type = "application/pdf";
            File out = new File(dir, System.currentTimeMillis() + "_" + name.replaceAll("[\\/:?*\\<>|]",
            ", "_));
            try (InputStream in = getContentResolver().openInputStream(uri);
                OutputStream os = new FileOutputStream(out)) {
                if (in == null) continue;
                byte[] buf = new byte[64 * 1024];
                int n;
                while ((n = in.read(buf)) > 0) os.write(buf, 0, n);
            }
            JSONObject f = new JSONObject();
            f.put("path", out.getAbsolutePath());
            f.put("name", name);
            f.put("type", type == null ? "" : type);
            files.put(f);
        } catch (Exception e) {
            try {
                JSONObject f = new JSONObject();
                f.put("error", "Could not read a file from the other app: " + e.getMessage());
                files.put(f);
            } catch (Exception ignored) {
                // nothing more to report
            }
        }
    }
    if (files.length() > 0) deliver(files.toString(), 0);
}).start();
}

private String displayName(Uri uri) {
    String name = null;
    try (Cursor c = getContentResolver().query(uri, new String[]{OpenableColumns.DISPLAY_NAME}, null, null
    , null)) {
        if (c != null && c.moveToFirst()) name = c.getString(0);
    } catch (Exception ignored) {
        // some providers do not answer; fall back to the path
    }
    if (name == null || name.isEmpty()) name = uri.getLastPathSegment();
    if (name == null || name.isEmpty()) name = "Shared file";
    return name;
}

private void deliver(final String json, final int attempt) {
    if (attempt >= DELIVER_MAX_TRIES) return;
    final String js =
        "(function(){var f=window.__pronotesReceiveFiles;if(!f)return 'wait';f(\" + json + \");return 'ok';})()";
    runOnUiThread(() -> {
        WebView webView = getBridge() != null ? getBridge().getWebView() : null;
        if (webView == null) {
            handler.postDelayed(() -> deliver(json, attempt + 1), DELIVER_RETRY_MS);
            return;
        }
        webView.evaluateJavascript(js, result -> {
            if (result == null || !result.contains("ok")) {

```

```

        handler.postDelayed(() -> deliver(json, attempt + 1), DELIVER_RETRY_MS);
    }
    });
});
}

@Override
public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions, grantResults);
    if (requestCode == REQUEST_RECORD_AUDIO_PERMISSION) {
        if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
            if (pendingAudioPermissionRequest != null) {
                pendingAudioPermissionRequest.grant(new String[]{PermissionRequest.RESOURCE_AUDIO_CAPTURE});
            }
            pendingAudioPermissionRequest = null;
        } else {
            if (pendingAudioPermissionRequest != null) {
                pendingAudioPermissionRequest.deny();
                pendingAudioPermissionRequest = null;
            }
        }
    }
}

/**
 * How long to wait after HOVER_EXIT before deciding the pen really left.
 * The exit that precedes a touch-down is followed by ACTION_DOWN within a
 * frame or two, so anything this side of ~80ms is the same gesture.
 */
private static final long HOVER_EXIT_GRACE_MS = 80;

/** Last state reported to the page; only changes are sent. */
private boolean reportedDown = false;

private final Handler handler = new Handler(Looper.getMainLooper());
private final Runnable penLeft = () -> setDown(false, 0, -1, "HOVER_EXIT timeout");

private static boolean isStylus(MotionEvent event) {
    if (event.getPointerCount() < 1) return false;
    int tool = event.getToolType(0);
    return tool == MotionEvent.TOOL_TYPE_STYLUS || tool == MotionEvent.TOOL_TYPE_ERASER;
}

/** True when the side button is held, per the button bits alone. */
private static boolean buttonHeld(int state) {
    return (state & MotionEvent.BUTTON_STYLUS_PRIMARY) != 0
        || (state & MotionEvent.BUTTON_STYLUS_SECONDARY) != 0
        || (state & MotionEvent.BUTTON_SECONDARY) != 0;
}

private void setDown(boolean down, int buttonState, int toolType, String because) {
    if (down == reportedDown) return;
    reportedDown = down;

    final String js =
        "window.dispatchEvent(new CustomEvent('pronotes:stylusbutton',{detail:{pressed:"
        + (down ? "true" : "false")
        + ",buttonState:"
        + buttonState
        + ",toolType:"
        + toolType
        + ",because:"
        + because
        + "}}));";

    runOnUiThread(() -> {
        try {
            WebView webView = getBridge() != null ? getBridge().getWebView() : null;
            if (webView != null) webView.evaluateJavascript(js, null);
        } catch (Exception ignored) {
            // A button press must never crash the app. If the bridge is not
            // up the press is simply lost, as it was before this existed.
        }
    });
}
}

```

```

private void observe(MotionEvent event) {
    final int action = event.getActionMasked();
    final int buttons = event.getButtonState();
    final int toolType = event.getToolType(0);
    // The eraser end of the pen is its own signal and needs no button.
    final boolean erasing = buttonHeld(buttons) || toolType == MotionEvent.TOOL_TYPE_ERASER;

    // Any real stylus activity means the pen did not leave.
    handler.removeCallbacks(penLeft);

    switch (action) {
        case MotionEvent.ACTION_BUTTON_PRESS:
            setDown(true, buttons, toolType, "BUTTON_PRESS");
            return;

        case MotionEvent.ACTION_BUTTON_RELEASE:
            // Authoritative: the button itself came up. The eraser TIP can
            // still hold the state on its own.
            setDown(toolType == MotionEvent.TOOL_TYPE_ERASER, buttons, toolType, "BUTTON_RELEASE");
            return;

        case MotionEvent.ACTION_UP:
        case MotionEvent.ACTION_CANCEL:
            // Android has cleared the button bits here. Reading them would
            // report a release the user never made – this is the flicker.
            return;

        case MotionEvent.ACTION_HOVER_EXIT:
            // Either the pen is leaving, or the nib is about to land. Let a
            // following event say which; if none comes, the pen left.
            handler.postDelayed(penLeft, HOVER_EXIT_GRACE_MS);
            return;

        default:
            // DOWN, MOVE, HOVER_ENTER, HOVER_MOVE: the state is meaningful.
            setDown(erasing, buttons, toolType, "action " + action);
    }
}

// -----
// Observing is not enough: the WebView must not SEE the button
// -----
// Measured on a Galaxy Tab S9: with the side button held, the kernel
// reported BTN_STYLUS DOWN -> touch -> lift -> BTN_STYLUS UP, and this
// bridge told the page to switch to the eraser - but the WebView delivered
// NO pen pointerdown for the stroke at all. Android turns stylus-button +
// contact into a context click, which reached the page as a `contextmenu`
// event (pointerType "mouse") while the stroke itself was swallowed. So the
// app switched to the eraser and then had nothing to erase with.
//
// The fix is to keep observing the real event (above) and then hand the
// WebView a copy with the button bits cleared: to the web layer it is an
// ordinary pen stroke, and the eraser choice travels separately through
// the pronotes:stylusbutton bridge. ACTION_BUTTON_PRESS / RELEASE are
// consumed outright - they exist only to drive that context click.

/**
 * A copy of the event with no buttons down, fully preserving all historical
 * 240Hz+ digitizer samples so erasing and hover tracking maintain maximum
 * coordinate density and smooth path geometry.
 * Caller is responsible for recycling the returned MotionEvent.
 */
private static MotionEvent withoutButtons(MotionEvent ev) {
    final int n = ev.getPointerCount();
    final MotionEvent.PointerProperties[] props = new MotionEvent.PointerProperties[n];
    for (int i = 0; i < n; i++) {
        props[i] = new MotionEvent.PointerProperties();
        ev.getPointerProperties(i, props[i]);
    }

    final int historySize = ev.getHistorySize();

    // Fast-path: When there are no historical samples (e.g. single sample or non-MOVE event),
    // perform a single standard obtain with buttonState = 0.
    if (historySize == 0) {
        final MotionEvent.PointerCoords[] coords = new MotionEvent.PointerCoords[n];
        for (int i = 0; i < n; i++) {
            coords[i] = new MotionEvent.PointerCoords();

```

```

        ev.getPointerCoords(i, coords[i]);
    }
    return MotionEvent.obtain(
        ev.getDownTime(), ev.getEventTime(), ev.getAction(), n, props, coords,
        ev.getMetaState(), 0 /* buttonState */, ev.getXPrecision(), ev.getYPrecision(),
        ev.getDeviceId(), ev.getEdgeFlags(), ev.getSource(), ev.getFlags());
}

// Full historical reconstruction:
// Initialize the event with the OLDEST historical sample (index 0),
// append intermediate historical samples (indices 1 .. historySize - 1),
// and finally append the current (latest) sample.
final MotionEvent.PointerCoords[] batchCoords = new MotionEvent.PointerCoords[n];
for (int i = 0; i < n; i++) {
    batchCoords[i] = new MotionEvent.PointerCoords();
}

// 1. Initial sample: oldest historical sample at index 0
for (int i = 0; i < n; i++) {
    ev.getHistoricalPointerCoords(i, 0, batchCoords[i]);
}
final MotionEvent clean = MotionEvent.obtain(
    ev.getDownTime(), ev.getHistoricalEventTime(0), ev.getAction(), n, props, batchCoords,
    ev.getMetaState(), 0 /* buttonState */, ev.getXPrecision(), ev.getYPrecision(),
    ev.getDeviceId(), ev.getEdgeFlags(), ev.getSource(), ev.getFlags());

// 2. Intermediate historical samples (indices 1 .. historySize - 1)
for (int h = 1; h < historySize; h++) {
    for (int i = 0; i < n; i++) {
        ev.getHistoricalPointerCoords(i, h, batchCoords[i]);
    }
    clean.addBatch(ev.getHistoricalEventTime(h), batchCoords, ev.getMetaState());
}

// 3. Current (latest) sample
for (int i = 0; i < n; i++) {
    ev.getPointerCoords(i, batchCoords[i]);
}
clean.addBatch(ev.getEventTime(), batchCoords, ev.getMetaState());

return clean;
}

/** True for the events that exist only to trigger a context click. */
private static boolean isButtonAction(MotionEvent ev) {
    final int a = ev.getActionMasked();
    return a == MotionEvent.ACTION_BUTTON_PRESS || a == MotionEvent.ACTION_BUTTON_RELEASE;
}

@Override
public boolean dispatchTouchEvent(MotionEvent event) {
    if (!isStylus(event)) return super.dispatchTouchEvent(event);
    observe(event);
    if (isButtonAction(event)) return true;
    if (!buttonHeld(event.getButtonState())) return super.dispatchTouchEvent(event);
    final MotionEvent clean = withoutButtons(event);
    try {
        return super.dispatchTouchEvent(clean);
    } finally {
        clean.recycle();
    }
}

@Override
public boolean dispatchGenericMotionEvent(MotionEvent event) {
    if (!isStylus(event)) return super.dispatchGenericMotionEvent(event);
    observe(event);
    if (isButtonAction(event)) return true;
    if (!buttonHeld(event.getButtonState())) return super.dispatchGenericMotionEvent(event);
    final MotionEvent clean = withoutButtons(event);
    try {
        return super.dispatchGenericMotionEvent(clean);
    } finally {
        clean.recycle();
    }
}

@Override

```

```
        public void onPause() {
            super.onPause();
            // Leaving the app with the button held must not strand the eraser.
            handler.removeCallbacks(penLeft);
            setDown(false, 0, -1, "onPause");
        }
    }
}

=====
// === END FILE: android/app/src/main/java/com/idrees/pronotes/MainActivity.java ===
=====

=====
// === START FILE: capacitor.config.ts ===
// File 3 of 194 | Lines: 23 | Size: 684 bytes
=====

import type { CapacitorConfig } from "@capacitor/cli";

/**
 * Capacitor wraps the STATIC export (`npm run build:apk` -> ./out) into a
 * native Android app. Everything runs from files inside the APK, so the app
 * works with no server and no internet – boards live in the WebView's local
 * storage exactly as they do in the browser.
 */
const config: CapacitorConfig = {
  appId: "com.idrees.pronotes",
  appName: "ProNotes",
  webDir: "out",
  android: {
    // Keeps the drawing surface responsive to stylus/touch without the
    // browser's 300ms tap delay heuristics.
    allowMixedContent: false,
  },
  server: {
    androidScheme: "https",
  },
};

export default config;

=====
// === END FILE: capacitor.config.ts ===
=====

=====
// === START FILE: components.json ===
// File 4 of 194 | Lines: 21 | Size: 430 bytes
=====

{
  "$schema": "https://ui.shadcn.com/schema.json",
  "style": "new-york",
  "rsc": true,
  "tsx": true,
  "tailwind": {
    "config": "",
    "css": "src/app/globals.css",
    "baseColor": "neutral",
    "cssVariables": true,
    "prefix": ""
  },
  "aliases": {
    "components": "@components",
    "utils": "@lib/utils",
    "ui": "@components/ui",
    "lib": "@lib",
    "hooks": "@/hooks"
  },
  "iconLibrary": "lucide"
}

=====
// === END FILE: components.json ===
=====
```

```

=====
// === START FILE: electron/main.js ===
// File 5 of 194 | Lines: 151 | Size: 5275 bytes
=====

/**
 * ProNotes – Windows desktop app (Electron)
 * =====
 * A real .exe: its own icon, its own window, no browser and no Node server.
 *
 * How the app is loaded
 * -----
 * The static export in ./out is served through a custom `pronotes://` scheme
 * rather than file://, because:
 *   - Next.js emits ABSOLUTE asset paths (/_next/...), which file:// resolves
 *     against the drive root and fails to find.
 *   - file:// is an opaque origin, so localStorage would not persist reliably –
 *     which is where every board is stored.
 * Registering the scheme as `standard: true, secure: true` gives it a stable,
 * secure origin, so storage persists across restarts exactly like a website.
 */

const { app, BrowserWindow, shell, protocol, net, Menu } = require("electron");
const path = require("node:path");
const { pathToFileURL } = require("node:url");

const SCHEME = "pronotes";
const WEB_ROOT = path.join(__dirname, "..", "out");

// Must be called before `app.ready`.
protocol.registerSchemesAsPrivileged([
  {
    scheme: SCHEME,
    privileges: {
      standard: true,
      secure: true,
      supportFetchAPI: true,
      stream: true,
    },
  },
]);

/** Resolve a request path to a file inside ./out, refusing to escape it. */
function resolveWebPath(urlPath) {
  const decoded = decodeURIComponent(urlPath.split("?")[0].split("#")[0]);
  const relative = decoded === "/" || decoded === "" ? "index.html" : decoded.replace(/^\/+/, "");
  const full = path.join(WEB_ROOT, relative);

  // Path-traversal guard: a crafted URL must never read outside the bundle.
  const normalizedRoot = path.resolve(WEB_ROOT) + path.sep;
  const normalizedFull = path.resolve(full);
  if (!normalizedFull.startsWith(normalizedRoot)) return null;

  return normalizedFull;
}

function createWindow() {
  const win = new BrowserWindow({
    width: 1440,
    height: 900,
    minWidth: 900,
    minHeight: 600,
    backgroundColor: "#09090b",
    title: "ProNotes",
    icon: path.join(__dirname, "..", "public", "pronotes.ico"),
    // Don't paint a white flash before the app renders.
    show: false,
    webPreferences: {
      contextIsolation: true,
      nodeIntegration: false,
      sandbox: true,
      // The canvas is the whole app; let it use the GPU.
      backgroundThrottling: false,
    },
  });
};

win.once("ready-to-show", () => win.show());

```

```
win.loadURL(`${SCHEME}://app/index.html`);

// Any external link (docs, a wiki link the user pasted) opens in the real
// browser instead of replacing the app inside its own window.
win.webContents.setWindowOpenHandler(({ url }) => {
  if (url.startsWith("http://") || url.startsWith("https://")) {
    shell.openExternal(url);
  }
  return { action: "deny" };
});
win.webContents.on("will-navigate", (event, url) => {
  if (url.startsWith(`${SCHEME}://`)) return;

  // Never let the window navigate away from the app.
  event.preventDefault();

  // Only real web links are handed to the browser. A dropped file arrives
  // here as file:// — passing that to openExternal would launch the user's
  // PDF reader as a side effect of dragging a document onto the app, which
  // is not what the gesture means. Anything that is not http(s) is simply
  // refused.
  if (url.startsWith("http://") || url.startsWith("https://")) {
    shell.openExternal(url);
  }
});

return win;
}

// One instance only — a second launch focuses the existing window rather than
// opening a rival copy with its own view of the same storage.
// Must match electron-builder's `appId` (package.json). Windows groups taskbar
// buttons, pins and jump lists by this id; without it the running window can
// fail to group with its own pinned shortcut.
if (process.platform === "win32") app.setAppUserModelId("com.idrees.pronotes");

if (!app.requestSingleInstanceLock()) {
  app.quit();
} else {
  let mainWindow = null;

  app.on("second-instance", () => {
    if (mainWindow) {
      if (mainWindow.isMinimized()) mainWindow.restore();
      mainWindow.focus();
    }
  });

  app.whenReady().then(() => {
    protocol.handle(SCHEME, (request) => {
      const url = new URL(request.url);
      const filePath = resolveWebPath(url.pathname);
      if (!filePath) {
        return new Response("Forbidden", { status: 403 });
      }
      // A missing file (a stale chunk after a partial rebuild, a typo'd path)
      // must answer 404, not throw: an exception here fails the request with a
      // bare network error, which on index.html is a silent blank window.
      return net.fetch(pathToFileURL(filePath).toString()).catch((err) => {
        console.error(`[pronotes://] ${url.pathname}:`, err && err.message);
        return new Response("Not found", { status: 404 });
      });
    });

    // A menu bar would just take vertical space from an infinite canvas;
    // every action already has a toolbar button and a shortcut.
    Menu.setApplicationMenu(null);

    mainWindow = createWindow();

    app.on("activate", () => {
      if (BrowserWindow.getAllWindows().length === 0) mainWindow = createWindow();
    });
  });

  app.on("window-all-closed", () => {
    if (process.platform !== "darwin") app.quit();
  });
}
```

```
}

=====
// === END FILE: electron/main.js ===
=====

=====
// === START FILE: eslint.config.mjs ===
// File 6 of 194 | Lines: 55 | Size: 1944 bytes
=====

import { dirname } from "node:path";
import { fileURLToPath } from "node:url";
import { FlatCompat } from "@eslint/eslintrc";

/**
 * ESLint for the app.
 *
 * This file used to contain ONLY an ignores list. With nothing extended there
 * were no rules at all, so `npm run lint` passed on anything - including the
 * react-hooks mistakes that are the most common real bug in this codebase's
 * history (stale closures in pointer handlers). eslint-config-next was already
 * installed; it just was never wired in.
 */
const __dirname = dirname(fileURLToPath(import.meta.url));
const compat = new FlatCompat({ baseDirectory: __dirname });

const eslintConfig = [
  {
    ignores: [
      "node_modules/**",
      ".next/**",
      ".next-dev/**",
      "out/**",
      "build/**",
      "dist-desktop/**",
      "next-env.d.ts",
      // Capacitor's native project: generated Java/Gradle plus vendored JS
      // (native-bridge.js) that ships its own lint directives for another config.
      "android/**",
      "ios/**",
      ".test-dist/**",
      // Generated/vendored: the pdf.js worker is copied in from node_modules.
      "public/pdf.worker.min.mjs",
    ],
  },
  ...compat.extends("next/core-web-vitals", "next/typescript"),
  {
    // Electron's main process, the build scripts and the e2e runner are real
    // CommonJS; `require` is correct there, not a style slip.
    files: ["**/*.js", "**/*.cjs"],
    rules: { "@typescript-eslint/no-require-imports": "off" },
  },
  {
    rules: {
      // Apostrophes in JSX text render correctly in React; the rule is about
      // typos in hand-written HTML entities and only produced noise here.
      "react/no-unescaped-entities": "off",
      // Existing `any`s are warnings to work down, not a reason for the gate to
      // fail on every run - a gate that always fails gets ignored.
      "@typescript-eslint/no-explicit-any": "warn",
    },
  },
];

export default eslintConfig;

=====
// === END FILE: eslint.config.mjs ===
=====

=====
// === START FILE: next.config.ts ===
// File 7 of 194 | Lines: 83 | Size: 3874 bytes
=====
```

```

import { readFileSync } from "node:fs";
import { join } from "node:path";
import type { NextConfig } from "next";
import { PHASE_DEVELOPMENT_SERVER } from "next/constants";

/**
 * The app's version, read from THIS FILE's directory rather than imported.
 *
 * `import pkg from "./package.json"` looks right and works when `next` is run
 * from the repository root - but Next transpiles this config and resolves that
 * specifier against the process's working directory, so starting the dev
 * server from anywhere else died on startup with MODULE_NOT_FOUND before it
 * printed a single useful line. Reading from __dirname is cwd-proof, and it
 * also keeps package.json out of the bundle.
 */
const pkg = JSON.parse(readFileSync(join(__dirname, "package.json"), "utf8")) as { version: string };

/**
 * The packaged apps need a fully static export: there is no Node server
 * inside an APK, and the Electron app loads ./out through its own
 * `pronotes://` scheme. Plain `next build` keeps the server build so that
 * `npm start` still works.
 *
 * So the export mode is opt-in via an env flag rather than always-on:
 *   npm run build    -> server build, for `npm start`
 *   npm run export   -> static export into ./out (scripts/build-static.mjs)
 *   npm run apk      -> export, then the Android APK
 *   npm run dist:win -> export, then the Windows installer
 */
const isStaticExport = process.env.PRONOTES_STATIC === "1";

const nextConfig = (phase: string): NextConfig => {
  return {
    // The DEV SERVER gets its own build directory; every build keeps `.next`.
    //
    // They used to share `.next`, so running an export while `npm run dev` was
    // up deleted the dev server's manifests under it and it served 500s with a
    // different "missing" file each reload - which looked exactly like a flaky
    // Windows race and was not one.
    //
    // It is the dev server that moves, not the export: with `output: "export"`
    // Next treats `distDir` as the EXPORT DESTINATION rather than the build
    // directory. Setting it on the export (tried first) wrote the site to that
    // folder instead of ./out - which Capacitor and Electron both read - and
    // still built into `.next`.
    distDir: phase === PHASE_DEVELOPMENT_SERVER ? ".next-dev" : ".next",
    ...(isStaticExport
      ? {
          output: "export" as const,
          // next/image's optimizer needs a server; the app ships its own assets.
          images: { unoptimized: true },
        }
      : {}),
    typescript: {
      // Next's own type pass runs ONLY when no wrapper already type-checked:
      // scripts/build.mjs (`npm run build`) and scripts/build-static.mjs (every
      // export: APK, Windows, desktop) run `npx tsc --noEmit` first and then set
      // PRONOTES_TYPECHECKED=1 for the child `next build`. A bare `npx next build`
      // never sets it, so broken types fail the build instead of shipping green.
      // (The emergency hatch PRONOTES_SKIP_TYPECHECK=1 deliberately leaves it
      // unset, so Next still type-checks that build.)
      ignoreBuildErrors: process.env.PRONOTES_TYPECHECKED === "1",
    },
  },
  /**
   * What this build IS, baked in so the running app can say so.
   *
   * Installing an APK and finding nothing had changed was impossible to tell
   * apart from a fix that did not work - there was no way, on the tablet, to
   * see which build was running. Settings now shows both numbers.
   *
   * The build code is Android's versionCode, handed down by
   * scripts/build-apk.mjs; a plain export has none and says so.
   */
  env: {
    NEXT_PUBLIC_APP_VERSION: pkg.version,
    NEXT_PUBLIC_BUILD_CODE: process.env.PRONOTES_BUILD_CODE || "",
  },
  // reactStrictMode is off; reason not recorded (present since the initial commit).

```

```
    reactStrictMode: false,
  };
};

export default nextConfig;

=====
// === END FILE: next.config.ts ===
=====

=====
// === START FILE: package.json ===
// File 8 of 194 | Lines: 120 | Size: 3415 bytes
=====

{
  "name": "pronotes",
  "version": "1.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev -p 3000",
    "build": "node scripts/build.mjs",
    "start": "next start -p 3000",
    "typecheck": "tsc --noEmit",
    "lint": "eslint .",
    "test": "node scripts/test-runner.mjs",
    "postinstall": "node scripts/sync-pdf-worker.mjs",
    "export": "node scripts/build-static.mjs",
    "apk": "node scripts/build-apk.mjs",
    "build:apk": "node scripts/build-apk.mjs",
    "apk:deploy": "node scripts/deploy-android.mjs",
    "desktop": "node scripts/build-static.mjs && electron .",
    "dist:win": "node scripts/build-desktop.mjs",
    "sync:pdf-worker": "node scripts/sync-pdf-worker.mjs",
    "audit": "node scripts/audit.mjs",
    "icons": "node scripts/generate-icons.js"
  },
  "dependencies": {
    "@capacitor/android": "^6.2.1",
    "@capacitor/core": "^6.2.1",
    "@capacitor/filesystem": "^6.0.4",
    "@capacitor/share": "^6.0.4",
    "@radix-ui/react-context-menu": "^2.2.15",
    "@radix-ui/react-dialog": "^1.1.14",
    "@radix-ui/react-dropdown-menu": "^2.1.15",
    "@radix-ui/react-popover": "^1.1.14",
    "@radix-ui/react-scroll-area": "^1.2.9",
    "@radix-ui/react-separator": "^1.1.7",
    "@radix-ui/react-slider": "^1.3.5",
    "@radix-ui/react-slot": "^1.2.3",
    "@radix-ui/react-toast": "^1.2.14",
    "@radix-ui/react-tooltip": "^1.2.7",
    "@supabase/supabase-js": "^2.112.3",
    "chart.js": "^4.5.1",
    "class-variance-authority": "^0.7.1",
    "clsx": "^2.1.1",
    "docx-preview": "^0.4.1",
    "fflate": "^0.8.3",
    "html-to-image": "^1.11.13",
    "html2canvas": "^1.4.1",
    "jspdf": "^4.2.1",
    "jszip": "^3.10.2",
    "lucide-react": "^0.525.0",
    "nanoid": "^6.0.1",
    "next": "^15.1.0",
    "next-themes": "^0.4.6",
    "pdfjs-dist": "^6.3.289",
    "perfect-freehand": "^1.2.3",
    "pptxviewjs": "^1.1.9",
    "react": "^19.0.0",
    "react-dom": "^19.0.0",
    "sharp": "^0.34.3",
    "tailwind-merge": "^3.3.1",
    "tailwindcss-animate": "^1.0.7",
    "zustand": "^5.0.6"
  },
  "devDependencies": {
```

```

"@capacitor/cli": "^6.2.1",
"@tailwindcss/postcss": "^4",
"@types/node": "^22.0.0",
"@types/react": "^19",
"@types/react-dom": "^19",
"electron": "^43.4.1",
"electron-builder": "^26.15.3",
"eslint": "^9",
"eslint-config-next": "^15.1.0",
"tailwindcss": "^4",
"tw-animate-css": "^1.3.5",
"typescript": "^5"
},
"main": "electron/main.js",
"build": {
  "appId": "com.idrees.pronotes",
  "productName": "ProNotes",
  "copyright": "ProNotes",
  "directories": {
    "output": "dist-desktop",
    "buildResources": "public"
  },
  "files": [
    "electron/**/*",
    "out/**/*",
    "public/pronotes.ico",
    "package.json",
    "!node_modules/**/*"
  ],
  "win": {
    "target": [
      {
        "target": "nsis",
        "arch": [
          "x64"
        ]
      },
      {
        "target": "portable",
        "arch": [
          "x64"
        ]
      }
    ],
    "icon": "public/pronotes.ico"
  },
  "nsis": {
    "oneClick": false,
    "perMachine": false,
    "allowToChangeInstallationDirectory": true,
    "createDesktopShortcut": true,
    "createStartMenuShortcut": true,
    "shortcutName": "ProNotes",
    "installerIcon": "public/pronotes.ico",
    "uninstallerIcon": "public/pronotes.ico"
  }
}
}

=====
// === END FILE: package.json ===
=====

=====
// === START FILE: postcss.config.mjs ===
// File 9 of 194 | Lines: 5 | Size: 81 bytes
=====

const config = {
  plugins: ["@tailwindcss/postcss"],
};

export default config;

=====
// === END FILE: postcss.config.mjs ===
=====

```

```
=====
// === START FILE: public/manifest.json ===
// File 10 of 194 | Lines: 75 | Size: 1680 bytes
=====

{
  "id": "?source=pwa",
  "name": "ProNotes — Infinite Whiteboard & Knowledge Canvas",
  "short_name": "ProNotes",
  "description": "Professional infinite whiteboard with Obsidian-style WikiLinks, smart connectors, freehand drawing, and touchscreen support.",
  "start_url": "/",
  "scope": "/",
  "display": "standalone",
  "display_override": [
    "window-controls-overlay",
    "standalone",
    "minimal-ui"
  ],
  "background_color": "#0f172a",
  "theme_color": "#4f46e5",
  "orientation": "any",
  "categories": [
    "productivity",
    "utilities",
    "education",
    "notes"
  ],
  "icons": [
    {
      "src": "/icon-192.png",
      "sizes": "192x192",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "/icon-512.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "/icon-192-maskable.png",
      "sizes": "192x192",
      "type": "image/png",
      "purpose": "maskable"
    },
    {
      "src": "/icon-512-maskable.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "maskable"
    }
  ],
  "shortcuts": [
    {
      "name": "New Whiteboard",
      "short_name": "New Board",
      "description": "Create a new blank whiteboard",
      "url": "?action=new",
      "icons": [
        {
          "src": "/icon-192.png",
          "sizes": "192x192"
        }
      ]
    },
    {
      "name": "Knowledge Graph",
      "short_name": "Graph View",
      "description": "Open interactive 2D graph view",
      "url": "?view=graph",
      "icons": [
        {
          "src": "/icon-192.png",
          "sizes": "192x192"
        }
      ]
    }
  ]
}
```

```

    }
  ]
}
}

=====
// === END FILE: public/manifest.json ===
=====

=====
// === START FILE: public/sw.js ===
// File 11 of 194 | Lines: 131 | Size: 3947 bytes
=====

/**
 * ProNotes Service Worker
 * =====
 * Provides offline caching, asset caching (Cache-First with SWR),
 * and navigation fallback (Network-First) for standalone PWA.
 */

// Bump on every deploy: 'activate' deletes every cache whose key differs,
// so stale hashed chunks and old entries never accumulate.
const CACHE_VERSION = 'pronotes-v4.3.0';
const STATIC_ASSETS = [
  '/',
  '/manifest.json',
  '/favicon.ico',
  '/favicon.png',
  '/favicon-32x32.png',
  '/favicon-16x16.png',
  '/icon-192.png',
  '/icon-512.png',
  '/icon-192-maskable.png',
  '/icon-512-maskable.png',
  '/apple-touch-icon.png',
  '/logo.png',
];

// Install: pre-cache core application shell
self.addEventListener('install', (event) => {
  event.waitUntil(
    caches
      .open(CACHE_VERSION)
      .then((cache) => cache.addAll(STATIC_ASSETS))
      .then(() => self.skipWaiting())
  );
});

// Activate: clean up old cache versions & claim clients immediately
self.addEventListener('activate', (event) => {
  event.waitUntil(
    caches
      .keys()
      .then((keys) =>
        Promise.all(
          keys.map((key) => {
            if (key !== CACHE_VERSION) {
              return caches.delete(key);
            }
          })
        )
      )
      .then(() => self.clients.claim())
  );
});

// Fetch: smart caching strategies
self.addEventListener('fetch', (event) => {
  const request = event.request;

  // Only handle GET requests
  if (request.method !== 'GET') {
    return;
  }

```

```

const url = new URL(request.url);

// Ignore API requests and browser extension requests
if (url.pathname.startsWith('/api') || !url.protocol.startsWith('http')) {
  return;
}

// Next.js App Router data (RSC payloads / prefetches) must NOT be served
// cache-first: after a deploy they would be inconsistent with freshly
// loaded hashed JS chunks. Let them go to the network (browser HTTP cache
// still applies) and never store them here.
if (request.headers.get('RSC') === '1' || request.headers.has('Next-Router-Prefetch')) {
  return;
}

// 1. Navigation requests (HTML pages): Network-First, fallback to cached '/'
if (request.mode === 'navigate') {
  event.respondWith(
    fetch(request)
      .then((networkResponse) => {
        if (networkResponse && networkResponse.status === 200) {
          const responseToCache = networkResponse.clone();
          event.waitUntil(
            caches.open(CACHE_VERSION).then((cache) => cache.put(request, responseToCache))
          );
        }
        return networkResponse;
      })
      .catch(() => caches.match('/'))
  );
  return;
}

// 2. Static assets (JS bundles, CSS, images, fonts): Cache-First with SWR
event.respondWith(
  caches.match(request).then((cachedResponse) => {
    if (cachedResponse) {
      // Stale-While-Revalidate: fetch in background to keep cache fresh
      // (kept alive via event.waitUntil so the SW isn't terminated first)
      event.waitUntil(
        fetch(request)
          .then((networkResponse) => {
            if (networkResponse && networkResponse.status === 200) {
              return caches
                .open(CACHE_VERSION)
                .then((cache) => cache.put(request, networkResponse));
            }
          })
          .catch(() => {
            // Offline or network error; ignore background update failure
          })
      );
      return cachedResponse;
    }

    // If not in cache, fetch from network and cache
    return fetch(request).then((networkResponse) => {
      if (!networkResponse || networkResponse.status !== 200) {
        return networkResponse;
      }
      const responseToCache = networkResponse.clone();
      event.waitUntil(
        caches.open(CACHE_VERSION).then((cache) => cache.put(request, responseToCache))
      );
      return networkResponse;
    });
  });
);
});
};

=====
// === END FILE: public/sw.js ===
=====

=====
// === START FILE: scripts/audit.mjs ===
// File 12 of 194 | Lines: 83 | Size: 3102 bytes

```

```

=====

/**
 * Static audit: unreachable modules and risky patterns.
 *
 *   node scripts/audit.mjs
 *
 * Reports files no entry point can reach, plus patterns worth a second look
 * (type suppressions, unguarded parses, stray console.log). It is a lead
 * generator, not a linter – every hit needs judging by hand.
 */
import fs from "node:fs";
import path from "node:path";

const ROOT = process.cwd();
const SRC = path.join(ROOT, "src");
const EXTS = [".tsx", ".ts"];
const rel = (f) => path.relative(ROOT, f).split(path.sep).join("/");

function* walk(dir) {
  for (const d of fs.readdirSync(dir, { withFileTypes: true })) {
    const p = path.join(dir, d.name);
    if (d.isDirectory()) yield* walk(p);
    else yield p;
  }
}

const files = [...walk(SRC)].filter((f) => EXTS.includes(path.extname(f)));

function resolveImport(spec, from) {
  let base;
  if (spec.startsWith("@/")) base = path.join(SRC, spec.slice(2));
  else if (spec.startsWith(".")) base = path.resolve(path.dirname(from), spec);
  else return null;
  for (const e of EXTS) if (fs.existsSync(base + e)) return base + e;
  for (const e of EXTS) {
    const idx = path.join(base, "index" + e);
    if (fs.existsSync(idx)) return idx;
  }
  return null;
}

// ---- reachability from the Next.js entry points -----
const entries = [path.join(SRC, "app/layout.tsx"), path.join(SRC, "app/page.tsx")].filter(fs.existsSync);
for (const f of files) if (/route\.tsx?$/i.test(f) && f.includes("app")) entries.push(f);

const reachable = new Set();
const stack = [...entries];
while (stack.length) {
  const f = stack.pop();
  if (reachable.has(f)) continue;
  reachable.add(f);
  const code = fs.readFileSync(f, "utf8");
  for (const m of code.matchAll(/(?:from\s+|import\s+)(["'])([^\s'"]+)(["']/g)) {
    const r = resolveImport(m[1], f);
    if (r) stack.push(r);
  }
}
const dead = files.filter((f) => !reachable.has(f));

// ---- risky patterns -----
const findings = [];
for (const f of files) {
  const lines = fs.readFileSync(f, "utf8").split("\n");
  lines.forEach((line, i) => {
    const at = `${rel(f)}:${i + 1}`;
    if (/^bas any\b/i.test(line)) findings.push(["as any", at]);
    if (/^@ts-ignore|^@ts-expect-error/i.test(line)) findings.push(["ts suppression", at]);
    if (/^console\.log$/i.test(line)) findings.push(["console.log", at]);
    if (/^dangerouslySetInnerHTML/i.test(line)) findings.push(["dangerouslySetInnerHTML", at]);
    // A bare await inside a component body cannot be caught by an error boundary.
    if (/^localStorage\.(getItem|setItem)/i.test(line)) findings.push(["direct localStorage", at]);
  });
}

console.log(`files ${files.length}   reachable ${reachable.size}   dead ${dead.length}`);
for (const d of dead) console.log(`  DEAD  ${rel(d)}`);

```

```

const grouped = {};
for (const [kind, at] of findings) (grouped[kind] ||= []).push(at);
console.log("\nrisk scan");
for (const kind of Object.keys(grouped).sort()) {
  console.log(` ${kind}: ${grouped[kind].length}`);
  grouped[kind].slice(0, 5).forEach((a) => console.log(`    ${a}`));
}

=====
// === END FILE: scripts/audit.mjs ===
=====

=====
// === START FILE: scripts/build-apk.mjs ===
// File 13 of 194 | Lines: 61 | Size: 2891 bytes
=====

/**
 * One-command Android build: static export -> Capacitor sync -> APK.
 *
 * npm run apk
 *
 * Produces android/app/build/outputs/apk/debug/app-debug.apk and copies it to
 * the project root as ProNotes.apk for easy transfer to a tablet.
 */
import { execSync } from "node:child_process";
import fs from "node:fs";
import path from "node:path";

const root = process.cwd();

// Version, computed rather than remembered.
//
// versionCode must strictly increase for Android to install an update over the
// old APK, and when it was a hand-edited number it stayed at 1 for months, so a
// tablet's build could not be identified. Minutes since 2026-01-01 UTC rises on
// every build, never repeats in practice, and stays far inside Android's limit
// (2,100,000,000) for thousands of years. The first value is ~375,000 - well
// above the 2 that the last hand-numbered APK used, so it installs over it.
const EPOCH = Date.UTC(2026, 0, 1);
const versionCode = Math.floor((Date.now() - EPOCH) / 60000);
const versionName = JSON.parse(fs.readFileSync(path.join(root, "package.json"), "utf8")).version;
if (!(versionCode > 2)) throw new Error(`Computed versionCode ${versionCode} is not above the last release (2)
  - is the clock right?`);
const run = (cmd, opts = {}) =>
  execSync(cmd, { stdio: "inherit", cwd: root, env: process.env, ...opts });

console.log("\n[1/3] Static export...");
// The web layer is told the version code so the app can show which build it
// is - the only way to tell "the fix did not work" from "the APK never
// installed" while holding the tablet.
run("node scripts/build-static.mjs", {
  env: { ...process.env, PRONOTES_BUILD_CODE: String(versionCode) },
});

console.log("\n[2/3] Syncing web assets into the Android project...");
run("npx cap sync android");
run("node scripts/make-android-icons.mjs");

console.log("\n[3/3] Building the APK (this takes a few minutes the first time)...");
// Absolute path: a bare "gradlew.bat" is not resolved when the command is
// spawned with a different cwd, and "./gradlew" is shell-specific.
const androidDir = path.join(root, "android");
const gradlew = path.join(androidDir, process.platform === "win32" ? "gradlew.bat" : "gradlew");
console.log(`    version ${versionName} (code ${versionCode})`);
run(`"${gradlew}" assembleDebug -PpronotesVersionCode=${versionCode} -PpronotesVersionName=${versionName}`, {
  cwd: androidDir,
});

const built = path.join(root, "android", "app", "build", "outputs", "apk", "debug", "app-debug.apk");
if (!fs.existsSync(built)) {
  console.error("APK not found at", built);
  process.exit(1);
}
const dest = path.join(root, "ProNotes.apk");
fs.copyFileSync(built, dest);
const mb = (fs.statSync(dest).size / 1048576).toFixed(1);

```

```
console.log(`\nDone. ${dest} (${mb} MB) - version ${versionName}, code ${versionCode}`);
console.log("Copy it to your tablet and open it to install.");
```

```
=====
// === END FILE: scripts/build-apk.mjs ===
=====
```

```
=====
// === START FILE: scripts/build-desktop.mjs ===
// File 14 of 194 | Lines: 31 | Size: 1008 bytes
=====
```

```
/**
 * Build the Windows desktop app: static export -> Electron package.
 *
 * npm run dist:win
 *
 * Produces, in ./dist-desktop:
 *   - ProNotes Setup <version>.exe (installer: Start menu + desktop shortcut)
 *   - ProNotes <version>.exe (portable: run it from anywhere, no install)
 */
import { execSync } from "node:child_process";
import fs from "node:fs";
import path from "node:path";

const root = process.cwd();
const run = (cmd) => execSync(cmd, { stdio: "inherit", cwd: root, env: process.env });

console.log("\n[1/2] Static export...");
run("node scripts/build-static.mjs");

console.log("\n[2/2] Packaging the Windows app...");
run("npx electron-builder --win --x64");

const out = path.join(root, "dist-desktop");
if (fs.existsSync(out)) {
  const exes = fs.readdirSync(out).filter((f) => f.endsWith(".exe"));
  console.log("\nBuilt:");
  for (const e of exes) {
    const mb = (fs.statSync(path.join(out, e)).size / 1048576).toFixed(1);
    console.log(`  ${e}  (${mb} MB)`);
  }
}
```

```
=====
// === END FILE: scripts/build-desktop.mjs ===
=====
```

```
=====
// === START FILE: scripts/build-static.mjs ===
// File 15 of 194 | Lines: 171 | Size: 7455 bytes
=====
```

```
/**
 * Static export into ./out, shared by BOTH packaged apps: Capacitor copies it
 * into the APK, and Electron serves it through the `pronotes://` scheme. Sets
 * PRONOTES_STATIC so next.config.ts switches to `output: "export"`. It is not
 * always on only because `npm run build` / `npm start` - the plain web server -
 * need the normal server build.
 *
 * Why the retry loop
 * -----
 * On Windows `next build` intermittently dies on a file it has just written,
 * with a DIFFERENT error each run:
 *
 *   Cannot find module for page: /_document
 *   ENOENT ... .next\server\pages-manifest.json
 *   ENOENT ... rename .next\export\500.html -> .next\server\pages\500.html
 *   EBUSY ... rmdir ...out
 *
 * The files are present immediately afterwards, and the same commit builds
 * fine on a retry, so nothing is genuinely missing. It is a file-handle race
 * on .next / out. Two things make it much worse and are checked for below:
 * a `next dev` server running against this project (it writes the same .next),
 * and an orphaned `next build` left over from a previous failed run.
 */
```

```

* Only that signature is retried. A real compile or type error fails on the
* first attempt, because retrying it would just bury the message.
*/
import { spawnSync, execSync } from "node:child_process";
import fs from "node:fs";
import path from "node:path";

const ATTEMPTS = 3;
const root = process.cwd();
// The build directory. Builds own `.next`; `npm run dev` now builds into
// `.next-dev` (see next.config.ts), so clearing this can no longer take a
// running dev server down with it.
//
// An earlier version of this script believed the export built into
// `.next-static`. Nothing ever set that, so it cleared a folder that did not
// exist while every export silently shared `.next` with the dev server.
const nextDir = path.join(root, ".next");
const outDir = path.join(root, "out");

/**
 * Transient Windows file-handle failure, as opposed to a real build error.
 * NOTE: the paths in these messages use BACKslashes. An earlier version of
 * this check used [\] and therefore never matched a single real failure.
 */
function isHandleRace(output) {
  const errno = /(ENOENT|EBUSY|EPERM|EACCES|MODULE_NOT_FOUND|Cannot find module|PageNotFoundError)/.test(output);
  const ourPaths = /(\\.next|\\[\\]out\\b|for page: \\_document)/.test(output);
  return errno && ourPaths;
}

/** rmSync with backoff – the handle is usually released within a second. */
function removeWithRetries(dir) {
  try {
    fs.rmSync(dir, { recursive: true, force: true, maxRetries: 10, retryDelay: 300 });
  } catch {
    /* Next will try again itself; a still-locked dir shows up as a retry. */
  }
}

/** A dev server or a stale build fights us over .next. Say so, don't guess. */
function warnAboutCompetingProcesses() {
  if (process.platform !== "win32") return;
  try {
    const ps =
      `Get-CimInstance Win32_Process | Where-Object { $_.Name -eq 'node.exe' -and ` +
      `$_ .CommandLine -match 'next.dist.bin.next' -and $_.CommandLine -match [regex]::Escape('${root}') } | ` +
      `ForEach-Object { "$($_.ProcessId) $($_.CommandLine)" }`;
    const found = execSync(`powershell -NoProfile -Command "${ps.replace(/"/g, '\\"')}"`, {
      encoding: "utf8",
      stdio: ["ignore", "pipe", "ignore"],
    }).trim();
    if (!found) return;
    const lines = found.split("\n").map((l) => l.trim()).filter(Boolean);
    const mine = String(process.pid);
    const others = lines.filter((l) => !l.startsWith(mine));
    if (!others.length) return;
    console.warn(
      `\\n[build-static] Another Next process is using this project's .next:\\n` +
      `  others.map((l) => ` +
      `    ${l.slice(0, 120)}).join("\\n")` +
      `  \\n A running \\npm run dev\\ writes the same .next as the build and is the\\n` +
      `  usual cause of these failures. Stop it, or kill the pid above, then retry.\\n`
    );
  } catch {
    /* Diagnostics only. */
  }
}

// -----
// Preflight, shared by EVERY packaged build (APK, Windows installer, desktop).
// -----
// Doing these here rather than in package.json is deliberate: the scripts had
// drifted - `desktop` skipped the worker sync while `dist:win` did it, and
// nothing type-checked at all - so one path could ship a stale pdf.js worker
// (a version-mismatch error on every PDF) or broken types. Every export goes
// through this file, so every export gets both.
function preflight(label, command) {

```

```

    console.log(`[build-static] ${label} ...`);
    const r = spawnSync(command, { cwd: root, stdio: "inherit", shell: true });
    if (r.status !== 0) {
        console.error(`\n[build-static] ${label} failed - not building.`);
        process.exit(r.status ?? 1);
    }
}
preflight("Syncing the pdf.js worker", "node scripts/sync-pdf-worker.mjs");
if (process.env.PRONOTES_SKIP_TYPECHECK === "1") {
    console.warn("[build-static] PRONOTES_SKIP_TYPECHECK=1 - shipping WITHOUT a type check.");
} else {
    preflight("Type-checking", "npx tsc --noEmit");
    // The check just passed in THIS process, so the child `next build` below may
    // skip Next's own type pass (see next.config.ts). A bare `npx next build`
    // never sets this, so it still gets full type errors.
    process.env.PRONOTES_TYPECHECKED = "1";
}

// The export ALWAYS starts from a clean build directory.
//
// A dev server leaves unminified fallback chunks in its .next/static, and the
// export copies static/ into out/ wholesale. That silently added 16 MB of
// development bundles to the APK (6.2 MB -> 10.5 MB) with no warning.
//
// Clearing `.next` is safe now that the dev server lives in `.next-dev`. When
// they shared `.next`, this wiped the running dev server's manifests and it
// served 500s with a different phantom-missing file every reload - which looks
// exactly like the file-handle race this script exists to survive.
removeWithRetries(nextDir);

let lastOutput = "";
for (let attempt = 1; attempt <= ATTEMPTS; attempt++) {
    // Next removes ./out itself but fails outright if a handle is still open.
    removeWithRetries(outDir);

    const res = spawnSync("npx", ["next", "build"], {
        cwd: root,
        // PRONOTES_TYPECHECKED is set above once the preflight type check passes
        // (and only then - the SKIP hatch leaves it unset, so Next type-checks).
        // Spread after the mutation so the child inherits it.
        env: { ...process.env, PRONOTES_STATIC: "1" },
        encoding: "utf8",
        shell: process.platform === "win32",
        maxBuffer: 64 * 1024 * 1024,
    });

    lastOutput = `${res.stdout ?? ""}${res.stderr ?? ""}`;
    process.stdout.write(lastOutput);

    if (res.status === 0) {
        if (attempt > 1) console.log(`\n[build-static] Succeeded on attempt ${attempt} of ${ATTEMPTS}.`);
        process.exit(0);
    }

    if (!isHandleRace(lastOutput)) {
        console.error(`\n[build-static] This is a real build error, not the file-handle race - not retrying.`);
        break;
    }
    if (attempt === ATTEMPTS) break;

    console.log(
        `\n[build-static] Attempt ${attempt} of ${ATTEMPTS} hit the Windows file-handle race ` +
        `(not a real build error). Clearing .next and retrying ...`
    );
    removeWithRetries(nextDir);
}

if (isHandleRace(lastOutput)) {
    console.error(`\n[build-static] Failed ${ATTEMPTS} times on the file-handle race.`);
    warnAboutCompetingProcesses();
}
process.exit(1);

=====
// === END FILE: scripts/build-static.mjs ===
=====

```

```

=====
// === START FILE: scripts/build.mjs ===
// File 16 of 194 | Lines: 33 | Size: 1162 bytes
=====

/**
 * Production server build: `npm run build`.
 *
 * Type-checks first, then runs `next build` with PRONOTES_TYPECHECKED=1 so
 * next.config.ts lets Next skip its own duplicate type pass (it just passed
 * here). A bare `npx next build` never sets the marker, so it still fails on
 * type errors instead of shipping them green.
 *
 * A script – not `tsc && next build` inline in package.json – because setting
 * an env var inline is not portable to Windows cmd. No new dependencies.
 */
import { spawnSync } from "node:child_process";

const root = process.cwd();

if (process.env.PRONOTES_SKIP_TYPECHECK === "1") {
  console.warn("[build] PRONOTES_SKIP_TYPECHECK=1 - building WITHOUT a type check.");
} else {
  const t = spawnSync("npx tsc --noEmit", { cwd: root, stdio: "inherit", shell: true });
  if (t.status !== 0) {
    console.error("\n[build] Type-checking failed - not building.");
    process.exit(t.status ?? 1);
  }
  process.env.PRONOTES_TYPECHECKED = "1";
}

const b = spawnSync("npx", ["next", "build"], {
  cwd: root,
  stdio: "inherit",
  shell: process.platform === "win32",
  env: { ...process.env },
});
process.exit(b.status ?? 1);

=====
// === END FILE: scripts/build.mjs ===
=====

=====
// === START FILE: scripts/deploy-android.mjs ===
// File 17 of 194 | Lines: 119 | Size: 4162 bytes
=====

/**
 * Build the APK and push it straight onto a connected tablet.
 *
 * npm run apk:deploy          build, install, launch
 * npm run apk:deploy -- --fast skip the build, install the existing APK
 *
 * Why this exists
 * -----
 * `npm run apk` stops at a file and tells you to copy it across by hand. Over
 * USB that is unnecessary: adb can replace the installed app in place. `-r`
 * keeps the app's data, so notebooks and imported PDFs survive the update –
 * an uninstall/reinstall would wipe them.
 */
import { execSync, execFileSync } from "node:child_process";
import fs from "node:fs";
import path from "node:path";

const root = process.cwd();
const args = process.argv.slice(2);
const fast = args.includes("--fast");
const APK = path.join(root, "ProNotes.apk");
const PKG = "com.idrees.pronotes";

const sdk = process.env.ANDROID_HOME || process.env.ANDROID_SDK_ROOT;
if (!sdk) {
  console.error("ANDROID_HOME is not set – cannot find adb.");
  process.exit(1);
}
const adb = path.join(sdk, "platform-tools", process.platform === "win32" ? "adb.exe" : "adb");

```

```

const sh = (args, opts = {}) => execFileSync(adb, args, { encoding: "utf8", ...opts }).trim();

// --- which device? -----
let devices = [];
try {
  devices = sh(["devices"])
    .split("\n")
    .slice(1)
    .map((l) => l.trim().split(/\s+/))
    .filter((p) => p.length >= 2 && p[1] === "device")
    .map((p) => p[0]);
} catch (e) {
  console.error("adb is not responding:", e.message);
  process.exit(1);
}

const unauthorized = (() => {
  try {
    return sh(["devices"]).includes("unauthorized");
  } catch {
    return false;
  }
})();

if (unauthorized) {
  console.error(
    "\nThe tablet is connected but NOT authorised.\n" +
    "Unlock it and tap \"Allow USB debugging\" on the prompt, ticking\n" +
    "\"Always allow from this computer\", then run this again."
  );
  process.exit(1);
}

if (devices.length === 0) {
  console.error(
    "\nNo device found. On the tablet:\n" +
    "  1. Settings > About tablet > tap \"Build number\" seven times\n" +
    "  2. Settings > System > Developer options > enable \"USB debugging\"\n" +
    "  3. Plug in the USB-C cable and accept the prompt\n" +
    "Then: adb devices (it should list one line ending in 'device')"
  );
  process.exit(1);
}

// Prefer a real tablet over a running emulator.
const physical = devices.filter((d) => !d.startsWith("emulator-"));
const target = (physical.length ? physical : devices)[0];
if (devices.length > 1) {
  console.log(`Several devices attached (${devices.join(", ")} – using ${target}.`);
}

// --- build -----
if (!fast) {
  execSync("node scripts/build-apk.mjs", { stdio: "inherit", cwd: root });
} else {
  if (!fs.existsSync(APK)) {
    console.error(`--fast given but ${APK} does not exist yet. Run without --fast once.`);
    process.exit(1);
  }
  const age = Math.round((Date.now() - fs.statSync(APK).mtimeMs) / 60000);
  console.log(`--fast: installing the existing APK (built ${age} min ago) without rebuilding.`);
}

// --- install -----
const mb = (fs.statSync(APK).size / 1048576).toFixed(1);
console.log(`\nInstalling ProNotes.apk (${mb} MB) on ${target} ...`);
let out;
try {
  out = execFileSync(adb, ["-s", target, "install", "-r", APK], { encoding: "utf8" });
} catch (e) {
  out = `${e.stdout || ""}${e.stderr || ""}`;
}

if (!/Success/.test(out)) {
  console.error(out.trim());
  if (/UPDATE_INCOMPATIBLE|signatures do not match/i.test(out)) {
    console.error(
      "\nThe installed copy was signed with a different key, so it cannot be\n" +

```

```

        "replaced in place. Uninstalling REMOVES ITS NOTEBOOKS — export a backup\n" +
        "from the app first, then:\n" +
        `adb -s ${target} uninstall ${PKG}`
    );
}
process.exit(1);
}

execFileSync(adb, ["-s", target, "shell", "monkey", "-p", PKG, "-c", "android.intent.category.LAUNCHER", "1"],
    {
        stdio: "ignore",
    });
console.log("Installed and launched. Existing notebooks were kept (-r).");

=====
// === END FILE: scripts/deploy-android.mjs ===
=====

=====
// === START FILE: scripts/generate-icons.js ===
// File 18 of 194 | Lines: 156 | Size: 5334 bytes
=====

/**
 * Icon Generation Pipeline for ProNotes
 * =====
 * Generates all multi-resolution brand assets, PWA icons, favicons,
 * and maskable adaptive icons using sharp.
 */

const path = require("path");
const fs = require("fs");
const sharp = require("sharp");

// Kept inside the project on purpose. This used to point at a file in an
// external scratch/brain folder, so the script broke the moment that folder was
// cleaned or the project was moved.
const SOURCE_ICON = path.join(__dirname, "..", "assets", "pronotes-icon-source.jpg");
const PUBLIC_DIR = path.resolve(__dirname, "../public");

// Ensure public directory exists
if (!fs.existsSync(PUBLIC_DIR)) {
    fs.mkdirSync(PUBLIC_DIR, { recursive: true });
}

// Background color for maskable icons and dark slate UI
const MASKABLE_BG = { r: 15, g: 23, b: 42, alpha: 1 }; // #0f172a

/**
 * Creates a valid binary ICO file embedding one or more PNG buffers.
 * @param {Array<{ width: number, height: number, buffer: Buffer }>} images
 * @returns {Buffer}
 */
function createIco(images) {
    const count = images.length;
    const headerSize = 6;
    const dirEntrySize = 16;
    let offset = headerSize + count * dirEntrySize;

    const header = Buffer.alloc(headerSize);
    header.writeUInt16LE(0, 0); // Reserved
    header.writeUInt16LE(1, 2); // Type 1 = ICO
    header.writeUInt16LE(count, 4); // Number of images

    const dirEntries = [];
    const imageBuffers = [];

    for (const img of images) {
        const entry = Buffer.alloc(dirEntrySize);
        entry.writeUInt8(img.width >= 256 ? 0 : img.width, 0);
        entry.writeUInt8(img.height >= 256 ? 0 : img.height, 1);
        entry.writeUInt8(0, 2); // Color palette
        entry.writeUInt8(0, 3); // Reserved
        entry.writeUInt16LE(1, 4); // Color planes
        entry.writeUInt16LE(32, 6); // Bits per pixel
        entry.writeUInt32LE(img.buffer.length, 8); // Image size in bytes
        entry.writeUInt32LE(offset, 12); // Image offset
    }
}

```

```
    dirEntries.push(entry);
    imageBuffers.push(img.buffer);
    offset += img.buffer.length;
  }

  return Buffer.concat([header, ...dirEntries, ...imageBuffers]);
}

async function run() {
  console.log(`[Icon Pipeline] Checking source icon: ${SOURCE_ICON}`);
  if (!fs.existsSync(SOURCE_ICON)) {
    throw new Error(`Source icon not found at: ${SOURCE_ICON}`);
  }

  console.log("[Icon Pipeline] Generating high-resolution PNG assets...");

  // 1. Standard PWA Icons
  const pwa192 = path.join(PUBLIC_DIR, "icon-192.png");
  await sharp(SOURCE_ICON)
    .resize(192, 192, { fit: "cover" })
    .png({ quality: 100 })
    .toFile(pwa192);
  console.log(` ✓ Created ${pwa192}`);

  const pwa512 = path.join(PUBLIC_DIR, "icon-512.png");
  await sharp(SOURCE_ICON)
    .resize(512, 512, { fit: "cover" })
    .png({ quality: 100 })
    .toFile(pwa512);
  console.log(` ✓ Created ${pwa512}`);

  // 2. High-res logo
  const logo512 = path.join(PUBLIC_DIR, "logo.png");
  await sharp(SOURCE_ICON)
    .resize(512, 512, { fit: "cover" })
    .png({ quality: 100 })
    .toFile(logo512);
  console.log(` ✓ Created ${logo512}`);

  // 3. Apple Touch Icon (iOS Safari)
  const appleTouch = path.join(PUBLIC_DIR, "apple-touch-icon.png");
  await sharp(SOURCE_ICON)
    .resize(180, 180, { fit: "cover" })
    .png({ quality: 100 })
    .toFile(appleTouch);
  console.log(` ✓ Created ${appleTouch}`);

  // 4. Favicon PNGs
  const fav32 = path.join(PUBLIC_DIR, "favicon-32x32.png");
  const fav32Buffer = await sharp(SOURCE_ICON)
    .resize(32, 32, { fit: "cover" })
    .png({ quality: 100 })
    .toBuffer();
  fs.writeFileSync(fav32, fav32Buffer);
  console.log(` ✓ Created ${fav32}`);

  const fav16 = path.join(PUBLIC_DIR, "favicon-16x16.png");
  const fav16Buffer = await sharp(SOURCE_ICON)
    .resize(16, 16, { fit: "cover" })
    .png({ quality: 100 })
    .toBuffer();
  fs.writeFileSync(fav16, fav16Buffer);
  console.log(` ✓ Created ${fav16}`);

  const favMain = path.join(PUBLIC_DIR, "favicon.png");
  fs.writeFileSync(favMain, fav32Buffer);
  console.log(` ✓ Created ${favMain}`);

  // 5. Maskable Adaptive Icons (Android safe zone = 80% with #0f172a background padding)
  const mask512 = path.join(PUBLIC_DIR, "icon-512-maskable.png");
  await sharp(SOURCE_ICON)
    .resize(410, 410, { fit: "contain", background: MASKABLE_BG })
    .extend({ top: 51, bottom: 51, left: 51, right: 51, background: MASKABLE_BG })
    .png({ quality: 100 })
    .toFile(mask512);
  console.log(` ✓ Created ${mask512}`);
```

```

const mask192 = path.join(PUBLIC_DIR, "icon-192-maskable.png");
await sharp(SOURCE_ICON)
  .resize(154, 154, { fit: "contain", background: MASKABLE_BG })
  .extend({ top: 19, bottom: 19, left: 19, right: 19, background: MASKABLE_BG })
  .png({ quality: 100 })
  .toFile(mask192);
console.log(` ✓ Created ${mask192}`);

// 6. Favicon .ICO (Multi-resolution: 32x32 and 16x16)
const favIco = path.join(PUBLIC_DIR, "favicon.ico");
const icoBuffer = createIco([
  { width: 32, height: 32, buffer: fav32Buffer },
  { width: 16, height: 16, buffer: fav16Buffer },
]);
fs.writeFileSync(favIco, icoBuffer);
console.log(` ✓ Created ${favIco}`);

console.log("[Icon Pipeline] All brand assets successfully generated!");
}

run().catch((err) => {
  console.error("[Icon Pipeline Error]", err);
  process.exit(1);
});

=====
// === END FILE: scripts/generate-icons.js ===
=====

=====
// === START FILE: scripts/make-android-icons.mjs ===
// File 19 of 194 | Lines: 51 | Size: 1779 bytes
=====

/**
 * Generate the Android launcher icons from public/icon-512.png.
 * Capacitor scaffolds Capacitor's own placeholder icon; this replaces it with
 * the ProNotes mark at every density, plus the round + foreground variants
 * Android's adaptive icons need.
 */
import sharp from "sharp";
import fs from "node:fs/promises";
import path from "node:path";

const SRC = path.join(process.cwd(), "public", "icon-512.png");
const RES = path.join(process.cwd(), "android", "app", "src", "main", "res");

// density -> launcher icon size in px
const DENSITIES = {
  "mipmap-mdpi": 48,
  "mipmap-hdpi": 72,
  "mipmap-xhdpi": 96,
  "mipmap-xxhdpi": 144,
  "mipmap-xxxhdpi": 192,
};

for (const [dir, size] of Object.entries(DENSITIES)) {
  const outDir = path.join(RES, dir);
  await fs.mkdir(outDir, { recursive: true });

  const square = await sharp(SRC).resize(size, size, { fit: "contain" }).png().toBuffer();
  await fs.writeFile(path.join(outDir, "ic_launcher.png"), square);
  await fs.writeFile(path.join(outDir, "ic_launcher_round.png"), square);

  // Adaptive-icon foreground: the system crops ~1/3, so the mark is inset
  // into a transparent canvas or it gets its edges shaved off.
  const inner = Math.round(size * 0.62);
  const pad = Math.round((size - inner) / 2);
  const foreground = await sharp({
    create: {
      width: size,
      height: size,
      channels: 4,
      background: { r: 0, g: 0, b: 0, alpha: 0 },
    },
  })
  .composite([

```

```

    { input: await sharp(SRC).resize(inner, inner, { fit: "contain" }).png().toBuffer(), top: pad, left: pad
    },
  ])
  .png()
  .toBuffer();
  await fs.writeFile(path.join(outDir, "ic_launcher_foreground.png"), foreground);
  console.log(`${dir}: ${size}px`);
}
console.log("Android launcher icons written.");

```

```

=====
// === END FILE: scripts/make-android-icons.mjs ===
=====

```

```

=====
// === START FILE: scripts/make-ico.mjs ===
// File 20 of 194 | Lines: 48 | Size: 1724 bytes
=====

```

```

/**
 * Build a multi-resolution Windows .ico from public/icon-512.png.
 *
 * The repo only shipped a 2KB favicon.ico (16/32px), which looks mushy as a
 * desktop-shortcut and taskbar icon. ICO files may embed PNGs directly, so we
 * resize with sharp and assemble the container by hand – no extra dependency.
 */
import sharp from "sharp";
import fs from "node:fs/promises";
import path from "node:path";

const SIZES = [16, 24, 32, 48, 64, 128, 256];
const src = path.join(process.cwd(), "public", "icon-512.png");
const out = path.join(process.cwd(), "public", "pronotes.ico");

const images = await Promise.all(
  SIZES.map(async (size) => ({
    size,
    data: await sharp(src).resize(size, size, { fit: "contain" }).png().toBuffer(),
  }))
);

const HEADER = 6;
const ENTRY = 16;
const header = Buffer.alloc(HEADER);
header.writeUInt16LE(0, 0); // reserved
header.writeUInt16LE(1, 2); // type 1 = icon
header.writeUInt16LE(images.length, 4);

let offset = HEADER + ENTRY * images.length;
const entries = [];
for (const img of images) {
  const e = Buffer.alloc(ENTRY);
  // 256 is stored as 0 in the single-byte width/height fields.
  e.writeUInt8(img.size === 256 ? 0 : img.size, 0);
  e.writeUInt8(img.size === 256 ? 0 : img.size, 1);
  e.writeUInt8(0, 2); // palette colours (0 = truecolour)
  e.writeUInt8(0, 3); // reserved
  e.writeUInt16LE(1, 4); // colour planes
  e.writeUInt16LE(32, 6); // bits per pixel
  e.writeUInt32LE(img.data.length, 8);
  e.writeUInt32LE(offset, 12);
  entries.push(e);
  offset += img.data.length;
}

await fs.writeFile(out, Buffer.concat([header, ...entries, ...images.map((i) => i.data)]));
console.log(`wrote ${out} (${SIZES.join("/")})px, ${await fs.stat(out).size} bytes`);

```

```

=====
// === END FILE: scripts/make-ico.mjs ===
=====

```

```

=====
// === START FILE: scripts/sync-pdf-worker.mjs ===
// File 21 of 194 | Lines: 32 | Size: 1298 bytes
=====

```

```

/**
 * Copies the pdf.js worker into public/ so it is served from the app's own
 * root in every package: the dev server, the static export, Electron's
 * `pronotes://app/` scheme, and the APK's asset root.
 *
 * It lives in public/ rather than being resolved with `new URL(...,
 * import.meta.url)` because that form cannot compile to CommonJS, which the
 * node test suite uses – and the store imports the PDF module, so the whole
 * suite failed to build. A CDN was never an option: the app must work offline.
 *
 * Run after any pdfjs-dist upgrade, or the app ships a worker from the old
 * version and rendering fails with a version-mismatch error.
 */
import fs from "node:fs";
import path from "node:path";

const from = path.join("node_modules", "pdfjs-dist", "build", "pdf.worker.min.mjs");
const to = path.join("public", "pdf.worker.min.mjs");

if (!fs.existsSync(from)) {
  console.error(`pdf.js worker not found at ${from} – is pdfjs-dist installed?`);
  process.exit(1);
}

const src = fs.readFileSync(from);
const same = fs.existsSync(to) && Buffer.compare(src, fs.readFileSync(to)) === 0;
if (same) {
  console.log("pdf.js worker already up to date");
} else {
  fs.writeFileSync(to, src);
  console.log(`pdf.js worker synced -> ${to} (${(src.length / 1024).toFixed(0)} KB)`);
}

=====
// === END FILE: scripts/sync-pdf-worker.mjs ===
=====

=====
// === START FILE: scripts/test-runner.mjs ===
// File 22 of 194 | Lines: 222 | Size: 6673 bytes
=====

import { execSync } from "node:child_process";
import fs from "node:fs";
import path from "node:path";

const rootDir = process.cwd();
const outDir = path.join(rootDir, ".test-dist");

if (!fs.existsSync(outDir)) {
  fs.mkdirSync(outDir, { recursive: true });
}

// Compile test and lib files using typescript
console.log("Compiling test files...");
const tscBin = path.join(rootDir, "node_modules", "typescript", "bin", "tsc");
execSync(`node "${tscBin}" -p tsconfig.test.json`, {
  stdio: "inherit",
  cwd: rootDir,
});

console.log("Running compiled tests...");
execSync(`node "${path.join(outDir, "tests", "milestone1.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "notebook.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "wheel_device.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "stylus_button.test.js")}"`, {
  stdio: "inherit",

```

```
    cwd: rootDir,
  });
execSync(`node "${path.join(outDir, "tests", "image_placement.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "notebook_camera.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "pens_bar_placement.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "notebook_store.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "export_geometry.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "share_insert.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "split_view.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "pointer_handoff.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "duplicate_notebook.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "workspace_store_v2.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "shape_tools.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "toolbar_fixes.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "text_editing.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "pro_text_tool.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "review_fixes.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "dictation_continuation.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "select_modes.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "onedrive.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "history.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
});
```

```
execSync(`node "${path.join(outDir, "tests", "challenger_stroke_stress.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "challenger_geometry_stress.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "adversarial_zoom_gestures.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "milestone2_tools_qa.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "milestone3_features_r3.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "adversarial_challenger1.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "regression_fix_pass.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
if (fs.existsSync(path.join(outDir, "tests", "challenger_2_empirical.test.js"))) {
  execSync(`node "${path.join(outDir, "tests", "challenger_2_empirical.test.js")}"`, {
    stdio: "inherit",
    cwd: rootDir,
  });
}
execSync(`node "${path.join(outDir, "tests", "pen_dynamics_and_stabilization.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "image_studio.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "dictation_robustness.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "office_import.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "toolbar_laser.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "eraser_settle.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "shape_recognition.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "capture.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "search_scopes.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "pdf_text.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "spen_review.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
});
```

```
execSync(`node "${path.join(outDir, "tests", "spen_deep_audit.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "m1_adversarial_spen_stress.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "interactive_knowledge_graph.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "top_tools_and_sidebar.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "sidebar_trash.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
execSync(`node "${path.join(outDir, "tests", "sidebar_batch.test.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});

// The tests/e2e suite has its own plain-JS runner and was never wired in here,
// so 258 tests sat outside the `npm test` gate and quietly went stale.
console.log("\nRunning E2E suite (tests/e2e)...");
execSync(`node "${path.join(rootDir, "tests", "e2e", "runner.js")}"`, {
  stdio: "inherit",
  cwd: rootDir,
});
```

```
=====
// === END FILE: scripts/test-runner.mjs ===
=====
```

```
=====
// === START FILE: src/app/globals.css ===
// File 23 of 194 | Lines: 195 | Size: 7525 bytes
=====
```

```
@import "tailwindcss";
@import "tw-animate-css";

@custom-variant dark (&:is(.dark *));

@theme inline {
  --color-background: var(--background);
  --color-foreground: var(--foreground);
  --font-sans: var(--font-geist-sans);
  --font-mono: var(--font-geist-mono);
  --color-sidebar-ring: var(--sidebar-ring);
  --color-sidebar-border: var(--sidebar-border);
  --color-sidebar-accent-foreground: var(--sidebar-accent-foreground);
  --color-sidebar-accent: var(--sidebar-accent);
  --color-sidebar-primary-foreground: var(--sidebar-primary-foreground);
  --color-sidebar-primary: var(--sidebar-primary);
  --color-sidebar-foreground: var(--sidebar-foreground);
  --color-sidebar: var(--sidebar);
  --color-chart-5: var(--chart-5);
  --color-chart-4: var(--chart-4);
  --color-chart-3: var(--chart-3);
  --color-chart-2: var(--chart-2);
  --color-chart-1: var(--chart-1);
  --color-ring: var(--ring);
  --color-input: var(--input);
  --color-border: var(--border);
  --color-destructive: var(--destructive);
  --color-accent-foreground: var(--accent-foreground);
  --color-accent: var(--accent);
  --color-muted-foreground: var(--muted-foreground);
  --color-muted: var(--muted);
  --color-secondary-foreground: var(--secondary-foreground);
  --color-secondary: var(--secondary);
  --color-primary-foreground: var(--primary-foreground);
  --color-primary: var(--primary);
```

```
--color-popover-foreground: var(--popover-foreground);
--color-popover: var(--popover);
--color-card-foreground: var(--card-foreground);
--color-card: var(--card);
--radius-sm: calc(var(--radius) - 4px);
--radius-md: calc(var(--radius) - 2px);
--radius-lg: var(--radius);
--radius-xl: calc(var(--radius) + 4px);
}

:root {
  --radius: 0.625rem;
  --background: oklch(1 0 0);
  --foreground: oklch(0.145 0 0);
  --card: oklch(1 0 0);
  --card-foreground: oklch(0.145 0 0);
  --popover: oklch(1 0 0);
  --popover-foreground: oklch(0.145 0 0);
  --primary: oklch(0.205 0 0);
  --primary-foreground: oklch(0.985 0 0);
  --secondary: oklch(0.97 0 0);
  --secondary-foreground: oklch(0.205 0 0);
  --muted: oklch(0.97 0 0);
  --muted-foreground: oklch(0.556 0 0);
  --accent: oklch(0.97 0 0);
  --accent-foreground: oklch(0.205 0 0);
  --destructive: oklch(0.577 0.245 27.325);
  --border: oklch(0.922 0 0);
  --input: oklch(0.922 0 0);
  --ring: oklch(0.708 0 0);
  --chart-1: oklch(0.646 0.222 41.116);
  --chart-2: oklch(0.6 0.118 184.704);
  --chart-3: oklch(0.398 0.07 227.392);
  --chart-4: oklch(0.828 0.189 84.429);
  --chart-5: oklch(0.769 0.188 70.08);
  --sidebar: oklch(0.985 0 0);
  --sidebar-foreground: oklch(0.145 0 0);
  --sidebar-primary: oklch(0.205 0 0);
  --sidebar-primary-foreground: oklch(0.985 0 0);
  --sidebar-accent: oklch(0.97 0 0);
  --sidebar-accent-foreground: oklch(0.205 0 0);
  --sidebar-border: oklch(0.922 0 0);
  --sidebar-ring: oklch(0.708 0 0);
}

.dark {
  --background: oklch(0.145 0 0);
  --foreground: oklch(0.985 0 0);
  --card: oklch(0.205 0 0);
  --card-foreground: oklch(0.985 0 0);
  --popover: oklch(0.205 0 0);
  --popover-foreground: oklch(0.985 0 0);
  --primary: oklch(0.922 0 0);
  --primary-foreground: oklch(0.205 0 0);
  --secondary: oklch(0.269 0 0);
  --secondary-foreground: oklch(0.985 0 0);
  --muted: oklch(0.269 0 0);
  --muted-foreground: oklch(0.708 0 0);
  --accent: oklch(0.269 0 0);
  --accent-foreground: oklch(0.985 0 0);
  --destructive: oklch(0.704 0.191 22.216);
  --border: oklch(1 0 0 / 10%);
  --input: oklch(1 0 0 / 15%);
  --ring: oklch(0.556 0 0);
  --chart-1: oklch(0.488 0.243 264.376);
  --chart-2: oklch(0.696 0.17 162.48);
  --chart-3: oklch(0.769 0.188 70.08);
  --chart-4: oklch(0.627 0.265 303.9);
  --chart-5: oklch(0.645 0.246 16.439);
  --sidebar: oklch(0.205 0 0);
  --sidebar-foreground: oklch(0.985 0 0);
  --sidebar-primary: oklch(0.488 0.243 264.376);
  --sidebar-primary-foreground: oklch(0.985 0 0);
  --sidebar-accent: oklch(0.269 0 0);
  --sidebar-accent-foreground: oklch(0.985 0 0);
  --sidebar-border: oklch(1 0 0 / 10%);
  --sidebar-ring: oklch(0.556 0 0);
}
```

```

@layer base {
  * {
    @apply border-border outline-ring/50;
  }
  html {
    /* The whole page must never zoom: double-tap zoom broke every double-tap
       gesture, and pinch-zooming the PAGE (not the canvas) slides content
       half off screen with names cut at the edge. The viewport
       maximum-scale/user-scalable lock IS emitted (see layout.tsx) but the
       Android WebView does not honor it, so this locks it at the gesture
       level instead. pan-x pan-y (scroll only): the canvas keeps its own
       JS-driven pinch/pan behind its touch-none surface, and every panel
       keeps scrolling. Nothing in the app needs browser-level zoom. */
    touch-action: pan-x pan-y;
  }
  html,
  body {
    /* The app is a fixed h-screen layout; every list scrolls inside its own
       container. If any floating element ever measures wider than the
       viewport, the page itself must still not pan – a panned page is exactly
       the "names cut off on the left" look, and no in-app scroller can undo
       it. Belt and suspenders next to the row-level min-w-0/truncate fixes. */
    overflow: hidden;
    overscroll-behavior: none;
  }
  body {
    @apply bg-background text-foreground;
  }
}

/*
 * Two rules that CANNOT be written as utility classes in this build.
 *
 * `overflow-x-clip` and the `pointer-coarse:` variant were both used in the
 * sidebar and both compiled to NOTHING - Tailwind drops utilities it does not
 * recognise without a word, so the fixes looked right in the source, passed
 * review, and did nothing on the device. Verified against the served CSS:
 * `overflow-x-hidden` is emitted, `overflow-x-clip` is not.
 *
 * These live outside @layer on purpose: unlayered rules beat layered utilities,
 * so `show-on-touch` wins against the `opacity-0` it has to override.
 */

/* Never let a container pan sideways. `clip` rather than `hidden` because
   hidden still creates a scroll container, and a focus or scrollIntoView can
   then scroll it programmatically - which is exactly the "names slide off to
   the left" bug this is guarding against. */
.no-pan-x {
  overflow-x: clip;
}

/* Row actions (the ... button) are revealed by hover on a mouse. A finger has
   no hover, so on a touch device they must simply always be visible. */
@media (pointer: coarse) {
  .show-on-touch {
    opacity: 1;
  }
}

/*
 * Radix ScrollArea: stop the viewport shrink-wrapping to its content.
 *
 * Radix injects its own wrapper inside the viewport with the inline style
 * `min-width: 100%; display: table`. `display: table` sizes to CONTENT, so in
 * a 255px sidebar a long board name made that wrapper 622px wide. Every
 * `truncate` and `min-w-0` inside it then had hundreds of pixels of room and
 * never truncated anything; the viewport's own `overflow-x: hidden` hid the
 * excess but still allowed PROGRAMMATIC scrolling, so focusing a row (tapping
 * it, or opening its menu) panned the list sideways and left the names cut
 * off at the left edge with the ... pushed out of reach.
 *
 * `display: table` only earns its keep when you want a wide child to scroll
 * horizontally. Every ScrollArea here is a vertical list - sidebar, search
 * results, backlinks, shortcuts, link manager - so block is right for all of
 * them, and the same latent truncation bug is fixed everywhere at once.
 *
 * !important is required: it is competing with an inline style.

```

```
*/
[data-radix-scroll-area-viewport] > div {
  display: block !important;
}

=====
// === END FILE: src/app/globals.css ===
=====

=====
// === START FILE: src/app/layout.tsx ===
// File 24 of 194 | Lines: 102 | Size: 2576 bytes
=====

import type { Metadata, Viewport } from "next";
import {
  Inter,
  Caveat,
  JetBrains_Mono,
  Playfair_Display,
  Plus_Jakarta_Sans,
} from "next/font/google";
import "./globals.css";
import { Toaster } from "@components/ui/toaster";
import { PWARegister } from "@components/pwa/PWARegister";
import { ThemeProvider } from "@components/theme-provider";

const inter = Inter({
  variable: "--font-inter",
  subsets: ["latin"],
});

const caveat = Caveat({
  variable: "--font-caveat",
  subsets: ["latin"],
});

const jetbrainsMono = JetBrains_Mono({
  variable: "--font-mono",
  subsets: ["latin"],
});

const playfair = Playfair_Display({
  variable: "--font-serif",
  subsets: ["latin"],
});

const plusJakarta = Plus_Jakarta_Sans({
  variable: "--font-display",
  subsets: ["latin"],
});

export const metadata: Metadata = {
  title: "ProNotes — Infinite Whiteboard & Knowledge Canvas",
  description:
    "ProNotes is an infinite whiteboard and visual knowledge canvas with Obsidian-style WikiLinks, smart connectors, freehand drawing, and touch support.",
  keywords: [
    "pronotes",
    "whiteboard",
    "drawing",
    "canvas",
    "infinite canvas",
    "knowledge graph",
    "backlinks",
    "mind map",
    "obsidian canvas",
  ],
  authors: [{ name: "ProNotes Team" }],
  applicationName: "ProNotes",
  manifest: "/manifest.json",
  appleWebApp: {
    capable: true,
    statusBarStyle: "black-translucent",
    title: "ProNotes",
  },
  icons: {
```

```
    icon: [
      { url: "/favicon.ico" },
      { url: "/favicon.png", sizes: "32x32", type: "image/png" },
      { url: "/icon-192.png", sizes: "192x192", type: "image/png" },
      { url: "/icon-512.png", sizes: "512x512", type: "image/png" },
    ],
    apple: [
      { url: "/apple-touch-icon.png", sizes: "180x180", type: "image/png" },
    ],
  },
};

export const viewport: Viewport = {
  themeColor: "#4f46e5",
  width: "device-width",
  initialScale: 1,
  maximumScale: 1,
  userScalable: false,
};

export default function RootLayout({
  children,
}: Readonly<{
  children: React.ReactNode;
}>) {
  return (
    <html lang="en" suppressHydrationWarning>
      <body
        className={` ${inter.variable} ${caveat.variable} ${jetbrainsMono.variable} ${playfair.variable} ${plusJakarta.variable} font-sans antialiased bg-background text-foreground`}
      >
        <ThemeProvider attribute="class" defaultTheme="system" enableSystem disableTransitionOnChange={false}>
          <PWARegister />
          {children}
          <Toaster />
        </ThemeProvider>
      </body>
    </html>
  );
}
```

```
=====
// === END FILE: src/app/layout.tsx ===
=====
```

```
=====
// === START FILE: src/app/page.tsx ===
// File 25 of 194 | Lines: 23 | Size: 560 bytes
=====
```

```
"use client";

import React, { useEffect, useState } from "react";
import { WhiteboardApp } from "@components/whiteboard/WhiteboardApp";

export default function Home() {
  const [mounted, setMounted] = useState(false);

  useEffect(() => {
    setMounted(true);
  }, []);

  if (!mounted) {
    return (
      <div className="w-screen h-screen flex items-center justify-center bg-background">
        <div className="text-muted-foreground text-sm font-medium animate-pulse">Loading whiteboard...</div>
      </div>
    );
  }

  return <WhiteboardApp />;
}
```

```
=====
// === END FILE: src/app/page.tsx ===
=====
```

```

=====

// === START FILE: src/components/pwa/PWARegister.tsx ===
// File 26 of 194 | Lines: 134 | Size: 5068 bytes
=====

"use client";

/**
 * PWARegister Component
 * =====
 * Registers the ProNotes Service Worker (/sw.js) - but ONLY where it helps -
 * and manages PWA install prompt lifecycle events.
 *
 * Where it must NOT run
 * -----
 * - Inside the Android APK. Every file is already on the device, so it adds no
 *   offline ability, and it is actively harmful: sw.js is cache-first for
 *   static files, and `pdf.worker.min.mjs` has no content hash in its name.
 *   After an APK update that bumps pdf.js, the first PDF opened would get the
 *   OLD cached worker and fail with a version mismatch.
 * - In the Electron app (`pronotes://`): service workers need http(s).
 * - Under `npm run dev`: dev chunks are not hashed either, so cache-first
 *   serves the previous code after an edit.
 *
 * Where it is not wanted it is also UNREGISTERED and its caches deleted,
 * because APKs built before this change already installed one.
 */

import React, { useEffect } from "react";
import { Capacitor } from "@capacitor/core";

/** Cache names written by sw.js all start with this. */
const SW_CACHE_PREFIX = "pronotes-";

function serviceWorkerWanted(): boolean {
  if (typeof window === "undefined" || !("serviceWorker" in navigator)) return false;
  if (Capacitor.isNativePlatform()) return false;
  if (!/^https?:$/.test(window.location.protocol ?? "")) return false;
  return process.env.NODE_ENV === "production";
}

/** Removes a service worker and its caches left by an earlier build. */
async function removeServiceWorker(): Promise<void> {
  try {
    if ("serviceWorker" in navigator) {
      const regs = await navigator.serviceWorker.getRegistrations();
      await Promise.all(regs.map((r) => r.unregister()));
      if (regs.length) console.log(`[PWA] Unregistered ${regs.length} service worker(s) - not used here.`);
    }
    if (typeof caches !== "undefined") {
      const keys = await caches.keys();
      await Promise.all(keys.filter((k) => k.startsWith(SW_CACHE_PREFIX)).map((k) => caches.delete(k)));
    }
  } catch (err) {
    console.warn("[PWA] Could not remove an old service worker:", err);
  }
}

// Global interface extension for BeforeInstallPromptEvent
declare global {
  interface WindowEventMap {
    beforeinstallprompt: BeforeInstallPromptEvent;
  }

  interface BeforeInstallPromptEvent extends Event {
    readonly platforms: string[];
    readonly userChoice: Promise<{
      outcome: "accepted" | "dismissed";
      platform: string;
    }>;
    prompt(): Promise<void>;
  }
}

export function PWARegister() {

```

```

useEffect(() => {
  // 1. Register Service Worker - or remove one an older build left behind.
  if (!serviceWorkerWanted()) {
    void removeServiceWorker();
  } else {
    // This effect runs after React mounts, which is often AFTER the window
    // `load` event. Waiting for `load` unconditionally meant the worker was
    // frequently never registered at all.
    const register = () => {
      navigator.serviceWorker
        .register("/sw.js")
        .then((registration) => {
          console.log("[PWA] Service Worker registered with scope:", registration.scope);

          // Check for updates
          registration.onupdatefound = () => {
            const installingWorker = registration.installing;
            if (installingWorker) {
              installingWorker.onstatechange = () => {
                if (installingWorker.state === "installed") {
                  if (navigator.serviceWorker.controller) {
                    console.log("[PWA] New content available; please refresh.");
                  } else {
                    console.log("[PWA] Content cached for offline use.");
                  }
                }
              }
            }
          };
        })
        .catch((error) => {
          console.warn("[PWA] Service Worker registration failed:", error);
        });
    };
    if (document.readyState === "complete") register();
    else window.addEventListener("load", register, { once: true });
  }

  // 2. Intercept BeforeInstallPrompt
  const handleBeforeInstallPrompt = (e: BeforeInstallPromptEvent) => {
    // Store event on window for UI triggers
    (window as any).__PWA_INSTALL_PROMPT__ = e;
    window.dispatchEvent(new CustomEvent("pronotes:can-install"));
    console.log("[PWA] App is installable. Prompt captured.");
  };

  // 3. Handle App Installed
  const handleAppInstalled = () => {
    (window as any).__PWA_INSTALL_PROMPT__ = null;
    window.dispatchEvent(new CustomEvent("pronotes:installed"));
    console.log("[PWA] ProNotes was successfully installed!");
  };

  window.addEventListener("beforeinstallprompt", handleBeforeInstallPrompt);
  window.addEventListener("appinstalled", handleAppInstalled);

  return () => {
    window.removeEventListener("beforeinstallprompt", handleBeforeInstallPrompt);
    window.removeEventListener("appinstalled", handleAppInstalled);
  };
}, []);

return null;
}

```

```

=====
// === END FILE: src/components/pwa/PWARegister.tsx ===
=====

```

```

=====
// === START FILE: src/components/theme-provider.tsx ===
// File 27 of 194 | Lines: 11 | Size: 299 bytes
=====

```

```

"use client";

import * as React from "react";

```

```
import { ThemeProvider as NextThemesProvider } from "next-themes";

export function ThemeProvider({
  children,
  ...props
}: React.ComponentProps<typeof NextThemesProvider>) {
  return <NextThemesProvider {...props}>{children}</NextThemesProvider>;
}

=====
// === END FILE: src/components/theme-provider.tsx ===
=====

=====
// === START FILE: src/components/ui/button.tsx ===
// File 28 of 194 | Lines: 59 | Size: 2123 bytes
=====

import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const buttonVariants = cva(
  "inline-flex items-center justify-center gap-2 whitespace-nowrap rounded-md text-sm font-medium transition-a
  ll disabled:pointer-events-none disabled:opacity-50 [&_svg]:pointer-events-none [&_svg]:not([class*='size
  -'])]:size-4 shrink-0 [&_svg]:shrink-0 outline-none focus-visible:border-ring focus-visible:ring-ring/50
  focus-visible:ring-[3px] aria-invalid:ring-destructive/20 dark:aria-invalid:ring-destructive/40 aria-in
  valid:border-destructive",
  {
    variants: {
      variant: {
        default:
          "bg-primary text-primary-foreground shadow-xs hover:bg-primary/90",
        destructive:
          "bg-destructive text-white shadow-xs hover:bg-destructive/90 focus-visible:ring-destructive/20 dark:
          focus-visible:ring-destructive/40 dark:bg-destructive/60",
        outline:
          "border bg-background shadow-xs hover:bg-accent hover:text-accent-foreground dark:bg-input/30 dark:b
          order-input dark:hover:bg-input/50",
        secondary:
          "bg-secondary text-secondary-foreground shadow-xs hover:bg-secondary/80",
        ghost:
          "hover:bg-accent hover:text-accent-foreground dark:hover:bg-accent/50",
        link: "text-primary underline-offset-4 hover:underline",
      },
      size: {
        default: "h-9 px-4 py-2 has- [>svg]:px-3",
        sm: "h-8 rounded-md gap-1.5 px-3 has- [>svg]:px-2.5",
        lg: "h-10 rounded-md px-6 has- [>svg]:px-4",
        icon: "size-9",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  }
)

function Button({
  className,
  variant,
  size,
  asChild = false,
  ...props
}: React.ComponentProps<"button"> &
VariantProps<typeof buttonVariants> & {
  asChild?: boolean
}) {
  const Comp = asChild ? Slot : "button"

  return (
    <Comp
      data-slot="button"
      className={cn(buttonVariants({ variant, size, className }))}
    >
```

```
    {...props}
  />
)
}

export { Button, buttonVariants }

=====
// === END FILE: src/components/ui/button.tsx ===
=====

=====
// === START FILE: src/components/ui/context-menu.tsx ===
// File 29 of 194 | Lines: 252 | Size: 8222 bytes
=====

"use client"

import * as React from "react"
import * as ContextMenuPrimitive from "@radix-ui/react-context-menu"
import { CheckIcon, ChevronRightIcon, CircleIcon } from "lucide-react"

import { cn } from "@lib/utils"

function ContextMenu({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Root>) {
  return <ContextMenuPrimitive.Root data-slot="context-menu" {...props} />
}

function ContextMenuTrigger({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Trigger>) {
  return (
    <ContextMenuPrimitive.Trigger data-slot="context-menu-trigger" {...props} />
  )
}

function ContextMenuGroup({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Group>) {
  return (
    <ContextMenuPrimitive.Group data-slot="context-menu-group" {...props} />
  )
}

function ContextMenuPortal({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Portal>) {
  return (
    <ContextMenuPrimitive.Portal data-slot="context-menu-portal" {...props} />
  )
}

function ContextMenuSub({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Sub>) {
  return <ContextMenuPrimitive.Sub data-slot="context-menu-sub" {...props} />
}

function ContextMenuRadioGroup({
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.RadioGroup>) {
  return (
    <ContextMenuPrimitive.RadioGroup
      data-slot="context-menu-radio-group"
      {...props}
    />
  )
}

function ContextMenuSubTrigger({
  className,
  inset,
  children,
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.SubTrigger> & {
```

```

    inset?: boolean
  }) {
    return (
      <ContextMenuPrimitive.SubTrigger
        data-slot="context-menu-sub-trigger"
        data-inset={inset}
        className={cn(
          "focus:bg-accent focus:text-accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground flex cursor-default items-center rounded-sm px-2 py-1.5 text-sm outline-hidden select-none data-[inset]:pl-8 [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg]:not([class*='size-'])",
          className
        )}
        {...props}
      >
        {children}
        <ChevronRightIcon className="ml-auto" />
      </ContextMenuPrimitive.SubTrigger>
    )
  }

function ContextMenuSubContent({
  className,
  ...props
}): React.ComponentProps<typeof ContextMenuPrimitive.SubContent> {
  return (
    <ContextMenuPrimitive.SubContent
      data-slot="context-menu-sub-content"
      className={cn(
        "bg-popover text-popover-foreground data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 z-50 min-w-[8rem] origin-(--radix-context-menu-content-transform-origin) overflow-hidden rounded-md border p-1 shadow-lg",
        className
      )}
      {...props}
    />
  )
}

function ContextMenuContent({
  className,
  ...props
}): React.ComponentProps<typeof ContextMenuPrimitive.Content> {
  return (
    <ContextMenuPrimitive.Portal>
      <ContextMenuPrimitive.Content
        data-slot="context-menu-content"
        className={cn(
          "bg-popover text-popover-foreground data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 z-50 max-h-(--radix-context-menu-content-available-height) min-w-[8rem] origin-(--radix-context-menu-content-transform-origin) overflow-x-hidden overflow-y-auto rounded-md border p-1 shadow-md",
          className
        )}
        {...props}
      />
    </ContextMenuPrimitive.Portal>
  )
}

function ContextMenuItem({
  className,
  inset,
  variant = "default",
  ...props
}): React.ComponentProps<typeof ContextMenuPrimitive.Item> & {
  inset?: boolean
  variant?: "default" | "destructive"
}) {
  return (
    <ContextMenuPrimitive.Item
      data-slot="context-menu-item"
      data-inset={inset}

```

```

    data-variant={variant}
    className={cn(
      "focus:bg-accent focus:text-accent-foreground data-[variant=destructive]:text-destructive data-[variant=destructive]:focus:bg-destructive/10 dark:data-[variant=destructive]:focus:bg-destructive/20 data-[variant=destructive]:focus:text-destructive data-[variant=destructive]:*:svg:text-destructive [&svg:not([class='text-'])]:text-muted-foreground relative flex cursor-default items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 data-[inset]:pl-8 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class='size-'])]:size-4",
      className
    )}
    {...props}
  ) />
}

function ContextMenuCheckboxItem({
  className,
  children,
  checked,
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.CheckboxItem>) {
  return (
    <ContextMenuPrimitive.CheckboxItem
      data-slot="context-menu-checkbox-item"
      className={cn(
        "focus:bg-accent focus:text-accent-foreground relative flex cursor-default items-center gap-2 rounded-sm py-1.5 pr-2 pl-8 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class='size-'])]:size-4",
        className
      )}
      checked={checked}
      {...props}
    >
      <span className="pointer-events-none absolute left-2 flex size-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <CheckIcon className="size-4" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.CheckboxItem>
  )
}

function ContextMenuRadioItem({
  className,
  children,
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.RadioItem>) {
  return (
    <ContextMenuPrimitive.RadioItem
      data-slot="context-menu-radio-item"
      className={cn(
        "focus:bg-accent focus:text-accent-foreground relative flex cursor-default items-center gap-2 rounded-sm py-1.5 pr-2 pl-8 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class='size-'])]:size-4",
        className
      )}
      {...props}
    >
      <span className="pointer-events-none absolute left-2 flex size-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <CircleIcon className="size-2 fill-current" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.RadioItem>
  )
}

function ContextMenuLabel({
  className,
  inset,
  ...props
}: React.ComponentProps<typeof ContextMenuPrimitive.Label> & {
  inset?: boolean
}) {

```

```
  }) {
    return (
      <ContextMenuPrimitive.Label
        data-slot="context-menu-label"
        data-inset={inset}
        className={cn(
          "text-foreground px-2 py-1.5 text-sm font-medium data-[inset]:pl-8",
          className
        )}
        {...props}
      />
    )
  }
}

function ContextMenuSeparator({
  className,
  ...props
}): React.ComponentProps<typeof ContextMenuPrimitive.Separator> {
  return (
    <ContextMenuPrimitive.Separator
      data-slot="context-menu-separator"
      className={cn("bg-border -mx-1 my-1 h-px", className)}
      {...props}
    />
  )
}

function ContextMenuShortcut({
  className,
  ...props
}): React.ComponentProps<"span"> {
  return (
    <span
      data-slot="context-menu-shortcut"
      className={cn(
        "text-muted-foreground ml-auto text-xs tracking-widest",
        className
      )}
      {...props}
    />
  )
}

export {
  ContextMenu,
  ContextMenuTrigger,
  ContextMenuContent,
  ContextMenuItem,
  ContextMenuCheckboxItem,
  ContextMenuRadioItem,
  ContextMenuLabel,
  ContextMenuSeparator,
  ContextMenuShortcut,
  ContextMenuGroup,
  ContextMenuPortal,
  ContextMenuSub,
  ContextMenuSubContent,
  ContextMenuSubTrigger,
  ContextMenuRadioGroup,
}

=====
// === END FILE: src/components/ui/context-menu.tsx ===
=====

=====
// === START FILE: src/components/ui/dialog.tsx ===
// File 30 of 194 | Lines: 143 | Size: 3982 bytes
=====

"use client"

import * as React from "react"
import * as DialogPrimitive from "@radix-ui/react-dialog"
import { XIcon } from "lucide-react"

import { cn } from "@/lib/utils"
```

```

function Dialog({
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Root> {
  return <DialogPrimitive.Root data-slot="dialog" {...props} />
}

function DialogTrigger({
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Trigger> {
  return <DialogPrimitive.Trigger data-slot="dialog-trigger" {...props} />
}

function DialogPortal({
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Portal> {
  return <DialogPrimitive.Portal data-slot="dialog-portal" {...props} />
}

function DialogClose({
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Close> {
  return <DialogPrimitive.Close data-slot="dialog-close" {...props} />
}

function DialogOverlay({
  className,
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Overlay> {
  return (
    <DialogPrimitive.Overlay
      data-slot="dialog-overlay"
      className={cn(
        "data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 fixed inset-0 z-50 bg-black/50",
        className
      )}
      {...props}
    />
  )
}

function DialogContent({
  className,
  children,
  showCloseButton = true,
  ...props
}): React.ComponentProps<typeof DialogPrimitive.Content> & {
  showCloseButton?: boolean
} {
  return (
    <DialogPortal data-slot="dialog-portal">
      <DialogOverlay />
      <DialogPrimitive.Content
        data-slot="dialog-content"
        className={cn(
          "bg-background data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 fixed top-[50%] left-[50%] z-50 grid w-full max-w-[calc(100%-2rem)] translate-x-[-50%] translate-y-[-50%] gap-4 rounded-lg border p-6 shadow-lg duration-200 sm:max-w-lg",
          className
        )}
        {...props}
      >
        {children}
        {showCloseButton && (
          <DialogPrimitive.Close
            data-slot="dialog-close"
            className="ring-offset-background focus:ring-ring data-[state=open]:bg-accent data-[state=open]:text-muted-foreground absolute top-4 right-4 rounded-xs opacity-70 transition-opacity hover:opacity-100 focus:ring-2 focus:ring-offset-2 focus:outline-hidden disabled:pointer-events-none [&_svg]:pointer-events-none [&_svg]:shrink-0 [&_svg]:not([class*='size-']):size-4"
            >
              <XIcon />
              <span className="sr-only">Close</span>
            </DialogPrimitive.Close>
          )}
      </DialogPrimitive.Content>
    </DialogPortal>
  )
}

```

```
    </DialogPortal>
  )
}

function DialogHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="dialog-header"
      className={cn("flex flex-col gap-2 text-center sm:text-left", className)}
      {...props}
    />
  )
}

function DialogFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="dialog-footer"
      className={cn(
        "flex flex-col-reverse gap-2 sm:flex-row sm:justify-end",
        className
      )}
      {...props}
    />
  )
}

function DialogTitle({
  className,
  ...props
}: React.ComponentProps<typeof DialogPrimitive.Title>) {
  return (
    <DialogPrimitive.Title
      data-slot="dialog-title"
      className={cn("text-lg leading-none font-semibold", className)}
      {...props}
    />
  )
}

function DialogDescription({
  className,
  ...props
}: React.ComponentProps<typeof DialogPrimitive.Description>) {
  return (
    <DialogPrimitive.Description
      data-slot="dialog-description"
      className={cn("text-muted-foreground text-sm", className)}
      {...props}
    />
  )
}

export {
  Dialog,
  DialogClose,
  DialogContent,
  DialogDescription,
  DialogFooter,
  DialogHeader,
  DialogOverlay,
  DialogPortal,
  DialogTitle,
  DialogTrigger,
}
```

```
=====
// === END FILE: src/components/ui/dialog.tsx ===
=====
```

```
=====
// === START FILE: src/components/ui/dropdown-menu.tsx ===
// File 31 of 194 | Lines: 257 | Size: 8284 bytes
=====
```

```
"use client"
```

```
import * as React from "react"
import * as DropdownMenuPrimitive from "@radix-ui/react-dropdown-menu"
import { CheckIcon, ChevronRightIcon, CircleIcon } from "lucide-react"

import { cn } from "@lib/utils"

function DropdownMenu({
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Root>) {
  return <DropdownMenuPrimitive.Root data-slot="dropdown-menu" {...props} />
}

function DropdownMenuPortal({
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Portal>) {
  return (
    <DropdownMenuPrimitive.Portal data-slot="dropdown-menu-portal" {...props} />
  )
}

function DropdownMenuTrigger({
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Trigger>) {
  return (
    <DropdownMenuPrimitive.Trigger
      data-slot="dropdown-menu-trigger"
      {...props}
    />
  )
}

function DropdownMenuContent({
  className,
  sideOffset = 4,
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Content>) {
  return (
    <DropdownMenuPrimitive.Portal>
      <DropdownMenuPrimitive.Content
        data-slot="dropdown-menu-content"
        sideOffset={sideOffset}
        className={cn(
          "bg-popover text-popover-foreground data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 z-50 max-h-(--radix-dropdown-menu-content-available-height) min-w-[8rem] origin-(--radix-dropdown-menu-content-transform-origin) overflow-x-hidden overflow-y-auto rounded-md border p-1 shadow-md",
          className
        )}
        {...props}
      />
    </DropdownMenuPrimitive.Portal>
  )
}

function DropdownMenuGroup({
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Group>) {
  return (
    <DropdownMenuPrimitive.Group data-slot="dropdown-menu-group" {...props} />
  )
}

function DropdownMenuItem({
  className,
  inset,
  variant = "default",
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Item> & {
  inset?: boolean
  variant?: "default" | "destructive"
}) {
  return (
    <DropdownMenuPrimitive.Item
      data-slot="dropdown-menu-item"
      data-inset={inset}
      data-variant={variant}
    />
  )
}
```

```

    className={cn(
      "focus:bg-accent focus:text-accent-foreground data-[variant=destructive]:text-destructive data-[variant=destructive]:focus:bg-destructive/10 dark:data-[variant=destructive]:focus:bg-destructive/20 data-[variant=destructive]:focus:text-destructive data-[variant=destructive]:*:svg:!text-destructive [&svg]:not([class*='text-']):text-muted-foreground relative flex cursor-default items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 data-[inset]:pl-8 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class*='size-']):size-4",
      className
    )}
  )}
  {...props}
)
}

function DropdownMenuCheckboxItem({
  className,
  children,
  checked,
  ...props
}): React.ComponentProps<typeof DropdownMenuPrimitive.CheckboxItem> {
  return (
    <DropdownMenuPrimitive.CheckboxItem
      data-slot="dropdown-menu-checkbox-item"
      className={cn(
        "focus:bg-accent focus:text-accent-foreground relative flex cursor-default items-center gap-2 rounded-sm py-1.5 pr-2 pl-8 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class*='size-']):size-4",
        className
      )}
      checked={checked}
      {...props}
    >
      <span className="pointer-events-none absolute left-2 flex size-3.5 items-center justify-center">
        <DropdownMenuPrimitive.ItemIndicator>
          <CheckIcon className="size-4" />
        </DropdownMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </DropdownMenuPrimitive.CheckboxItem>
  )
}

function DropdownMenuRadioGroup({
  ...props
}): React.ComponentProps<typeof DropdownMenuPrimitive.RadioGroup> {
  return (
    <DropdownMenuPrimitive.RadioGroup
      data-slot="dropdown-menu-radio-group"
      {...props}
    />
  )
}

function DropdownMenuRadioItem({
  className,
  children,
  ...props
}): React.ComponentProps<typeof DropdownMenuPrimitive.RadioItem> {
  return (
    <DropdownMenuPrimitive.RadioItem
      data-slot="dropdown-menu-radio-item"
      className={cn(
        "focus:bg-accent focus:text-accent-foreground relative flex cursor-default items-center gap-2 rounded-sm py-1.5 pr-2 pl-8 text-sm outline-hidden select-none data-[disabled]:pointer-events-none data-[disabled]:opacity-50 [&svg]:pointer-events-none [&svg]:shrink-0 [&svg]:not([class*='size-']):size-4",
        className
      )}
      {...props}
    >
      <span className="pointer-events-none absolute left-2 flex size-3.5 items-center justify-center">
        <DropdownMenuPrimitive.ItemIndicator>
          <CircleIcon className="size-2 fill-current" />
        </DropdownMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </DropdownMenuPrimitive.RadioItem>
  )
}

```

```
    </DropdownMenuPrimitive.RadioItem>
  )
}

function DropdownMenuLabel({
  className,
  inset,
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Label> & {
  inset?: boolean
}) {
  return (
    <DropdownMenuPrimitive.Label
      data-slot="dropdown-menu-label"
      data-inset={inset}
      className={cn(
        "px-2 py-1.5 text-sm font-medium data-[inset]:pl-8",
        className
      )}
      {...props}
    />
  )
}

function DropdownMenuSeparator({
  className,
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Separator>) {
  return (
    <DropdownMenuPrimitive.Separator
      data-slot="dropdown-menu-separator"
      className={cn("bg-border -mx-1 my-1 h-px", className)}
      {...props}
    />
  )
}

function DropdownMenuShortcut({
  className,
  ...props
}: React.ComponentProps<"span">) {
  return (
    <span
      data-slot="dropdown-menu-shortcut"
      className={cn(
        "text-muted-foreground ml-auto text-xs tracking-widest",
        className
      )}
      {...props}
    />
  )
}

function DropdownMenuSub({
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.Sub>) {
  return <DropdownMenuPrimitive.Sub data-slot="dropdown-menu-sub" {...props} />
}

function DropdownMenuSubTrigger({
  className,
  inset,
  children,
  ...props
}: React.ComponentProps<typeof DropdownMenuPrimitive.SubTrigger> & {
  inset?: boolean
}) {
  return (
    <DropdownMenuPrimitive.SubTrigger
      data-slot="dropdown-menu-sub-trigger"
      data-inset={inset}
      className={cn(
        "focus:bg-accent focus:text-accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground flex cursor-default items-center rounded-sm px-2 py-1.5 text-sm outline-hidden select-none data-[inset]:pl-8",
        className
      )}
      {...props}
    />
  )
}
```

```

    >
    {children}
    <ChevronRightIcon className="ml-auto size-4" />
  </DropdownMenuPrimitive.SubTrigger>
)
}

function DropdownMenuSubContent({
  className,
  ...props
}): React.ComponentProps<typeof DropdownMenuPrimitive.SubContent> {
  return (
    <DropdownMenuPrimitive.SubContent
      data-slot="dropdown-menu-sub-content"
      className={cn(
        "bg-popover text-popover-foreground data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 z-50 min-w-[8rem] origin-(--radix-dropdown-menu-content-transform-origin) overflow-hidden rounded-md border p-1 shadow-lg",
        className
      )}
      {...props}
    />
  )
}

export {
  DropdownMenu,
  DropdownMenuPortal,
  DropdownMenuTrigger,
  DropdownMenuContent,
  DropdownMenuGroup,
  DropdownMenuLabel,
  DropdownMenuItem,
  DropdownMenuCheckboxItem,
  DropdownMenuRadioGroup,
  DropdownMenuRadioItem,
  DropdownMenuSeparator,
  DropdownMenuShortcut,
  DropdownMenuSub,
  DropdownMenuSubTrigger,
  DropdownMenuSubContent,
}

```

```

=====
// === END FILE: src/components/ui/dropdown-menu.tsx ===
=====

```

```

=====
// === START FILE: src/components/ui/input.tsx ===
// File 32 of 194 | Lines: 21 | Size: 967 bytes
=====

```

```

import * as React from "react"

import { cn } from "@lib/utils"

function Input({ className, type, ...props }: React.ComponentProps<"input">) {
  return (
    <input
      type={type}
      data-slot="input"
      className={cn(
        "file:text-foreground placeholder:text-muted-foreground selection:bg-primary selection:text-primary-foreground dark:bg-input/30 border-input flex h-9 w-full min-w-0 rounded-md border bg-transparent px-3 py-1 text-base shadow-xs transition-[color,box-shadow] outline-none file:inline-flex file:h-7 file:border-0 file:bg-transparent file:text-sm file:font-medium disabled:pointer-events-none disabled:cursor-not-allowed disabled:opacity-50 md:text-sm",
        "focus-visible:border-ring focus-visible:ring-ring/50 focus-visible:ring-[3px]",
        "aria-invalid:ring-destructive/20 dark:aria-invalid:ring-destructive/40 aria-invalid:border-destructive",
        className
      )}
      {...props}
    />
  )
}

```

```
)
}

export { Input }

=====
// === END FILE: src/components/ui/input.tsx ===
=====

=====
// === START FILE: src/components/ui/popover.tsx ===
// File 33 of 194 | Lines: 48 | Size: 1635 bytes
=====

"use client"

import * as React from "react"
import * as PopoverPrimitive from "@radix-ui/react-popover"

import { cn } from "@lib/utils"

function Popover({
  ...props
}: React.ComponentProps<typeof PopoverPrimitive.Root>) {
  return <PopoverPrimitive.Root data-slot="popover" {...props} />
}

function PopoverTrigger({
  ...props
}: React.ComponentProps<typeof PopoverPrimitive.Trigger>) {
  return <PopoverPrimitive.Trigger data-slot="popover-trigger" {...props} />
}

function PopoverContent({
  className,
  align = "center",
  sideOffset = 4,
  ...props
}: React.ComponentProps<typeof PopoverPrimitive.Content>) {
  return (
    <PopoverPrimitive.Portal>
      <PopoverPrimitive.Content
        data-slot="popover-content"
        align={align}
        sideOffset={sideOffset}
        className={cn(
          "bg-popover text-popover-foreground data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 z-50 w-72 origin-(--radix-popover-content-transform-origin) rounded-md border p-4 shadow-md outline-hidden",
          className
        )}
        {...props}
      />
    </PopoverPrimitive.Portal>
  )
}

function PopoverAnchor({
  ...props
}: React.ComponentProps<typeof PopoverPrimitive.Anchor>) {
  return <PopoverPrimitive.Anchor data-slot="popover-anchor" {...props} />
}

export { Popover, PopoverTrigger, PopoverContent, PopoverAnchor }

=====
// === END FILE: src/components/ui/popover.tsx ===
=====

=====
// === START FILE: src/components/ui/scroll-area.tsx ===
// File 34 of 194 | Lines: 58 | Size: 1645 bytes
=====
```

```
"use client"

import * as React from "react"
import * as ScrollAreaPrimitive from "@radix-ui/react-scroll-area"

import { cn } from "@lib/utils"

function ScrollArea({
  className,
  children,
  ...props
}: React.ComponentProps<typeof ScrollAreaPrimitive.Root>) {
  return (
    <ScrollAreaPrimitive.Root
      data-slot="scroll-area"
      className={cn("relative", className)}
      {...props}>
      <ScrollAreaPrimitive.Viewport
        data-slot="scroll-area-viewport"
        className="focus-visible:ring-ring/50 size-full rounded-[inherit] transition-[color,box-shadow] outline-none focus-visible:ring-[3px] focus-visible:outline-1"
        >
        {children}
      </ScrollAreaPrimitive.Viewport>
      <ScrollBar />
      <ScrollAreaPrimitive.Corner />
    </ScrollAreaPrimitive.Root>
  )
}

function ScrollBar({
  className,
  orientation = "vertical",
  ...props
}: React.ComponentProps<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>) {
  return (
    <ScrollAreaPrimitive.ScrollAreaScrollbar
      data-slot="scroll-area-scrollbar"
      orientation={orientation}
      className={cn(
        "flex touch-none p-px transition-colors select-none",
        orientation === "vertical" &&
        "h-full w-2.5 border-l border-l-transparent",
        orientation === "horizontal" &&
        "h-2.5 flex-col border-t border-t-transparent",
        className
      )}>
      <ScrollAreaPrimitive.ScrollAreaThumb
        data-slot="scroll-area-thumb"
        className="bg-border relative flex-1 rounded-full"
        />
    </ScrollAreaPrimitive.ScrollAreaScrollbar>
  )
}

export { ScrollArea, ScrollBar }
```

```
=====
// === END FILE: src/components/ui/scroll-area.tsx ===
=====
```

```
=====
// === START FILE: src/components/ui/separator.tsx ===
// File 35 of 194 | Lines: 28 | Size: 699 bytes
=====
```

```
"use client"

import * as React from "react"
import * as SeparatorPrimitive from "@radix-ui/react-separator"

import { cn } from "@lib/utils"

function Separator({
```

```

    className,
    orientation = "horizontal",
    decorative = true,
    ...props
  }: React.ComponentProps<typeof SeparatorPrimitive.Root>) {
    return (
      <SeparatorPrimitive.Root
        data-slot="separator"
        decorative={decorative}
        orientation={orientation}
        className={cn(
          "bg-border shrink-0 data-[orientation=horizontal]:h-px data-[orientation=horizontal]:w-full data-[orientation=vertical]:h-full data-[orientation=vertical]:w-px",
          className
        )}
      >{...props}</>
    )
  }
}

```

```
export { Separator }
```

```

=====
// === END FILE: src/components/ui/separator.tsx ===
=====

```

```

=====
// === START FILE: src/components/ui/slider.tsx ===
// File 36 of 194 | Lines: 63 | Size: 2001 bytes
=====

```

```
"use client"
```

```

import * as React from "react"
import * as SliderPrimitive from "@radix-ui/react-slider"

```

```
import { cn } from "@lib/utils"
```

```

function Slider({
  className,
  defaultValue,
  value,
  min = 0,
  max = 100,
  ...props
}: React.ComponentProps<typeof SliderPrimitive.Root>) {
  const _values = React.useMemo(
    () =>
      Array.isArray(value)
        ? value
        : Array.isArray(defaultValue)
          ? defaultValue
          : [min, max],
    [value, defaultValue, min, max]
  )

  return (
    <SliderPrimitive.Root
      data-slot="slider"
      defaultValue={defaultValue}
      value={value}
      min={min}
      max={max}
      className={cn(
        "relative flex w-full touch-none items-center select-none data-[disabled]:opacity-50 data-[orientation=vertical]:h-full data-[orientation=vertical]:min-h-44 data-[orientation=vertical]:w-auto data-[orientation=vertical]:flex-col",
        className
      )}
    >{...props}</>
    <SliderPrimitive.Track
      data-slot="slider-track"
      className={cn(
        "bg-muted relative grow overflow-hidden rounded-full data-[orientation=horizontal]:h-1.5 data-[orientation=horizontal]:w-full data-[orientation=vertical]:h-full data-[orientation=vertical]:w-1.5"
      )}
    >

```

```

    >
    <SliderPrimitive.Range
      data-slot="slider-range"
      className={cn(
        "bg-primary absolute data-[orientation=horizontal]:h-full data-[orientation=vertical]:w-full"
      )}
    />
  </SliderPrimitive.Track>
  {Array.from({ length: _values.length }, (_, index) => (
    <SliderPrimitive.Thumb
      data-slot="slider-thumb"
      key={index}
      className="border-primary bg-background ring-ring/50 block size-4 shrink-0 rounded-full border shado
        w-sm transition-[color,box-shadow] hover:ring-4 focus-visible:ring-4 focus-visible:outline-hidde
        n disabled:pointer-events-none disabled:opacity-50"
    />
  ))}
</SliderPrimitive.Root>
)
}

```

```
export { Slider }
```

```
=====
// === END FILE: src/components/ui/slider.tsx ===
=====
```

```
=====
// === START FILE: src/components/ui/toast.tsx ===
// File 37 of 194 | Lines: 129 | Size: 4829 bytes
=====
```

```
"use client"
```

```
import * as React from "react"
import * as ToastPrimitives from "@radix-ui/react-toast"
import { cva, type VariantProps } from "class-variance-authority"
import { X } from "lucide-react"
```

```
import { cn } from "@lib/utils"
```

```
const ToastProvider = ToastPrimitives.Provider
```

```
const ToastViewport = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Viewport>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Viewport>
>(({ className, ...props }, ref) => (
  <ToastPrimitives.Viewport
    ref={ref}
    className={cn(
      "fixed top-0 z-[100] flex max-h-screen w-full flex-col-reverse p-4 sm:bottom-0 sm:right-0 sm:top-auto sm
        :flex-col md:max-w-[420px]",
      className
    )}
    {...props}
  />
))
ToastViewport.displayName = ToastPrimitives.Viewport.displayName
```

```
const toastVariants = cva(
  "group pointer-events-auto relative flex w-full items-center justify-between space-x-2 overflow-hidden round
    ed-md border p-4 pr-6 shadow-lg transition-all data-[swipe=cancel]:translate-x-0 data-[swipe=end]:transl
    ate-x-[var(--radix-toast-swipe-end-x)] data-[swipe=move]:translate-x-[var(--radix-toast-swipe-move-x)] d
    ata-[swipe=move]:transition-none data-[state=open]:animate-in data-[state=closed]:animate-out data-[swip
    e=end]:animate-out data-[state=closed]:fade-out-80 data-[state=closed]:slide-out-to-right-full data-[sta
    te=open]:slide-in-from-top-full data-[state=open]:sm:slide-in-from-bottom-full",
  {
    variants: {
      variant: {
        default: "border bg-background text-foreground",
        destructive:
          "destructive group border-destructive bg-destructive text-destructive-foreground",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  },
)
```

```
}
)

const Toast = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Root>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Root> &
  VariantProps<typeof toastVariants>
>(({ className, variant, ...props }, ref) => {
  return (
    <ToastPrimitives.Root
      ref={ref}
      className={cn(toastVariants({ variant }), className)}
      {...props}
    />
  )
})
Toast.displayName = ToastPrimitives.Root.displayName

const ToastAction = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Action>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Action>
>(({ className, ...props }, ref) => (
  <ToastPrimitives.Action
    ref={ref}
    className={cn(
      "inline-flex h-8 shrink-0 items-center justify-center rounded-md border bg-transparent px-3 text-sm font
      -medium transition-colors hover:bg-secondary focus:outline-none focus:ring-1 focus:ring-ring disable
      d:pointer-events-none disabled:opacity-50 group-[.destructive]:border-muted/40 group-[.destructive]:
      hover:border-destructive/30 group-[.destructive]:hover:bg-destructive group-[.destructive]:hover:tex
      t-destructive-foreground group-[.destructive]:focus:ring-destructive",
      className
    )}
    {...props}
  />
))
ToastAction.displayName = ToastPrimitives.Action.displayName

const ToastClose = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Close>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Close>
>(({ className, ...props }, ref) => (
  <ToastPrimitives.Close
    ref={ref}
    className={cn(
      "absolute right-1 top-1 rounded-md p-1 text-foreground/50 opacity-0 transition-opacity hover:text-foregr
      ound focus:opacity-100 focus:outline-none focus:ring-1 group-hover:opacity-100 group-[.destructive]:
      text-red-300 group-[.destructive]:hover:text-red-50 group-[.destructive]:focus:ring-red-400 group-[.
      destructive]:focus:ring-offset-red-600",
      className
    )}
    toast-close=""
    {...props}
  >
    <X className="h-4 w-4" />
  </ToastPrimitives.Close>
))
ToastClose.displayName = ToastPrimitives.Close.displayName

const ToastTitle = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Title>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Title>
>(({ className, ...props }, ref) => (
  <ToastPrimitives.Title
    ref={ref}
    className={cn("text-sm font-semibold [&+div]:text-xs", className)}
    {...props}
  />
))
ToastTitle.displayName = ToastPrimitives.Title.displayName

const ToastDescription = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Description>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Description>
>(({ className, ...props }, ref) => (
  <ToastPrimitives.Description
    ref={ref}
    className={cn("text-sm opacity-90", className)}
    {...props}
  />
))
```

```
    />
  ))
  ToastDescription.displayName = ToastPrimitives.Description.displayName

  type ToastProps = React.ComponentPropsWithoutRef<typeof Toast>

  type ToastActionElement = React.ReactElement<typeof ToastAction>

  export {
    type ToastProps,
    type ToastActionElement,
    ToastProvider,
    ToastViewport,
    Toast,
    ToastTitle,
    ToastDescription,
    ToastClose,
    ToastAction,
  }

=====
// === END FILE: src/components/ui/toast.tsx ===
=====

=====
// === START FILE: src/components/ui/toaster.tsx ===
// File 38 of 194 | Lines: 35 | Size: 785 bytes
=====

"use client"

import { useToast } from "@hooks/use-toast"
import {
  Toast,
  ToastClose,
  ToastDescription,
  ToastProvider,
  ToastTitle,
  ToastViewport,
} from "@components/ui/toast"

export function Toaster() {
  const { toasts } = useToast()

  return (
    <ToastProvider>
      {toasts.map(function ({ id, title, description, action, ...props }) {
        return (
          <Toast key={id} {...props}>
            <div className="grid gap-1">
              {title && <ToastTitle>{title}</ToastTitle>}
              {description && (
                <ToastDescription>{description}</ToastDescription>
              )}
            </div>
            {action}
            <ToastClose />
          </Toast>
        )
      })}
    <ToastViewport />
  </ToastProvider>
  )
}

=====
// === END FILE: src/components/ui/toaster.tsx ===
=====

=====
// === START FILE: src/components/ui/tooltip.tsx ===
// File 39 of 194 | Lines: 61 | Size: 1891 bytes
=====

"use client"
```

```

import * as React from "react"
import * as TooltipPrimitive from "@radix-ui/react-tooltip"

import { cn } from "@lib/utils"

function TooltipProvider({
  delayDuration = 0,
  ...props
}: React.ComponentProps<typeof TooltipPrimitive.Provider>) {
  return (
    <TooltipPrimitive.Provider
      data-slot="tooltip-provider"
      delayDuration={delayDuration}
      {...props}
    />
  )
}

function Tooltip({
  ...props
}: React.ComponentProps<typeof TooltipPrimitive.Root>) {
  return (
    <TooltipProvider>
      <TooltipPrimitive.Root data-slot="tooltip" {...props} />
    </TooltipProvider>
  )
}

function TooltipTrigger({
  ...props
}: React.ComponentProps<typeof TooltipPrimitive.Trigger>) {
  return <TooltipPrimitive.Trigger data-slot="tooltip-trigger" {...props} />
}

function TooltipContent({
  className,
  sideOffset = 0,
  children,
  ...props
}: React.ComponentProps<typeof TooltipPrimitive.Content>) {
  return (
    <TooltipPrimitive.Portal>
      <TooltipPrimitive.Content
        data-slot="tooltip-content"
        sideOffset={sideOffset}
        className={cn(
          "bg-primary text-primary-foreground animate-in fade-in-0 zoom-in-95 data-[state=closed]:animate-out
          data-[state=closed]:fade-out-0 data-[state=closed]:zoom-out-95 data-[side=bottom]:slide-in-from-
          top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=t
          op]:slide-in-from-bottom-2 z-50 w-fit origin(--radix-tooltip-content-transform-origin) rounded-
          md px-3 py-1.5 text-xs text-balance",
          className
        )}
        {...props}
      >
        {children}
      <TooltipPrimitive.Content>
        <TooltipPrimitive.Arrow className="bg-primary fill-primary z-50 size-2.5 translate-y-[calc(-50%_-_2px)] rotate-45 rounded-[2px]" />
      </TooltipPrimitive.Content>
    </TooltipPrimitive.Portal>
  )
}

export { Tooltip, TooltipTrigger, TooltipContent, TooltipProvider }

=====
// === END FILE: src/components/ui/tooltip.tsx ===
=====

=====
// === START FILE: src/components/whiteboard/AccountSection.tsx ===
// File 40 of 194 | Lines: 243 | Size: 8097 bytes
=====

"use client";

/**

```

```

* Account section of the Sync panel: self-service sign-up, sign-in, password
* reset, and two-way sync of the workspace for the signed-in account.
*/

import React, { useEffect, useState } from "react";
import { useWhiteboard } from "@store/whiteboard";
import {
  getCurrentUser,
  isConfigured,
  onAuthChange,
  pullWorkspace,
  pushWorkspace,
  sendPasswordReset,
  signIn,
  signOut,
  signUp,
  type AccountUser,
} from "@lib/whiteboard/account";
import { Loader2, Logout, RefreshCw, UserPlus } from "lucide-react";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { cn } from "@lib/utils";

type Mode = "signin" | "signup" | "reset";

interface Props {
  deviceLabel: string;
  onStatus: (s: { kind: "busy" | "ok" | "error"; message: string }) => void;
  busy: boolean;
}

export function AccountSection({ deviceLabel, onStatus, busy }: Props) {
  const getWorkspaceSnapshot = useWhiteboard((s) => s.getWorkspaceSnapshot);
  const mergeWorkspace = useWhiteboard((s) => s.mergeWorkspace);

  const [user, setUser] = useState<AccountUser | null>(null);
  const [mode, setMode] = useState<Mode>("signin");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [confirm, setConfirm] = useState("");

  useEffect(() => {
    let alive = true;
    getCurrentUser()
      .then((u) => alive && setUser(u))
      .catch(() => {});
    const off = onAuthChange((u) => alive && setUser(u));
    return () => {
      alive = false;
      off();
    };
  }, []);

  if (!isConfigured()) {
    return (
      <div className="rounded-lg border border-amber-500/40 bg-amber-500/10 p-3 text-xs leading-relaxed">
        <p className="font-medium text-amber-700 dark:text-amber-400">
          Accounts are not set up in this build.
        </p>
        <p className="mt-1 text-muted-foreground">
          Add <code className="font-semibold">NEXT_PUBLIC_SUPABASE_URL</code> and{" "}
          <code className="font-semibold">NEXT_PUBLIC_SUPABASE_ANON_KEY</code>, then rebuild.
          See <code>ACCOUNTS_SETUP.md</code> – free, about five minutes.
        </p>
      </div>
    );
  }

  const submit = async () => {
    const mail = email.trim();
    if (!mail) return onStatus({ kind: "error", message: "Enter your email." });

    try {
      if (mode === "reset") {
        onStatus({ kind: "busy", message: "Sending reset email..." });
        await sendPasswordReset(mail);
        onStatus({ kind: "ok", message: `Reset link sent to ${mail}.` });
        setMode("signin");
      }
    }
  }

```

```

    return;
  }

  if (password.length < 6) {
    return onStatus({ kind: "error", message: "Password must be at least 6 characters." });
  }

  if (mode === "signup") {
    // Catching the mismatch here, rather than after the account exists,
    // saves the user from an account with a password they mistyped.
    if (password !== confirm) {
      return onStatus({ kind: "error", message: "The two passwords do not match." });
    }
    onStatus({ kind: "busy", message: "Creating your account..." });
    const res = await signUp(mail, password);
    if (res.needsEmailConfirmation) {
      onStatus({
        kind: "ok",
        message: `Account created. Click the link sent to ${mail}, then sign in.`
      });
      setMode("signin");
    } else {
      onStatus({ kind: "ok", message: "Account created and signed in." });
    }
    setPassword("");
    setConfirm("");
    return;
  }

  onStatus({ kind: "busy", message: "Signing in..." });
  await signIn(mail, password);
  setPassword("");
  onStatus({ kind: "ok", message: "Signed in." });
} catch (e) {
  onStatus({ kind: "error", message: (e as Error).message });
}
};

const doSync = async () => {
  onStatus({ kind: "busy", message: "Syncing..." });
  try {
    const remote = await pullWorkspace();
    let summary: string;
    if (!remote) {
      summary = "First sync – uploaded this device's boards.";
    } else {
      const r = mergeWorkspace(remote.snapshot);
      summary =
        `Merged: ${r.added} new, ${r.updatedFromRemote} updated from your account, ` +
        `${r.keptLocal} kept from this device.`;
    }
    await pushWorkspace(getWorkspaceSnapshot(), deviceLabel);
    onStatus({ kind: "ok", message: summary });
  } catch (e) {
    onStatus({ kind: "error", message: (e as Error).message });
  }
};

const doSignOut = async () => {
  await signOut();
  onStatus({ kind: "ok", message: "Signed out. Your boards stay on this device." });
};

if (user) {
  return (
    <>
    <div className="rounded-lg border border-border p-3 text-xs">
      <div className="font-medium">{user.email}</div>
      <div className="text-muted-foreground">
        {user.confirmed ? "Signed in" : "Email not confirmed yet"}
      </div>
    </div>
    <div className="flex gap-2">
      <Button size="sm" className="flex-1" onClick={doSync} disabled={busy}>
        {busy ? (
          <Loader2 size={14} className="mr-1.5 animate-spin" />
        ) : (
          <RefreshCw size={14} className="mr-1.5" />
        )}
      </Button>
    </div>
  );
}

```

```

    })
    Sync now
  </Button>
  <Button size="sm" variant="outline" onClick={doSignOut} disabled={busy}>
    <Logout size={14} className="mr-1.5" />
    Sign out
  </Button>
</div>
<p className="text-[11px] leading-snug text-muted-foreground">
  Sign in with the same email on your tablet to get the same boards.
  Boards merge one by one – newest edit wins.
</p>
</>
  );
}

return (
  <>
    <div className="flex gap-1 rounded-lg bg-muted p-0.5 text-xs">
      {[["signin", "signup"] as const].map((m) => (
        <button
          key={m}
          onClick={() => setMode(m)}
          className={cn(
            "flex-1 rounded-md px-2 py-1.5 font-medium transition-colors",
            mode === m ? "bg-background shadow-sm" : "text-muted-foreground hover:text-foreground"
          )}
        >
          {m === "signin" ? "Sign in" : "Create account"}
        </button>
      )]}
    </div>

    <div className="space-y-2">
      <Input
        type="email"
        autoComplete="email"
        placeholder="you@example.com"
        value={email}
        onChange={(e) => setEmail(e.target.value)}
        className="h-9 text-xs"
      />
      {mode !== "reset" && (
        <Input
          type="password"
          autoComplete={mode === "signup" ? "new-password" : "current-password"}
          placeholder="Password"
          value={password}
          onChange={(e) => setPassword(e.target.value)}
          onKeyDown={(e) => e.key === "Enter" && void submit()}
          className="h-9 text-xs"
        />
      )}
      {mode === "signup" && (
        <Input
          type="password"
          autoComplete="new-password"
          placeholder="Confirm password"
          value={confirm}
          onChange={(e) => setConfirm(e.target.value)}
          onKeyDown={(e) => e.key === "Enter" && void submit()}
          className="h-9 text-xs"
        />
      )}
      <Button size="sm" className="w-full" onClick={submit} disabled={busy}>
        {busy ? (
          <Loader2 size={14} className="mr-1.5 animate-spin" />
        ) : (
          <UserPlus size={14} className="mr-1.5" />
        )}
        {mode === "signup" ? "Create account" : mode === "reset" ? "Send reset link" : "Sign in"}
      </Button>

      <button
        onClick={() => setMode(mode === "reset" ? "signin" : "reset")}
        className="w-full text-center text-[11px] text-muted-foreground underline-offset-2 hover:underline"
      >

```

```

        {mode === "reset" ? "Back to sign in" : "Forgot your password?"}
      </button>
    </div>
  </>
);
}

=====
// === END FILE: src/components/whiteboard/AccountSection.tsx ===
=====

=====
// === START FILE: src/components/whiteboard/BacklinksPanel.tsx ===
// File 41 of 194 | Lines: 207 | Size: 8497 bytes
=====

"use client";

/**
 * Backlinks & Relationship Panel (v3 – Obsidian Canvas Grade)
 * =====
 * Aggregates both [[WikiLinks]] and typed directed arrow connections.
 */

import React, { useMemo } from "react";
import { useWhiteboard, useActiveBoard } from "@store/whiteboard";
import { getBacklinks, getBoardLinks, getBoardTags } from "@lib/whiteboard/links";
import { Link2, Tag, ArrowRight, Hash, Plus, Sparkles, X } from "lucide-react";
import { Button } from "@components/ui/button";
import { ScrollArea } from "@components/ui/scroll-area";
import { cn } from "@lib/utils";

interface BacklinksPanelProps {
  /** Close the panel. Without this the panel could cover the only button that
   * toggles it, leaving no way out except the Ctrl+B shortcut. */
  onClose?: () => void;
}

export function BacklinksPanel({ onClose }: BacklinksPanelProps) {
  const boards = useWhiteboard((s) => s.boards);
  const customLinkTypes = useWhiteboard((s) => s.customLinkTypes);
  const activeBoard = useActiveBoard();
  const setActiveBoard = useWhiteboard((s) => s.setActiveBoard);
  const findOrCreateBoardByName = useWhiteboard((s) => s.findOrCreateBoardByName);

  const backlinks = useMemo(() => {
    if (!activeBoard) return [];
    return getBacklinks(boards, activeBoard.name, activeBoard.id, customLinkTypes);
  }, [boards, activeBoard, customLinkTypes]);

  const outgoingLinks = useMemo(() => {
    if (!activeBoard) return [];
    return getBoardLinks(activeBoard);
  }, [activeBoard]);

  const tags = useMemo(() => {
    if (!activeBoard) return [];
    return getBoardTags(activeBoard);
  }, [activeBoard]);

  if (!activeBoard) return null;

  const navigateToBoard = (name: string) => {
    const id = findOrCreateBoardByName(name);
    setActiveBoard(id);
  };

  return (
    <div className="w-72 h-full bg-sidebar border-r border-sidebar-border flex flex-col select-none">
      <div className="px-4 py-3 border-b border-sidebar-border">
        <div className="flex items-center justify-between">
          <div className="flex items-center gap-2">
            <Link2 size={16} className="text-primary" />
            <h3 className="text-xs font-bold uppercase tracking-wider text-foreground">
              Links & Connections
            </h3>
          </div>
        </div>
      </div>
    </div>
  );
}

```

```

</div>
<div className="flex items-center gap-1">
  <span className="text-[11px] px-2 py-0.5 rounded-full bg-primary/10 text-primary font-semibold">
    {backlinks.length + outgoingLinks.length}
  </span>
  {onClose && (
    <Button
      variant="ghost"
      size="icon"
      className="h-6 w-6 text-muted-foreground hover:text-foreground"
      onClick={onClose}
      title="Close links panel (Ctrl+B)"
      aria-label="Close links panel"
    >
      <X size={14} />
    </Button>
  )}
</div>
</div>
/* Say what this panel is for – it opens from a button labelled only
   "Links", which does not explain itself. */
<p className="mt-1.5 text-[11px] leading-snug text-muted-foreground">
  Every board that <strong>this</strong> board points to, and every board
  pointing back at it.
</p>
</div>

<ScrollArea className="flex-1">
  <div className="p-3.5 space-y-4">
    /* Outgoing links */
    <div>
      <div className="text-xs font-semibold text-muted-foreground mb-2 flex items-center justify-between">
        <span className="flex items-center gap-1.5">
          <ArrowRight size={13} className="text-blue-500" />
          Outgoing References
        </span>
        <span className="tabular-nums font-bold text-[11px]">{outgoingLinks.length}</span>
      </div>

      {outgoingLinks.length === 0 ? (
        <div className="text-[11px] text-muted-foreground italic px-1 py-1">
          No outgoing links. Type <code className="text-primary font-semibold">[[Board Name]]</code> to
          link ideas.
        </div>
      ) : (
        <div className="space-y-1">
          {outgoingLinks.map((target) => {
            const exists = boards.some(
              (b) => b.name.toLowerCase() === target.toLowerCase()
            );
            return (
              <button
                key={target}
                onClick={() => navigateToBoard(target)}
                className="w-full text-left text-xs px-2.5 py-1.5 rounded-lg hover:bg-sidebar-accent text-
                t-foreground flex items-center justify-between transition-colors"
              >
                <span className="font-medium truncate text-blue-600 dark:text-blue-400">
                  {target}
                </span>
                {!exists && (
                  <span className="text-[10px] text-amber-500 font-semibold">(new)</span>
                )}
              </button>
            );
          })}
        </div>
      )}
    </div>

    /* Backlinks */
    <div>
      <div className="text-xs font-semibold text-muted-foreground mb-2 flex items-center justify-between">
        <span className="flex items-center gap-1.5">
          <Link2 size={13} className="text-primary" />
          Incoming Backlinks
        </span>
      </div>
    </div>
  </div>
</ScrollArea>

```

```

    </span>
    <span className="tabular-nums font-bold text-[11px]">{backlinks.length}</span>
  </div>

  {backlinks.length === 0 ? (
    <div className="text-[11px] text-muted-foreground italic px-1 py-1">
      No boards link to this one yet.
    </div>
  ) : (
    <div className="space-y-1.5">
      {backlinks.map((bl, i) => (
        <button
          key={` ${bl.sourceBoardId}-${bl.elementId}-${i}`}
          onClick={() => setActiveBoard(bl.sourceBoardId)}
          className="w-full text-left p-2 rounded-lg border border-sidebar-border hover:border-primary/50 hover:bg-sidebar-accent transition-all group"
        >
          <div className="flex items-center justify-between mb-1">
            <span className="font-semibold text-xs text-foreground group-hover:text-primary transition-on-colors truncate">
              {bl.sourceBoardName}
            </span>
            <span
              {bl.relationship} && (
                <span
                  className="text-[10px] px-1.5 py-0.5 rounded font-medium text-white"
                  style={{ backgroundColor: bl.color || "#6366f1" }}
                >
                  {bl.relationship}
                </span>
              )
            </span>
          </div>
            {bl.snippet} && (
              <p className="text-[11px] text-muted-foreground line-clamp-2 italic">
                "{bl.snippet}"
              </p>
            )
          </button>
        </div>
      )
    </div>
  )
}

/* Tags */
<div>
  <div className="text-xs font-semibold text-muted-foreground mb-2 flex items-center justify-between">
    ">
    <span className="flex items-center gap-1.5">
      <Hash size={13} className="text-emerald-500" />
      Tags
    </span>
    <span className="tabular-nums font-bold text-[11px]">{tags.length}</span>
  </div>

  {tags.length === 0 ? (
    <div className="text-[11px] text-muted-foreground italic px-1 py-1">
      No tags yet. Use <code className="text-emerald-500 font-semibold">#tag</code> in text.
    </div>
  ) : (
    <div className="flex flex-wrap gap-1">
      {tags.map((tag) => (
        <span
          key={tag}
          className="text-[11px] px-2 py-0.5 rounded-md bg-emerald-500/10 text-emerald-600 dark:text-emerald-400 font-medium"
        >
          #{tag}
        </span>
      )
    </div>
  )
}
</div>
</div>
</ScrollArea>
</div>
);
}

```

```
=====
```

```
// === END FILE: src/components/whiteboard/BacklinksPanel.tsx ===
=====

// === START FILE: src/components/whiteboard/CanvasContextMenu.tsx ===
// File 42 of 194 | Lines: 102 | Size: 3377 bytes
=====

"use client";

/**
 * Canvas Context Menu
 * =====
 * Right-click menu for canvas elements.
 * Provides quick access to: bring forward/back, duplicate, delete, edit style.
 *
 * `disabled` exists because Radix opens this menu on a 700ms LONG PRESS for
 * any pointer that is not a mouse – which on a tablet means a hand resting on
 * the glass opens it uninvited, and a stylus paused mid-thought does too
 * (Radix's own test is `pointerType !== "mouse"`, so it lumps pens in with
 * fingers). The trigger sits directly on the canvas <svg>, so the canvas'
 * palm rejection cannot reach it; the canvas disarms it through this prop
 * instead. Radix clears any pending timer the moment `disabled` flips true.
 */

import React from "react";
import {
  ContextMenu,
  ContextMenuContent,
  ContextMenuItem,
  ContextMenuSeparator,
  ContextMenuTrigger,
  ContextMenuSub,
  ContextMenuSubContent,
  ContextMenuSubTrigger,
} from "@components/ui/context-menu";
import { useWhiteboard } from "@store/whiteboard";
import {
  BringToFront,
  SendToBack,
  ArrowUp,
  ArrowDown,
  Copy,
  Trash2,
  Pencil,
} from "lucide-react";

export function CanvasContextMenu({
  children,
  elementId,
  disabled = false,
}: {
  children: React.ReactNode;
  elementId: string | null;
  /** Suppress the menu entirely – see the note above on long-press opening. */
  disabled?: boolean;
}) {
  const bringToFront = useWhiteboard((s) => s.bringToFront);
  const sendToBack = useWhiteboard((s) => s.sendToBack);
  const bringForward = useWhiteboard((s) => s.bringForward);
  const sendBackward = useWhiteboard((s) => s.sendBackward);
  const deleteElements = useWhiteboard((s) => s.deleteElements);
  const duplicateBoard = useWhiteboard((s) => s.duplicateBoard);
  const selectedIds = useWhiteboard((s) => s.selectedIds);

  // Use either the right-clicked element or the current selection
  const targetIds = elementId ? [elementId] : selectedIds;

  return (
    <ContextMenu>
      <ContextMenuTrigger asChild disabled={disabled}>
        {children}
      </ContextMenuTrigger>
      <ContextMenuContent className="w-48">
        <ContextMenuItem
          disabled={targetIds.length === 0}
          onClick={() => bringForward(targetIds)}
        />
      </ContextMenuContent>
    </ContextMenu>
  );
}
```

```

    >
    <ArrowUp size={14} className="mr-2" /> Bring Forward
  </ContextMenuItem>
  <ContextMenuItem
    disabled={targetIds.length === 0}
    onClick={() => sendBackward(targetIds)}
  >
    <ArrowDown size={14} className="mr-2" /> Send Backward
  </ContextMenuItem>
  <ContextMenuItem
    disabled={targetIds.length === 0}
    onClick={() => bringToFront(targetIds)}
  >
    <BringToFront size={14} className="mr-2" /> Bring to Front
  </ContextMenuItem>
  <ContextMenuItem
    disabled={targetIds.length === 0}
    onClick={() => sendToBack(targetIds)}
  >
    <SendToBack size={14} className="mr-2" /> Send to Back
  </ContextMenuItem>
  <ContextMenuSeparator />
  <ContextMenuItem
    disabled={targetIds.length === 0}
    onClick={() => deleteElements(targetIds, true)}
    className="text-destructive focus:text-destructive"
  >
    <Trash2 size={14} className="mr-2" /> Delete
  </ContextMenuItem>
</ContextMenuContent>
</ContextMenu>
);
}

=====
// === END FILE: src/components/whiteboard/CanvasContextMenu.tsx ===
=====

=====
// === START FILE: src/components/whiteboard/CaptureHud.tsx ===
// File 43 of 194 | Lines: 112 | Size: 5066 bytes
=====

"use client";

/**
 * The chip that says "screenshot mode is on".
 *
 * A mode that changes what the select tool does must be impossible to miss,
 * or Select appears broken: you drag a box and nothing gets selected. So
 * while screenshot mode is on and Select or Lasso is the tool, this chip sits
 * at the top of the canvas saying what the next drag does, with the box /
 * shape switch and an Off button right there. (The select button wears a
 * camera badge too, and the cursor is a crosshair.)
 *
 * The mode stays on across captures - taking three pictures of three
 * questions should not mean re-arming it three times - until you switch it
 * off here, in the Select menu, or with Esc.
 */

import React, { useEffect } from "react";
import { Camera, Lasso, SquareDashedMousePointer, X } from "lucide-react";
import { useWhiteboard } from "@/store/whiteboard";
import { screenshotShapeFor, useCapture } from "@/store/capture";
import { cn } from "@/lib/utils";

export function CaptureHud() {
  const screenshotMode = useCapture((s) => s.screenshotMode);
  const panelOpen = useCapture((s) => s.request !== null);
  const tool = useWhiteboard((s) => s.tool);
  const setTool = useWhiteboard((s) => s.setTool);
  // With the tool bar along the top, the hint goes to the bottom: at the top
  // it covered the tools and the page controls.
  const dockTop = useWhiteboard((s) => s.toolbarPosition === "top");
  const shape = screenshotShapeFor(screenshotMode, tool);

  // Esc leaves the mode - but not while the panel is up; there Esc closes the

```

```

// panel and you are still in screenshot mode for the next one.
useEffect(() => {
  if (!screenshotMode) return;
  const onKey = (e: KeyboardEvent) => {
    if (e.key !== "Escape" || useCapture.getState().request) return;
    // An Esc meant for something else - finishing a text box, closing
    // search or another window - is not "leave screenshot mode".
    const t = e.target as HTMLInputElement | null;
    if (t && (t.isContentEditable || /^(INPUT|TEXTAREA|SELECT)$/.test(t.tagName))) return;
    if (document.querySelector('[data-wb-modal], [role="dialog"], [data-wb-searchbar]')) return;
    if (useWhiteboard.getState().editingId) return;
    useCapture.getState().setScreenshotMode(false);
  };
  // Capture phase: the dialog closes itself on Esc from a document listener,
  // and by the time a bubbling listener ran the panel would already look
  // closed - the same Esc would then also switch the mode off.
  window.addEventListener("keydown", onKey, true);
  return () => window.removeEventListener("keydown", onKey, true);
}, [screenshotMode]);

if (!shape || panelOpen) return null;

const shapeBtn = (s: "rectangle" | "lasso", Icon: typeof Camera, label: string) => (
  <button
    type="button"
    onClick={() => setTool(s === "rectangle" ? "select" : "lasso")}
    aria-pressed={shape === s}
    title={label}
    className={cn(
      "flex h-8 items-center gap-1.5 rounded-full px-2.5 text-[12px] font-medium transition",
      shape === s ? "bg-white/15 text-white" : "text-white/60 hover:text-white"
    )}
  >
    <Icon size={15} />
    <span className="hidden @min-[34rem]:inline">{label}</span>
  </button>
);

return (
  <div
    className={cn(
      "@container pointer-events-none absolute inset-x-0 z-40 flex justify-center px-3",
      dockTop ? "bottom-6" : "top-[72px]"
    )}
  >
    <div
      role="status"
      className={cn(
        "pointer-events-auto flex max-w-full items-center gap-1 rounded-full border border-white/10 bg-zinc-900/95 py-1 pl-1 pr-1 text-white shadow-2xl shadow-black/30 backdrop-blur-md animate-in fade-in"
      )}
      dockTop ? "slide-in-from-bottom-2" : "slide-in-from-top-2"
    >
      onPointerDown={(e) => e.stopPropagation()}
      >
        <span className="flex h-8 w-8 shrink-0 items-center justify-center rounded-full bg-gradient-to-br from-indigo-500 to-violet-500">
          <Camera size={15} />
        </span>
        <span className="min-w-0 truncate px-1.5 text-[12.5px] font-medium">
          <span className="hidden @min-[26rem]:inline">Screenshot mode – </span>
          {shape === "lasso" ? "draw around what to capture" : "drag over what to capture"}
        </span>
        <span className="mx-0.5 h-5 w-px shrink-0 bg-white/15" />
        {shapeBtn("rectangle", SquareDashedMousePointer, "Box")}
        {shapeBtn("lasso", Lasso, "Freeform")}
        <span className="mx-0.5 h-5 w-px shrink-0 bg-white/15" />
        <button
          type="button"
          onClick={() => useCapture.getState().setScreenshotMode(false)}
          aria-label="Turn screenshot mode off"
          title="Back to selecting (Esc)"
          className="flex h-8 shrink-0 items-center gap-1 rounded-full px-2.5 text-[12px] font-medium text-white/75 transition hover:bg-white/10 hover:text-white"
        >
          <X size={15} />
          <span className="hidden @min-[30rem]:inline">Off</span>
        </button>
      </div>
    </div>
  );

```

```

    </div>
  </div>
);
}

=====
// === END FILE: src/components/whiteboard/CaptureHud.tsx ===
=====

=====
// === START FILE: src/components/whiteboard/CaptureMarquee.tsx ===
// File 44 of 194 | Lines: 122 | Size: 4244 bytes
=====

"use client";

/**
 * What the canvas shows while you draw a capture area. Screen space: it is
 * drawn outside the zoomed world group so its lines stay one crisp width.
 *
 * The look is a VIEWFINDER, the capture feature's own motif (the panel's
 * preview wears the same corner brackets): everything outside the area dims,
 * the area stays at full brightness with a thin indigo edge, and a small
 * readout gives its size in page units - the size it will have if you drop
 * it into a notebook.
 */

import React from "react";
import { worldToScreen, type Viewport } from "@lib/whiteboard/geometry";
import { rectFromCorners, type CaptureShape } from "@lib/whiteboard/capture";

const ACCENT = "#6366f1";

/** A corner bracket's two arms, as one path, for each corner of a box. */
function bracketsPath(x: number, y: number, w: number, h: number, arm: number): string {
  const a = Math.max(4, Math.min(arm, w / 3, h / 3));
  return [
    `M ${x} ${y + a} V ${y} H ${x + a}`,
    `M ${x + w - a} ${y} H ${x + w} V ${y + a}`,
    `M ${x + w} ${y + h - a} V ${y + h} H ${x + w - a}`,
    `M ${x + a} ${y + h} H ${x} V ${y + h - a}`,
  ].join(" ");
}

export function CaptureMarquee({
  shape,
  points,
  camera,
  viewport,
}): {
  shape: CaptureShape;
  /** World points: a box's two corners, or a loop's trail. */
  points: Array<{ x: number; y: number }>;
  camera: { x: number; y: number; zoom: number };
  viewport: Viewport;
} {
  if (points.length < 2) return null;
  const sp = points.map((p) => worldToScreen(p.x, p.y, camera, viewport));
  const W = viewport.width;
  const H = viewport.height;

  let area: string;
  let box: { x: number; y: number; width: number; height: number };
  if (shape === "rectangle") {
    box = rectFromCorners(sp[0], sp[sp.length - 1]);
    area = `M ${box.x} ${box.y} h ${box.width} v ${box.height} h ${-box.width} Z`;
  } else {
    area = `M ${sp.map((p) => `${p.x.toFixed(1)} ${p.y.toFixed(1)}`).join(" L ")} Z`;
    const xs = sp.map((p) => p.x);
    const ys = sp.map((p) => p.y);
    box = {
      x: Math.min(...xs),
      y: Math.min(...ys),
      width: Math.max(...xs) - Math.min(...xs),
      height: Math.max(...ys) - Math.min(...ys),
    };
  }
}

```

```

const z = Math.max(camera.zoom, 0.01);
const label = `${Math.round(box.width / z)} × ${Math.round(box.height / z)}`;
const labelW = label.length * 6.6 + 18;
const labelX = Math.min(Math.max(box.x + box.width - labelW, 6), W - labelW - 6);
const below = box.y + box.height + 8;
const labelY = below + 22 < H - 6 ? below : Math.max(6, box.y - 30);

return (
  <g pointerEvents="none">
    /* Dim everything outside the area: evenodd cuts the area out. */
    <path d={`M 0 0 H ${W} V ${H} H 0 Z ${area}`} fillRule="evenodd" fill="rgba(9, 9, 11, 0.36)" />
    /* A white under-stroke keeps the edge visible on dark ink and dark paper. */
    <path d={area} fill="none" stroke="rgba(255, 255, 255, 0.9)" strokeOpacity={0.9} strokeWidth={3} strokeLinejoin="round" />
    <path
      d={area}
      fill="none"
      stroke={ACCENT}
      strokeWidth={1.5}
      strokeLinejoin="round"
      strokeDasharray={shape === "lasso" ? "6 4" : undefined}
    />
    {shape === "rectangle" && box.width > 8 && box.height > 8 && (
      <path
        d={bracketsPath(box.x - 2, box.y - 2, box.width + 4, box.height + 4, 18)}
        fill="none"
        stroke="rgba(255, 255, 255, 0.9)"
        strokeWidth={5.5}
        strokeLinecap="round"
        strokeLinejoin="round"
      />
      <path
        d={bracketsPath(box.x - 2, box.y - 2, box.width + 4, box.height + 4, 18)}
        fill="none"
        stroke={ACCENT}
        strokeWidth={3}
        strokeLinecap="round"
        strokeLinejoin="round"
      />
    )}
    <g transform={`translate(${labelX}, ${labelY})`}>
      <rect width={labelW} height={22} rx={11} fill="rgba(24, 24, 27, 0.88)" />
      <text
        x={labelW / 2}
        y={11}
        textAnchor="middle"
        fontSize={11.5}
        fontWeight={600}
        fill="white"
        style={{ fontVariantNumeric: "tabular-nums" }}
      >
        {label}
      </text>
    </g>
  </g>
);
}

```

```

// === END FILE: src/components/whiteboard/CaptureMarquee.tsx ===

```

```

// === START FILE: src/components/whiteboard/CapturePanel.tsx ===
// File 45 of 194 | Lines: 834 | Size: 33802 bytes

```

```
"use client";
```

```

/**
 * The capture panel: what happens to the picture you just drew around.
 * =====
 * One screen, three questions, in the order people ask them:
 *

```

```

* 1. IS THIS RIGHT? A large live preview, on a checkerboard wherever the
* picture is transparent, framed by the same viewfinder corners the
* canvas drew while you dragged - so the picture reads as "the thing I
* just caught" rather than a file.
* 2. HOW SHOULD IT LOOK? Background (the page as you saw it / clean white /
* none) and sharpness (1x-3x, starting at what the screen showed). Every
* change re-renders the preview; nothing is applied blind.
* 3. WHERE DOES IT GO? Out of the app (Copy, Share, Save) or into your notes
* (this page, a new page, another notebook). Two groups because they are
* two different intents - sending something away and keeping it.
*
* Compared with the apps this was modelled on: Samsung's smart select has
* Copy / Share / Add to note but no look controls; GoodNotes' lasso screenshot
* is copy-and-share only; Excalidraw's export has background and scale but
* nowhere to put the result. This keeps all three and adds nothing else.
*
* Honesty rules, as in ExportModal: every outcome is reported as what it was.
* A copy the Android clipboard refused says so (and is still pasteable inside
* ProNotes); a dismissed share sheet is not called a failure; a picture or PDF
* page that could not be loaded is named, not silently left blank.
*/

import React, { useCallback, useEffect, useMemo, useRef, useState } from "react";
import { useTheme } from "next-themes";
import {
  ArrowLeft,
  BookOpen,
  Check,
  ChevronRight,
  Copy,
  Download,
  FilePlus2,
  Images as GalleryIcon,
  ImagePlus,
  LayoutGrid,
  Loader2,
  NotebookPen,
  Scan,
  Search,
  Share2,
  X,
} from "lucide-react";
import { useWhiteboard } from "@/store/whiteboard";
import { useCapture, type CaptureRequest } from "@/store/capture";
import {
  CAPTURE_SCALES,
  captureFilename,
  capturePixelSize,
  collectLiveText,
  defaultCaptureScale,
  renderCapture,
  resolveCaptureAssets,
  type CaptureAssets,
  type CaptureBackground,
  type CaptureScale,
  type CaptureSource,
} from "@/lib/whiteboard/capture";
import { blobToDataURL, isNativeApp, shareBlob, canChooseApp } from "@/lib/whiteboard/share";
import { GALLERY_ALBUM, saveImageToGallery } from "@/lib/whiteboard/gallery";
import type { ExportReport } from "@/lib/whiteboard/export";
import { putAppClipImage, writeImageToSystemClipboard } from "@/lib/whiteboard/appClipboard";
import { storeImage } from "@/lib/whiteboard/imagestore";
import { getActivePageIndex, getPages, isNotebook } from "@/lib/whiteboard/notebook";
import { toast } from "@/hooks/use-toast";
import { ToastAction } from "@/components/ui/toast";
import { Dialog, DialogContent, DialogDescription, DialogTitle } from "@/components/ui/dialog";
import { cn } from "@/lib/utils";

const BG_KEY = "pronotes:captureBackground";

const BACKGROUNDS: Array<{ id: CaptureBackground; label: string; hint: string }> = [
  { id: "page", label: "Page", hint: "As you see it: paper, ruling and the PDF page." },
  { id: "white", label: "White", hint: "Clean white behind it. Ruling and paper colour left out; PDF pages stay." },
  { id: "transparent", label: "None", hint: "Only your ink and objects, on transparency." },
];

function readBackground(): CaptureBackground {

```

```

    try {
      const v = localStorage.getItem(BG_KEY);
      if (v === "page" || v === "white" || v === "transparent") return v;
    } catch {
      /* a remembered preference is a convenience, never a requirement */
    }
    return "page";
  }

function formatBytes(n: number): string {
  if (n < 1024) return `${n} B`;
  if (n < 1024 * 1024) return `${Math.round(n / 1024)} KB`;
  return `${(n / (1024 * 1024)).toFixed(1)} MB`;
}

/** A sentence naming what the picture had to leave out, or null. */
function describeMissing(r: ExportReport): string | null {
  const parts: string[] = [];
  if (r.missingImages > 0) {
    parts.push(`${r.missingImages} picture${r.missingImages === 1 ? " is" : "s are"} missing from this device
      and shown as grey`);
  }
  if (r.missingPdfPages > 0) {
    parts.push(`${r.missingPdfPages} PDF page${r.missingPdfPages === 1 ? "" : "s"} could not be loaded`);
  }
  return parts.length ? `${parts.join("; ")}` : null;
}

interface Rendered {
  url: string;
  blob: Blob;
  width: number;
  height: number;
  report: ExportReport;
}

/** The little background swatch drawn inside each Background option. */
function Swatch({ kind }: { kind: CaptureBackground }) {
  if (kind === "transparent") {
    return <span className="pn-checker h-3.5 w-3.5 rounded-[4px] border border-black/10 dark:border-white/15"
      />;
  }
  if (kind === "white") {
    return <span className="h-3.5 w-3.5 rounded-[4px] border border-black/15 bg-white" />;
  }
  return (
    <span className="relative h-3.5 w-3.5 overflow-hidden rounded-[4px] border border-black/15 bg-[#fdfaf3]">
      <span className="absolute inset-x-0 top-[4px] h-px bg-sky-300" />
      <span className="absolute inset-x-0 top-[8px] h-px bg-sky-300" />
    </span>
  );
}

/** A labelled row of mutually exclusive choices. */
function Segmented<T extends string | number>({
  label,
  value,
  options,
  onChange,
}: {
  label: string;
  value: T;
  options: Array<{ id: T; label: React.ReactNode; title?: string }>;
  onChange: (v: T) => void;
}) {
  return (
    <div>
      <div className="mb-1 text-[10px] font-semibold uppercase tracking-wider text-muted-foreground">{label}</div>
      <div role="radiogroup" aria-label={label} className="flex rounded-xl bg-muted/70 p-0.5 dark:bg-zinc-800/70">
        {options.map((o) => {
          const on = o.id === value;
          return (
            <button
              key={String(o.id)}
              type="button"
              role="radio"

```

```

        aria-checked={on}
        title={o.title}
        onClick={() => onChange(o.id)}
        className={cn(
          "flex h-8 flex-1 items-center justify-center gap-1.5 rounded-[10px] text-[12px] font-medium transition-all",
          on
            ? "bg-white text-foreground shadow-sm ring-1 ring-black/5 dark:bg-zinc-600 dark:ring-white/10"
            : "text-muted-foreground hover:text-foreground"
        )}
      >
        {o.label}
      </button>
    );
  }
}
</div>
</div>
);
}

/** One "send it out" button: icon over label, a finger-sized target. */
function SendTile({
  icon: Icon,
  label,
  onClick,
  busy,
  done,
  primary,
  disabled,
}): {
  icon: typeof Copy;
  label: string;
  onClick: () => void;
  busy?: boolean;
  done?: boolean;
  primary?: boolean;
  disabled?: boolean;
}) {
  return (
    <button
      type="button"
      onClick={onClick}
      disabled={disabled}
      className={cn(
        "flex h-[62px] flex-col items-center justify-center gap-1 rounded-2xl text-[12px] font-semibold transition active:scale-[0.97] disabled:opacity-50 disabled:active:scale-100",
        primary
          ? "bg-gradient-to-br from-indigo-500 to-violet-500 text-white shadow-md shadow-indigo-500/25 hover:from-indigo-400 hover:to-violet-400"
          : "border border-border bg-white text-foreground hover:border-indigo-300 hover:bg-indigo-50/60 dark:bg-zinc-800/60 dark:hover:border-indigo-400/50 dark:hover:bg-zinc-800"
      )}
    >
      {busy ? <Loader2 size={19} className="animate-spin" /> : done ? <Check size={19} /> : <Icon size={19} />}
      {label}
    </button>
  );
}

/** One "keep it in my notes" row. */
function AddRow({
  icon: Icon,
  title,
  hint,
  onClick,
  disabled,
  chevron,
}): {
  icon: typeof Copy;
  title: string;
  hint: string;
  onClick: () => void;
  disabled?: boolean;
  chevron?: boolean;
}) {
  return (

```

```

    <button
      type="button"
      onClick={onClick}
      disabled={disabled}
      className="group flex w-full items-center gap-3 rounded-xl px-2.5 py-2 text-left transition hover:bg-muted/80 disabled:opacity-50 dark:hover:bg-zinc-800"
    >
      <span className="flex h-8 w-8 shrink-0 items-center justify-center rounded-lg bg-indigo-500/10 text-indigo-600 transition group-hover:bg-indigo-500/15 dark:text-indigo-300">
        <Icon size={16} />
      </span>
      <span className="min-w-0 flex-1">
        <span className="block truncate text-[13px] font-medium leading-tight">{title}</span>
        <span className="block truncate text-[11px] leading-tight text-muted-foreground">{hint}</span>
      </span>
      {chevron && <ChevronRight size={15} className="shrink-0 text-muted-foreground" />}
    </button>
  );
}

/** The viewfinder corners that frame the preview (the canvas marquee's motif). */
function Corners() {
  const c = "pointer-events-none absolute h-4 w-4 border-indigo-500";
  return (
    <>
      <span className={cn(c, "-left-2 -top-2 rounded-tl-md border-l-[3px] border-t-[3px]}") />
      <span className={cn(c, "-right-2 -top-2 rounded-tr-md border-r-[3px] border-t-[3px]}") />
      <span className={cn(c, "-bottom-2 -right-2 rounded-br-md border-b-[3px] border-r-[3px]}") />
      <span className={cn(c, "-bottom-2 -left-2 rounded-bl-md border-b-[3px] border-l-[3px]}") />
    </>
  );
}

export function CapturePanel() {
  const request = useCapture((s) => s.request);
  const close = useCapture((s) => s.close);
  // Keyed by the request so every capture starts from a clean slate.
  return (
    <Dialog open={!request} onOpenChange={(o) => !o && close()}>
      {request && <CaptureDialogBody key={request.source.region.x + ":" + request.source.region.y} request={request} onClose={close} />}
    </Dialog>
  );
}

function CaptureDialogBody({ request, onClose }: { request: CaptureRequest; onClose: () => void }) {
  const { resolvedTheme } = useTheme();
  const dark = resolvedTheme === "dark";
  const customLinkTypes = useWhiteboard((s) => s.customLinkTypes);
  const boards = useWhiteboard((s) => s.boards);
  const board = request.source.board;
  const notebook = isNotebook(board);

  const [background, setBackground] = useState<CaptureBackground>("page");
  const [scale, setScale] = useState<CaptureScale>(() => defaultCaptureScale(request.zoom, request.dpr));
  const [onlySelection, setOnlySelection] = useState(false);
  const [rendered, setRendered] = useState<Rendered | null>(null);
  const [rendering, setRendering] = useState(true);
  const [renderError, setRenderError] = useState<string | null>(null);
  const [busy, setBusy] = useState<null | "copy" | "share" | "save" | "add">(null);
  const [copied, setCopied] = useState(false);
  const [view, setView] = useState<"main" | "pick">("main");
  const [query, setQuery] = useState("");

  useEffect(() => setBackground(readBackground()), []);
  const chooseBackground = (b: CaptureBackground) => {
    setBackground(b);
    try {
      localStorage.setItem(BG_KEY, b);
    } catch {
      /* a remembered preference is a convenience, never a requirement */
    }
  };

  const source: CaptureSource = useMemo(
    () => ({
      ...request.source,
      onlyIds: onlySelection && request.selectionIds ? request.selectionIds : undefined,
    })
  );
}

```

```

    }},
    [request, onlySelection]
  );

  // The canvas' own text boxes, read once, while the canvas still shows them.
  const liveText = useMemo(() => collectLiveText(request.source), [request.source]);

  // Pictures and PDF pages are fetched once per sharpness; switching the
  // background or the selection filter only re-draws.
  const assetsRef = useRef(new Map<number, Promise<CaptureAssets>>());
  const assetsFor = useCallback(
    (s: number) => {
      let p = assetsRef.current.get(s);
      if (!p) {
        p = resolveCaptureAssets(request.source, s);
        assetsRef.current.set(s, p);
        p.catch(() => assetsRef.current.delete(s));
      }
      return p;
    },
    [request.source]
  );

  // The live preview. A newer setting wins over a render still in flight, and
  // the previous picture stays up (dimmed) until its replacement is ready, so
  // the preview never flashes empty.
  const renderSeq = useRef(0);
  useEffect(() => {
    const seq = ++renderSeq.current;
    setRendering(true);
    setRenderError(null);
    (async () => {
      try {
        const [assets, live] = await Promise.all([assetsFor(scale), liveText]);
        const out = await renderCapture(source, { background, dark }, scale, customLinkTypes, assets, live);
        if (seq !== renderSeq.current) return;
        const url = URL.createObjectURL(out.blob);
        setRendered((prev) => {
          if (prev) URL.revokeObjectURL(prev.url);
          return { url, ...out };
        });
      } catch (err) {
        if (seq !== renderSeq.current) return;
        console.error("[capture] render failed", err);
        setRenderError((err as Error).message || "The picture could not be made.");
      } finally {
        if (seq === renderSeq.current) setRendering(false);
      }
    })();
  }, [source, background, dark, scale, customLinkTypes, assetsFor, liveText]);

  // Free the last preview when the panel goes.
  const lastUrl = useRef<string | null>(null);
  lastUrl.current = rendered?.url ?? null;
  useEffect(() => () => {
    if (lastUrl.current) URL.revokeObjectURL(lastUrl.current);
  }, []);

  // Fit the preview into its stage by measurement: percentage max-heights do
  // not resolve inside a flexible stage, and the corners must hug the image.
  // A callback ref, because the dialog's portal mounts its content a render
  // after this component, so an effect reading a ref would find nothing.
  const [stageEl, stageRef] = useState<HTMLDivElement | null>(null);
  const [stage, setStage] = useState({ width: 0, height: 0 });
  useEffect(() => {
    const el = stageEl;
    if (!el) return;
    const ro = new ResizeObserver(([entry]) => {
      const r = entry.contentRect;
      setStage({ width: r.width, height: r.height });
    });
    ro.observe(el);
    return () => ro.disconnect();
  }, [stageEl]);

  const region = request.source.region;
  const pixel = capturePixelSize(region, scale);
  const fitted = useMemo(() => {

```

```

    if (!stage.width || !stage.height) return null;
    const pad = 28;
    const k = Math.min((stage.width - pad * 2) / region.width, (stage.height - pad * 2) / region.height, 4);
    return { width: Math.max(8, region.width * k), height: Math.max(8, region.height * k) };
  }, [stage, region.width, region.height]);

const filename = useMemo(() => captureFilename(board.name), [board.name]);
const native = isNativeApp();
const missing = rendered ? describeMissing(rendered.report) : null;
const ready = !!rendered && !rendering;
const pageNo = notebook ? request.source.anchorIndex + 1 : null;

// ---- Actions -----

const copy = async () => {
  if (!rendered || busy) return;
  setBusy("copy");
  try {
    // First, while the tap still counts as a user gesture.
    const system = await writeImageToSystemClipboard(rendered.blob);
    const dataUrl = await blobToDataUrl(rendered.blob);
    putAppClipImage({
      dataUrl,
      width: rendered.width,
      height: rendered.height,
      worldSize: { width: region.width, height: region.height },
      systemCopied: system,
    });
    setCopied(true);
    setTimeout(() => setCopied(false), 1800);
    toast(
      system
        ? { title: "Copied", description: "Paste it into any app, or back into ProNotes." }
        : {
            title: "Copied inside ProNotes",
            description: native
              ? "Android doesn't let ProNotes put pictures on the system clipboard. Paste it here (Add picture → Paste), or use Share to send it to another app."
              : "This browser blocked the system clipboard. Paste it here with Ctrl+V, or use Save or Share."
          },
    );
  } finally {
    setBusy(null);
  }
};

const share = async () => {
  if (!rendered || busy) return;
  setBusy("share");
  try {
    const outcome = await shareBlob(rendered.blob, filename, board.name);
    if (outcome === "cancelled") {
      toast({ title: "Not sent", description: "The share sheet was closed; the picture went nowhere." });
      return;
    }
    toast(
      outcome === "downloaded"
        ? { title: "Saved instead", description: `No app chooser is available here, so it was saved as ${filename}.` }
        : { title: "Shared", description: filename },
    );
    onClose();
  } catch (err) {
    toast({ title: "That didn't work", description: (err as Error).message, variant: "destructive" });
  } finally {
    setBusy(null);
  }
};

const save = async () => {
  if (!rendered || busy) return;
  setBusy("save");
  try {
    const outcome = await saveImageToGallery(rendered.blob, filename);
    if (outcome === "cancelled") {
      toast({ title: "Not saved", description: "The share sheet was closed; the picture was not saved." });
      return;
    }
  }
};

```

```

    }
    toast(
      outcome === "gallery"
        ? { title: "Saved to Gallery", description: `In the ${GALLERY_ALBUM} album.` }
        : outcome === "downloaded"
          ? { title: "Saved", description: `Downloaded as ${filename}.` }
          : {
              title: "Choose where to save it",
              description: "This version of the app can't write to the Gallery directly; pick Save or Gallery in the share sheet.",
            }
    );
    onClose();
  } catch (err) {
    toast({ title: "That didn't work", description: (err as Error).message, variant: "destructive" });
  } finally {
    setBusy(null);
  }
};

const size = { width: region.width, height: region.height };

const addHere = async () => {
  if (!rendered || busy) return;
  setBusy("add");
  try {
    const dataUrl = await blobToDataURL(rendered.blob);
    onClose();
    // The canvas owns placement (it knows the view and the sheet): centred
    // in view, at the size it was captured, selected, and one Undo away.
    window.dispatchEvent(new CustomEvent("pronotes:insertimage", { detail: { dataUrl, size } }));
    toast({ title: notebook ? "Added to this page" : "Added to this board", description: "It's selected: drag it where you want it." });
  } finally {
    setBusy(null);
  }
};

const addTo = async (targetId: string, afterCurrentPage: boolean) => {
  if (!rendered || busy) return;
  setBusy("add");
  try {
    const src = await storeImage(await blobToDataURL(rendered.blob));
    const st = useWhiteboard.getState();
    const target = st.boards.find((b) => b.id === targetId);
    if (!target) return;
    const res = st.addCaptureToBoard(targetId, {
      src,
      naturalWidth: rendered.width,
      naturalHeight: rendered.height,
      size,
      afterPageId: afterCurrentPage ? target.activePageId : undefined,
    });
    onClose();
    if (!res) return;
    const vpHeight = () => useWhiteboard.getState().canvasViewport.height;
    const open = () => {
      const s = useWhiteboard.getState();
      if (s.activeBoardId !== targetId) s.setActiveBoard(targetId);
      if (res.pageId) {
        const pageId = res.pageId;
        requestAnimationFrame(() => useWhiteboard.getState().goToPage(pageId, vpHeight()));
      }
    };
  };
  if (targetId === st.activeBoardId) {
    open();
    toast({ title: `Added as page ${res.pageNumber}`, description: "Undo takes it back out." });
    return;
  }
  const where = res.pageNumber ? `as page ${res.pageNumber}` : "beside its content";
  const t = toast({
    title: `Added to "${target.name}"`,
    description: `${where[0].toUpperCase()}${where.slice(1)}. To take it out, open it and Undo.`,
    action: (
      <ToastAction
        altText="Open it"
        onClick={() => {
          open();
        }}
      />
    )
  });
};

```

```

        t.dismiss();
      }}
    >
    Open
  </ToastAction>
),
});
} catch (err) {
  toast({ title: "Couldn't add it", description: (err as Error).message, variant: "destructive" });
} finally {
  setBusy(null);
}
};

// Ctrl/Cmd+C copies, the way it does everywhere else a picture is on screen.
const copyRef = useRef(copy);
copyRef.current = copy;
useEffect(() => {
  const onKey = (e: KeyboardEvent) => {
    if ((e.ctrlKey || e.metaKey) && e.code === "KeyC" && !(e.target instanceof HTMLInputElement)) {
      e.preventDefault();
      void copyRef.current();
    }
  };
  window.addEventListener("keydown", onKey);
  return () => window.removeEventListener("keydown", onKey);
}, []);

// ---- Where it can go -----

const others = useMemo(() => {
  const q = query.trim().toLowerCase();
  return boards
    .filter(b => b.id !== board.id && (!q || (b.name || "").toLowerCase().includes(q)))
    .sort((a, b) => {
      // Notebooks first: that is what "another notebook" means; then recent.
      const k = Number(isNotebook(b)) - Number(isNotebook(a));
      return k || (b.updatedAt ?? 0) - (a.updatedAt ?? 0);
    })
    .slice(0, 60);
}, [boards, board.id, query]);

const liveBoard = boards.find((b) => b.id === board.id);
const livePage = liveBoard && isNotebook(liveBoard) ? getActivePageIndex(liveBoard) + 1 : null;

const origin =
  request.origin === "selection" ? "Selection" : request.shape === "lasso" ? "Freeform" : "Area";

// ---- Layout -----

return (
  <DialogContent
    showCloseButton={false}
    onOpenAutoFocus={(e) => e.preventDefault()}
    className="flex max-h-[calc(100dvh-1.5rem)] w-[min(880px,calc(100vw-1.5rem))] max-w-none flex-col gap-0
      overflow-hidden rounded-3xl border border-border/80 bg-white p-0 shadow-2xl sm:max-w-none dark:bg-zinc-900"
  >
    <style>{`
      .pn-checker { background-color: #ffffff; background-image: conic-gradient(#e4e4e7 25%, transparent 0 50%, #e4e4e7 0 75%, transparent 0); background-size: 14px 14px; }
      .dark .pn-checker { background-color: #27272a; background-image: conic-gradient(#3f3f46 25%, transparent 0 50%, #3f3f46 0 75%, transparent 0); }
    `}</style>

    {/* Header */}
    <div className="flex items-center gap-3 border-b border-border/60 px-4 py-3">
      <span className="flex h-9 w-9 shrink-0 items-center justify-center rounded-xl bg-gradient-to-br from-indigo-500 to-violet-500 text-white shadow-md shadow-indigo-500/30">
        <Scan size={18} />
      </span>
      <div className="min-w-0 flex-1">
        <DialogTitle className="text-[15px] font-semibold leading-tight">Capture</DialogTitle>
        <DialogDescription className="truncate text-[12px] text-muted-foreground">
          {origin} from {board.name || "Untitled"}
          {pageNo ? ` · page ${pageNo}` : ""}
        </DialogDescription>
      </div>
    </div>
  >

```

```

<button
  type="button"
  onClick={onClose}
  aria-label="Close"
  className="flex h-9 w-9 items-center justify-center rounded-full text-muted-foreground transition ho
    ver:bg-muted hover:text-foreground"
>
  <X size={18} />
</button>
</div>

<div className="@container min-h-0 flex-1 overflow-y-auto">
  <div className="flex flex-col @[40rem]:h-full @[40rem]:flex-row">
    {/* Preview: the light table */}
    <div className="relative min-w-0 flex-1 bg-zinc-100 dark:bg-zinc-950/70">
      <div
        ref={stageRef}
        className="relative h-[36vh] min-h-[190px] @[40rem]:h-full @[40rem]:min-h-[420px]"
      >
        {fitted && (rendered || renderError) && (
          <div className="absolute inset-0 flex items-center justify-center">
            {rendered ? (
              <div className="relative" style={{ width: fitted.width, height: fitted.height }}>
                <Corners />
                {/* eslint-disable-next-line @next/next/no-img-element -- a local blob, not a page asset
                  */}
                <img
                  src={rendered.url}
                  alt="Preview of the capture"
                  draggable={false}
                  className={cn(
                    "pn-checker h-full w-full rounded-[3px] object-contain shadow-[0_8px_30px_rgba(0,0,0
                      ,0.18)] ring-1 ring-black/10 transition-opacity dark:ring-white/10",
                    rendering && "opacity-60"
                  )}
                />
              </div>
            ) : (
              <p className="max-w-xs px-6 text-center text-[13px] text-red-600 dark:text-red-400">{rende
                rError}</p>
            )}
          </div>
        )}
        {!rendered && !renderError && (
          <div className="absolute inset-0 flex items-center justify-center text-muted-foreground">
            <Loader2 size={22} className="animate-spin" />
          </div>
        )}
      </div>
      {/* Readouts */}
      <div className="pointer-events-none absolute bottom-2.5 left-2.5 flex flex-wrap gap-1.5">
        <span className="rounded-full bg-zinc-900/80 px-2.5 py-1 text-[11px] font-medium tabular-nums
          text-white backdrop-blur">
          {pixel.width} × {pixel.height} px
        </span>
        {rendered && (
          <span className="rounded-full bg-zinc-900/80 px-2.5 py-1 text-[11px] font-medium tabular-num
            s text-white/85 backdrop-blur">
            PNG · {formatBytes(rendered.blob.size)}
          </span>
        )}
      </div>
      {rendering && rendered && (
        <span className="pointer-events-none absolute right-2.5 top-2.5 flex items-center gap-1.5 roun
          ded-full bg-zinc-900/80 px-2.5 py-1 text-[11px] font-medium text-white backdrop-blur">
          <Loader2 size={12} className="animate-spin" /> Updating
        </span>
      )}
    </div>
  </div>

  {/* Controls rail */}
  <div className="flex shrink-0 flex-col gap-4 border-t border-border/60 p-4 @[40rem]:w-[18.5rem] @[40
    rem]:overflow-y-auto @[40rem]:border-l @[40rem]:border-t-0">
    {view === "main" ? (
      <>
        <section className="space-y-3">
          <Segmented
            label="Background"

```